

# Bonds of Love: Patriotism and the Rise of Modern Banks

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## Abstract

This study examines the role of patriotism in the rise of modern banks when they emerged as a new form of financial institution. In China, the loss of Western financial support during WWI (1914) prompted the Republican government to issue patriotic domestic bonds to address fiscal shortfalls. Modern banks underwrote these bonds and expanded into regions with stronger patriotic sentiment to attract subscriptions. These banks thus established themselves by achieving strong profitability and gaining long-term public familiarity and trust. Historical evidence suggests that patriotism similarly fueled early banking development in other countries.

Keywords: Patriotism, social contract, trust, modern banks, government bonds JEL Classification: G21, N25, N45, P16

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# 1 Introduction

Trust in government is a fragile foundation for banking. When state credibility is strong, government endorsement stabilizes financial systems, such as during bank runs ([Diamond and Dybvig, 1983](#)). Yet, the state's credibility is often most fragile precisely during the crises that necessitate its intervention. This tension is particularly acute during the formative stages of modern banking. In many countries, the earliest modern banks were established by the state to address fiscal shortages following wars and revolutions.<sup>1</sup> How did the nascent modern banks establish themselves and grow when public confidence in the state was weak? This paper proposes that patriotism substituted for formal institutional credibility and fueled the rise of early modern banks.

In this paper, patriotism refers to citizens' willingness to engage in a reciprocal relationship specifically with the state, distinct from broader social capital or collective action capacity, which may facilitate coordination among citizens without reference to the state. Our empirical analysis focuses on early twentieth-century China, a period that provides a unique setting to test our hypothesis. The genesis of modern banking in China emerged from a half-century of crisis. Following the Opium Wars of the 1840s, the Chinese government (both Qing and its Republican successor) struggled with persistent fiscal deficits and political instability, relying heavily on Western countries and foreign banks for financing ([Xu, 1962](#)). The outbreak of World War I in 1914 created an exogenous fiscal shock as foreign creditors withdrew their support. In response, the Chinese Republican government turned to domestic bond issuance, appealing to patriotism to encourage subscriptions. This shift directly spurred the rise of modern Chinese banks, many of which first earned profits by underwriting and trading these patriotic bonds ([Du, 2012](#); [Yan, 2015](#)). As underwriters, banks purchased these bonds from the government at deep discounts and resold them at market prices; the profitability of this intermediation depended critically on public willingness to subscribe, which in turn was driven by patriotic sentiment.

This historical setting supports a Difference-in-Differences (DiD) strategy to gauge the causal effects of patriotism on modern banking development. The first difference is the fiscal shock of 1914 that spurred the domestic bond market, and the second difference is the cross-sectional variations in patriotic sentiment. We find that, following the 1914 fiscal shock, regions with stronger patriotic sentiment saw larger government bond subscriptions and, consequently, greater growth in modern banks.<sup>2</sup> Appendix Figure B1 illustrates this historical timeline.

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<sup>1</sup>For instance, the Bank of England, the earliest modern bank in Great Britain, was founded in 1694 as a private bank but served as the government's banker to finance the Nine Years' War (1688–1697) and to manage the escalating national debt ([North and Weingast, 1989](#)). The public's initial interactions with the bank involved patriotic financial instruments, which included subscriptions to bank shares to support government lending and the purchase of government bonds underwritten by the bank. During the Nine Years' War, the public viewed supporting the government through monetary contributions as a "patriotic duty" ([Murphy, 2006](#)).

<sup>2</sup>Rich historical anecdotes support the idea that the rise of the domestic government bond market largely fueled the emergence of modern Chinese banks, as many were initially established to underwrite and trade government

Defining and measuring regional patriotism is challenging due to its intangible nature. To define it, we use a social contract theory framework, which posits that citizens enter into an “agreement” with the state, exchanging civic engagement for rights and protection (Locke, 1689; Rousseau, 1762). Civilians dedicate themselves to state-building through tax contributions, military service, and other forms of voluntary support for the state (Besley, 2020; Laudenbach, Malmendier, and Niessen-Ruenzi, 2025; Qian and Tabellini, 2025). Patriotism can thus be viewed as civilians’ willingness to engage in a form of social contract. Note that patriotism drives both devotion and active civic participation in state affairs; the same strength of the social contract can manifest differently depending on the legitimacy of the regime. When the state underperforms, civilians exhibit constructive criticism, participating in public discussions or protests to constrain the government’s conduct (Schatz, Staub, and Lavine, 1999).

In our empirical setting, the DiD fiscal shock occurs in 1914, shortly after China’s regime change from the Qing to the Republic in 1911. To construct sensible ex-ante, cross-regional predictors of patriotism (or measures of social contract strength), we exploit this transition and focus on two dimensions: (1) oppositional anti-tax unrest against the delegitimized late Qing state (1902–1911), and (2) supportive civic engagements under the new Republican regime (post-1912). In the DiD design, we use anti-tax unrest as the main treatment variable (measured entirely before 1912, making it a genuinely pre-treatment measure of civic capital that predicts patriotic engagement after the regime transition, free from contamination by the 1914 fiscal shock), and employ either the principal component or average standardized ( $z$ -score) index of all patriotism measures before 1914 as robustness checks.

While it may seem counterintuitive that resistance to Qing taxation predicts patriotic engagement with the Republican state, the logic is straightforward: those who resisted Qing taxation were precisely those most attuned to the reciprocal obligations between state and citizen, and therefore most prepared to honor those obligations toward a state they deemed legitimate.<sup>3</sup> We develop three supporting arguments, illustrated with anecdotal evidence, in Section 2.4.

In Section 4, we empirically validate our proxy by showing that prefectures with more anti-tax unrest under the Qing subsequently exhibited stronger patriotic civic engagement under the Republic, measured by (1) military participation, i.e., the number of cadets enrolled at the *Baoding Military Academy*, (2) political participation, i.e., the number of associations promoting political ideologies, (3) social participation, i.e., the number of associations supporting social welfare, and (4) financial participation, i.e., subscription amounts to the 1914 national Patri-

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bonds as well as to issue banknotes backed by government bonds (Du, 2012; Yan, 2015). Naturally, these banks expanded in areas where public support for government bonds was strong. Amid the uncertainties of the time, people’s patriotic sentiment acted as a catalyst for their trust in the nascent Republican government and for their support of government bonds (Zhu, 1993).

<sup>3</sup>In social contract theory, tax compliance represents a fundamental form of citizens’ consent to state legitimacy (Locke, 1689). Therefore, resistance to taxation in the late Qing China reflected an awareness of the principles of the social contract, although perhaps not formally articulated.

otic Bond. At the province level, we further find that regions with stronger anti-tax resistance raised significantly more funds through government bond issuance following the outbreak of WWI, providing additional evidence that anti-tax unrest predicts the patriotic sentiment successfully mobilized by the Republican government. These findings affirm that anti-tax unrest in the late Qing, far from being mere disorder, captured the strength of civic consciousness that later expressed itself as patriotic state-building under the Republic.

In Section 5, we formally examine the extent to which the growth of modern banks was influenced by patriotic sentiment. The dependent variable, banking development, is proxied by the number of modern Chinese banks in each prefecture-year. This dataset was manually compiled from more than 2,000 local gazetteers. The analysis is based on the DiD strategy. Consistent with our hypothesis, prefectures with more anti-tax unrest between 1902 and 1911 saw a greater increase in modern banks after 1914. Each additional anti-tax unrest correlates with a 21% increase in the number of banks when evaluated at the mean. The results are robust when we use either the principal component or the average standardized  $z$ -scores of anti-tax unrest and civic engagement before 1914 as proxies for overall patriotism. A one-standard-deviation increase in patriotism is associated with a 0.5-standard-deviation increase in the number of banks. Dynamic DiD analysis shows no parallel trends and a sharp jump in the number of banks immediately after 1914.

Even with the DiD strategy, one might be concerned that endogeneity issues remain due to omitted variables. For instance, regions with stronger economic foundations may have had more exposure to modern ideologies, which could have fostered civic awareness and banking development after 1914. To address this concern, we employ an instrumental variable (IV) approach, using early Qing massacres (1644–1661) to instrument late Qing anti-tax unrest. These brutal massacres, carried out during the Qing conquest, significantly reduced populations and left a legacy of collective mistrust toward the government. We hypothesize that regions affected by these massacres were more likely to see unrest resisting tax payment when central control weakened. The exclusion restriction is plausibly satisfied, as the locations of the massacres were exogenously determined by Qing military campaigns, and the economic devastation caused by these events was unlikely to directly promote banking development. The instrumented results support our hypothesis, demonstrating a causal relationship between anti-tax unrest and banking growth.

To account for other confounding factors that may have influenced banking development around WWI, we introduce additional controls: (1) number of industrial firms, because industrialization drives financial demand; (2) treaty ports, because foreign exposure was a key driver of economic modernization; (3) import/export duties, to distinguish trade shocks from fiscal shocks, both induced by WWI; (4) prefectural civil examination quotas  $\times$  post, proxying human capital supply; (5) Confucian clan  $\times$  post, considering the influence of internal and informal

financial markets; (6) telegraph and railway stations, to capture modern communication infrastructure; and (7) local civil wars and natural disasters (floods and droughts), which could create needs for government bond issuance. After incorporating these variables, the estimated effect of patriotism on banking remains robust, with a minimal coefficient change.

Next, we conduct falsification tests to rule out alternative explanations. First, it is possible that anti-tax unrest merely reflects high tax burdens. To address this, we compile data on (1) per capita land tax in each prefecture for the year 1820 (log) and (2) the number of commercial tax bureaus in each prefecture, and use them as alternative treatment variables in the DiD regressions. The interactions of these tax burden measures with the post-1914 indicator show no significant impact on banking development.

Second, we differentiate anti-tax unrest from radical nationalism or xenophobia (Schatz et al., 1999). Unlike anti-tax protests, which aim to uphold social contract principles, xenophobia directed anger toward foreigners residing in China and overlooked the Qing government's responsibility for domestic issues. We use an indicator variable for the occurrence of either the Boxer Rebellion<sup>4</sup>, or anti-foreigner unrest (e.g., attacks on foreigners, missionaries, or churches) between 1902 and 1911. Our DiD regression shows that these xenophobic measures did not significantly impact banking development post-1914.

Finally, to ensure that anti-tax unrest does not simply capture general discontent, we examine other types of unrest that were not explicitly directed at the Qing government: (1) anti-gentry protests, (2) peasant riots, (3) banditry activities, (4) secret societies, and (5) strikes. These forms of unrest showed weak or no significant impact on banking development, in particular after excluding economically developed Jiangsu and Zhejiang provinces. Crucially, these alternative forms of unrest required comparable or even greater collective action capacity than anti-tax protests. Their insignificance rules out the possibility that our results merely capture a region's organizational capacity. What distinguishes anti-tax unrest is its target: the state's claim to fiscal authority. This specificity reflects the hallmark of social-contract-based patriotism. Overall, the falsification tests offer robust justification for using anti-tax unrest as a plausible measure of social-contract strength that influenced the development of modern banks.

Section 6 proposes two channels through which patriotic sentiment fostered banking growth. The first is a direct profitability channel: modern banks increased their earnings by underwriting and trading government bonds, strengthening their financial foundations. We use bank-level balance sheet data to show that banks operating in regions with stronger patriotic sentiment held more bonds and achieved higher performance.

The second is a longer-term public confidence channel: repeated interactions between citizens and modern banks cultivated familiarity-and-trust in these nascent institutions. We provide

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<sup>4</sup>The Boxer Movement was an anti-foreign and anti-Christian riot between 1899 and 1901. The Boxers were particularly hostile towards Christian missionaries and Chinese converts, whom they blamed for the state's crisis (Esherick, 1987).

suggestive evidence on this channel by employing two indirect tests. First, we compare the development of official and private banks. Historical records indicate that government bonds were initially primarily underwritten by official banks. As official banks demonstrated operational reliability, the public became more familiar with modern banking as an institution, making the public more receptive to engaging with private banks offering similar services. Dynamic DiD analyses show that while anti-tax unrest spurred the growth of official banks immediately after 1914, its impact on private banks grew only gradually, indicating a time-lagged spillover consistent with the slow buildup of public familiarity and trust. Second, we exploit the 1921 banking crisis as a revealed-preference test of public confidence. Bank runs, by their nature, reflect depositors' mistrust in financial institutions; regions where branches survived are thus regions where public confidence proved resilient. During this crisis, widespread bank runs occurred. After controlling for bank fixed effects, we find that branches in prefectures with stronger patriotic sentiment were significantly more likely to survive, suggesting that patriotic engagement contributed to banking resilience beyond what bank-level fundamentals alone can explain.

In Section 7, we present anecdotes suggesting that our findings may generalize to other countries. In nations such as Great Britain, the United States, and Japan, governments established early modern banks to address public finance shortages following revolutions and wars. These governments and banks often appealed to patriotism when raising funds from the populace. By contributing financially, citizens not only supported their nations, but also contributed to the growth and legitimacy of the emerging modern banks.

While our study examines the nascent period of banking systems, it offers broader implications for financial development. Capable financial systems depend heavily on trust in public institutions; for instance, government backing is a common approach to mitigate financial instability such as bank runs. Maintaining public trust is crucial for the resilience of financial systems. However, trust in governments is not innate and often deteriorates during financial crises (Algan, Guriev, Papaioannou, and Passari, 2017; Doerr, Gissler, Peydró, and Voth, 2022). When that happens, patriotism can function as a last-resort safeguard, boosting public confidence and ensuring financial stability. While previous research has identified various trust determinants, including culture (Guiso, Sapienza, and Zingales, 2006; Algan and Cahuc, 2010) and historical contingencies (Stevenson and Wolfers, 2011; Nunn and Wantchekon, 2011; An, Hou, and Lin, 2022), patriotism has received little attention. This paper proposes patriotism as an additional and crucial source of trust and examines its role in fostering banking development.

Our study contributes to three strands of literature. First, it complements the literature on the state-building dynamics within the framework of the social contract theory (Besley, 2020; Caprettini and Voth, 2023; Qian and Tabellini, 2025). We expand this literature by showing that social contracts can synergistically foster financial development, as patriotic sentiment promotes public confidence in government (Gangl, Torgler, and Kirchler, 2016; Depetris-Chauvin,

Durante, and Campante, 2020), which encourages government bond subscriptions and in turn drives banking growth.

Second, this paper sheds light on the determinants of modern banking development. Existing research has emphasized the roles of laws and institutions (Levine, Lin, Ma, and Xu, 2023), technological advances (Atack, Jaremski, and Rousseau, 2014; Lin, Ma, Sun, and Xu, 2021), human capital (Lin, Ma, Sun, and Xu, 2025), culture (Guiso et al., 2006; Chen, Ma, and Sinclair, 2022), and historical endowments (Pascali, 2016; D’Acunto, Prokopczuk, and Weber, 2019; An et al., 2022). To the best of our knowledge, patriotism has not been systematically examined as a driver of banking development.

Third, our study speaks to the literature on the influence of government bonds on banking development (Hilt, Jaremski, and Rahn, 2022; Tang and Basco, 2023), particularly within China’s historical context (He, 2013; Ho and Li, 2014). We propose that patriotic sentiment shapes the relationship between government bonds and banks, especially during periods when government creditworthiness is in question. Our analysis indicates that patriotism can mitigate the “credible commitment problem” (North and Weingast, 1989; Ma and Rubin, 2019), thereby enhancing financial stability and growth. More broadly, our findings suggest that non-pecuniary factors, specifically the civic foundations of public trust, can substitute for formal institutional credibility in financial market development, with implications for emerging markets today.

The paper is structured as follows: Section 2 gives an overview of the historical background. Section 3 elaborates on the data employed. Section 4 establishes that anti-tax unrest predicts patriotic civic engagement after the regime transition, including bond subscriptions. Section 5 shows that patriotism drove banking development, with the potential mechanisms discussed in Section 6. Section 7 expands the geographic scope of our research and discusses the broader applicability of our findings. Section 8 concludes.

## 2 Historical background

### 2.1 Development of modern banks in China

After China’s opening in the 1840s, a few foreign banks (e.g., HSBC) established their business in China (Cheng, 2003). In 1897, the Imperial Bank of China, which mimicked foreign banks, was founded as the first modern Chinese-owned bank. While foreign banks in China remained few throughout the twentieth century, modern Chinese banks experienced significant growth. By 1926, there were 866 Chinese banks (254 headquarters and 612 branches) operating in 165, or 57.5%, of Chinese prefectures within our study sample.

Before the emergence of modern banks, China’s financial system was dominated by traditional banks, in particular *qianzhuang* (“money houses”) and *piaohao* (“draft banks”). Their

operations relied heavily on informal “soft” information, for example, individuals’ reputation within the local community (Huang, 2005). Modern banks differed from traditional banks in several dimensions. Modern banks had limited liability, allowing them to attract more capital and shareholders. They adopted modern governance structures, including shareholder meetings, independent boards, and modern accounting standards (Brett, 2003).

The birth of modern Chinese banks was tightly linked to the central government. While proposing the establishment of the first modern Chinese bank to the Qing court, Sheng Xuanhuai <sup>5</sup> advocated in his memorandum in 1896:

*China should emulate Western countries by consolidating financial resources in modern banks, which are crucial for commercial and industrial development.... The establishment of Chinese banks would contribute to building trust within domestic businessmen and commoners, **facilitating government bond issuance. The objective is to shift from relying on foreign debt**, alleviating the burden of substantial interest rates and exchange rate losses. (Sheng 1974, vol. 2, 27, emphasis added)*

In the early twentieth century, a few other official banks, mainly the Bank of China and the Bank of Communications, were founded to address fiscal needs and reduce reliance on foreign financial support. However, as we will discuss in Section 2.2, modern banks saw little growth in their first decade, likely because the Qing government was unable to mobilize the domestic bond market. Modern banks had yet to establish direct ties with the general public.

Since the 1910s, official banks gained momentum to grow alongside the expansion of domestic bond markets. As official banks matured and the public became familiar with modern banks, private banks developed as well. These private banks actively engaged in bond trading, focusing on secondary markets. In 1912, private banks comprised 22% of all banks in China; by 1926, this share had risen to 43%.

## 2.2 Government bonds in twentieth-century China and the outbreak of WWI

Since the 1850s, the Qing government had relied heavily on high-interest foreign loans, underwritten by foreign banks and backed by customs revenues, to finance military expenses and war indemnities after successive defeats.

Before 1914, China’s domestic government bond market was virtually non-existent. In 1894, the first national debt, *Xijie Shangkuan* (“commercial interest-bearing loans”), functioned more as forced taxation and collapsed within a year due to corruption. Before the Qing dynasty fell in 1911, only two other bonds were issued: *Zhaoxin Gupiao* (“government-issued trust stocks” in 1898) and *Aiguo Gongzhai* (“patriotic bond” in 1911); both failed to meet issuance targets

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<sup>5</sup>Sheng Xuanhuai (1844–1916) was a prominent politician active during the Qing dynasty. He played a pivotal role in modernizing China’s banking, communication, transportation, and education systems.

and resembled compulsory levies rather than market-driven sales (Qian, 1984).<sup>6</sup>

The new Republic of China, established in 1912, inherited the Qing's fiscal shortage; tax revenues were nearly depleted, and the government relied on foreign loans to cover indemnities and military expenses.<sup>7</sup> The outbreak of WWI in 1914 disrupted this reliance, as Western creditors, particularly the United Kingdom and France, redirected funds to the war effort. As shown in Figure 1, this exogenous fiscal shock forced China to shift to domestic bond issuance. Between 1914 and 1926, the total targeted (face) amount of government bonds, denoted *Gongzhai Piao* ("public debt tickets"), reached 756 million yuan (Ling, 1928; Qian, 1984).<sup>8</sup>

[Insert Figure 1 about here]

Government bonds were issued via direct public offerings or indirect bank-led sales, with the latter being dominant. Banks underwrote entire issuances at deep discounts, and they immediately resold the bonds at higher market prices for substantial profits.<sup>9</sup> Banks were also permitted or required to use these bonds as reserves to support banknote issuance, thereby expanding the credit supply. Additionally, banks profited from trading in secondary markets. During the early twentieth century, China's other capital markets remained underdeveloped, making government bonds one of the few viable financial instruments and a significant revenue source for banks (Yan, 2015).<sup>10</sup>

## 2.3 The role of patriotism in government bond issuance

The 1914 Patriotic Bond was the first Chinese government bond to be successfully issued. Initially targeting 16 million yuan at a 7.25% annual interest rate, it exceeded this goal within two months, prompting an increase to 24 million yuan, with final subscriptions reaching 25 million

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<sup>6</sup>For example, the "*Zhaoxin Gupiao*" were intended to raise 100 million taels, but the actual amount collected was estimated to be less than one-fifth of the intended amount (Qian, 1984).

<sup>7</sup>There is a sharp jump in foreign debt around 1911–1913 (Figure 1), which is attributable to the 1913 *Shanhou dajiekuan* ("Reorganization Loan"), contracted by the new government through a Five-Power Consortium of foreign banks (HSBC, Deutsch-Asiatische Bank, Banque de l'Indo-Chine, Russo-Asiatic Bank, and Yokohama Specie Bank) representing Britain, France, Germany, Russia, and Japan, to meet outstanding indemnity obligations. The issue totaled £25 million and was secured primarily by revenues from the Salt Gabelle, China's highly lucrative salt-tax monopoly. The pledging of tax revenues as collateral provoked public outrage and was denounced as a "national humiliation" (Research Institute for Fiscal Science and The Second Historical Archives of China, 1990).

<sup>8</sup>To give an idea of the size of the bonds, we use a Purchasing Power Parity (PPP) approach to estimate their present-day value. The average price of pork in Peking was approximately 15.5 silver yuan per 50 kg during 1914–1924 (Meng and Gamble, 1926, pp. 38–39), and the contemporary retail price for pork is approximately 30 RMB per kg. Therefore, the 756 million yuan would be equivalent to 73.3 billion constant RMB, or approximately 10.25 billion USD.

<sup>9</sup>In practice, the Chinese government used newly issued government bonds as collateral to obtain loans from banks. These bonds were discounted 50%–70% of their nominal value (Qian, 1984).

<sup>10</sup>It is estimated that, before the 1930s, trading government bonds in the secondary market could yield annual returns exceeding 20%, sometimes even reaching 30%–40%. This high yield was attributed to the credit risks associated with the prevailing social instability (Qian, 1984).

yuan.<sup>11</sup> While factors such as government efforts to build credibility contributed,<sup>12</sup> patriotic sentiment played a crucial role. Zhu (1993, p. 57) provides anecdotes detailing the circumstances surrounding the issuance of bonds in 1914 (emphasis added):

*All sectors of society are in desperate need of social stability. Consequently, they support the government's efforts to 'maintain law and order and consolidate the foundations of the state,' leading them to **actively favor the issuance of public debt**....*

*Zhou Jinzhen, the president of the Shanghai Chamber of Commerce, delivered a public speech in Shanghai: 'As **businessmen with a patriotic fervor**, and being part of the financial sector, I urge you to **lead the way by subscribing to this debt** above all other industries.' The businessmen present at the event enthusiastically pledged subscriptions amounting to over 100,000 yuan....*

*At the debt mobilization meeting held by the Hankow Chamber of Commerce, Wang Qinfu remarked, 'Since the devastating fire in Hankow, we have yet to settle our existing debts. Acknowledging **new public debts is certainly challenging**. However, **your willingness to contribute today is a testament to your patriotic spirit**'....*

The 1914 Patriotic Bonds were subscribed by a wide range of social strata, including local officials, merchants, and grassroots such as “tea attendants” and “kitchen staff” who purchased bonds with their wages (Shenbao, 1914).

After 1914, local authorities, including provinces, prefectures, and counties, emulated the successful experience and actively issued government bonds. Between 1914 and 1926, more than 80% of targeted bonds were successfully issued (613 million of 756 million yuan raised), though success rates varied significantly across regions (Qian, 1984).

## 2.4 From anti-tax unrest to patriotism: a social contract interpretation

Repeated defeats by Western powers in the late nineteenth century led China to face heavy indemnities and economic decline, eroding Qing legitimacy. Public frustration and demands for reform fueled widespread support for regime change. As C. E. S. Wakefield, an official at the Changsha Customs, observed on the eve of the collapse of the Qing court in 1911: “*Undoubtedly, most Chinese civilians are calling for a regime change. They are not necessarily revolutionaries, but they sincerely support overthrowing the Qing government*” (Modern Chinese Economic History Materials, 1964, v.13, pp.88).

Wakefield's observation stems from the unprecedentedly high frequency of civilian unrest (“*minbian*”) during the final decade of the Qing dynasty. Zhang and Ding (1982) document 1,293 events of unrest across eighteen provinces during the years 1902–1911, an average of

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<sup>11</sup>The 1914 Patriotic Bond was distributed through two channels: direct public subscriptions and underwriting by the Bank of China and the Bank of Communications. Direct subscriptions accounted for the majority of total proceeds, as documented in Section 3.4.

<sup>12</sup>To enhance bond creditworthiness, the government founded the National Loan Bureau (NLB) and provincial Bonds Promotion Associations to oversee issuance. The NLB's board included representatives from the Bank of China, Bank of Communications, and Maritime Customs officials (Qian, 1984).

one every three days. The unrest targeted a wide range of issues, such as resistance to taxation, to foreigners, and to the gentry class; it took various forms, including protests, strikes, and violent activities; and it involved participants from various social classes, including commoners (peasants and citizens), elites (scholars and wealthy individuals), and even local officials. These actions were believed to have eroded Qing authority, thus fueling popular support for the *Xinhai Revolution* in 1911 (Chen, 1992).

We posit that unrest for the purpose of tax resistance reflects an awareness of a social contract, and the frequency of unrest captures the strength of the social contract, or tie, between people and their government. Unlike anti-gentry protests targeting local elites or xenophobic unrest targeting foreigners, anti-tax resistance was uniquely directed at the state's claim to legitimate authority: those most attuned to the reciprocal obligations between state and citizen were therefore also most prepared to honor those obligations toward a state they deemed legitimate. We propose three reasons why this same civic consciousness later expressed itself as patriotic support for the Republic.

First, within social contract theory, tax compliance represents a fundamental form of citizens' consent to state legitimacy (Rousseau, 1762), and tax resistance reflects collective rejection of the ruling government's legitimacy. Therefore, resistance to taxation in late Qing China reflected an awareness—although perhaps not formally articulated—of the principles of the social contract.<sup>13</sup> A notable example is the *Sichuan Baolu (Railway Protection) Movement*. As a major protest involving tax refusal, it triggered the 1911 *Xinhai Revolution* which toppled the Qing court.<sup>14</sup>

Globally, resistance to taxation was a significant catalyst of several major revolutions. Notable examples include the Boston Tea Party (1773), a protest against taxation without representation that triggered the American Revolutionary War (Smith, 2013), and the French Revolution (1789–1799), in which the nation's taxation burden was a significant cause of revolutionary sentiment (Lefebvre, Palmer, and Tackett, 1947).

Second, resistance to taxation is costly and entails significant personal risk. It signifies not only civic awareness but also citizens' determination and capacity to hold their government accountable. Only when individuals possess a profound sense of civic rights are they compelled to challenge a government they perceive as inadequate or corrupt (Schatz et al., 1999). In the *Sichuan Railway Protection Movement*, the Qing government responded with a violent crackdown, including the use of gunfire (Appendix Figure B1, Panel A, illustrates the scene), which resulted in several dozen deaths (*Xinhai Revolution and Wuchang Uprising Memorial Museum*, 1991).

Third, opposition to an illegitimate state and support for a legitimate one are two expres-

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<sup>13</sup>Lin, Sun, and Wang (2026) also posits that anti-tax unrest was a key determinant for the Qing's collapse.

<sup>14</sup>The movement was sparked by the Qing government's plan to nationalize a locally crowdfunded railway as collateral for foreign loans (*Xinhai Revolution and Wuchang Uprising Memorial Museum*, 1991).

sions of the same underlying civic disposition, not opposing forces (Laudenbach et al., 2025). Crucially, what persisted across the regime transition was not loyalty to any particular government, but the civic disposition itself —the willingness to hold the state accountable and, when the state proved worthy, to actively support it. In the language of social contract theory, regions exhibiting stronger resistance to the Qing regime indicated a heightened desire to “tear up” the old “contract”. This shift also signified a readiness to engage in a new “contract” that embraced the promise of patriotic participation in state-building during the subsequent Republican era. In other words, the same civic consciousness that drove resistance to Qing authority became the wellspring of enthusiastic support for the Republican state.

For example, honoring the victims of the *Sichuan Railway Protection Movement* as heroes has led to lasting commemoration. A monument was erected in 1913 and remains preserved today (Appendix Figure B1, Panel B). The role-model effects reinforced the persistence of civic awareness, which transitioned into state-building efforts under the new Republican regime. On June 3, 1912, Sichuan military and civic leaders submitted a proposal titled “*Sichuan Military Discusses Central Finance*” to the Republican government, published in the *Minli Daily*.<sup>15</sup> The proposal states (emphasis added):

*The Republic relies on its military for strength and on its finances for survival....However, we recognize that the foundations of the Republic are still fragile....There is no solution except for all four hundred million compatriots to shoulder this burden together....*

*Sichuan took the lead in resisting the Qing government’s foreign railway loans, and **thanks to the efforts of patriots nationwide**, the imperial court was overthrown within months....If the entire nation does not unite and contribute to financial self-sufficiency, we will inevitably become subjugated once again.*

*As military officers, we feel this responsibility deeply. Thus, we have collectively decided to contribute a portion of our salaries to address the immediate crisis....Together, we can eliminate the existing three-million-tael loan, and subsequently, **discuss further cost-saving measures, including salary reductions and promoting national bonds, a national bank, and voluntary citizen donations**. Only by stabilizing the central government’s finances can we ensure national sovereignty and deter foreign interference.*

In sum, we posit that late Qing anti-tax unrest events embody collective resistance to unjust policies and a breakdown of the social contract with the Qing court. Unlike other forms of unrest, which targeted local elites, foreigners, or specific grievances, anti-tax protests were uniquely directed at the state’s fundamental claim to legitimate authority, making them the most direct expression of social contract awareness (as confirmed empirically in Section 5.4). The frequency of this unrest reflects the strength of people’s awareness of such a contract, which was manifested in patriotic state-building under the new regime. As one outcome, the domestic patriotic bond market flourished, driving the early development of modern banks.

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<sup>15</sup> Appendix Figure B1, Panel C, presents images of the newspaper article and its full translation.

### 3 Data

Our sample comprises 286 prefectures from eighteen provinces in China Proper.<sup>16</sup> *Shuntian* prefecture (Beijing) is excluded from the regression sample as its status as the national capital introduces fundamentally different incentives along all key dimensions of our analysis: residents faced distinct pressures regarding civil unrest, political proximity to the Republican government likely inflated bond subscription figures beyond what patriotic sentiment alone would predict, and the concentration of state-backed financial institutions makes banking development in the capital not comparable to that in other prefectures. The sample period begins in 1912, when the Qing court collapsed and China entered the Republican Era, and ends in 1926, on the eve of the Northern Expedition. We obtain records of the prefectures in 1911 from the Harvard China Historical Geographic Information System ([CHGIS, 2016](#)).

#### 3.1 Anti-tax unrest

We use the total number of anti-tax unrest events taking place in each prefecture during 1902–1911 as a proxy for the strength of the social contract across prefectures. The data are drawn from *The Chronology of Civil Unrest in the Late Qing Dynasty* ([Zhang and Ding, 1982](#)), which provides a comprehensive survey of all recorded instances of civilian unrest and details the reasons, timing, and locations of these events. Of the 1,293 recorded unrest events within China Proper, we identify 311 as tax-related which contain keywords such as *Kangjuan* or *Kangshui* (tax resistance).<sup>17</sup> As shown in Table 1, an average of 1.14 anti-tax unrest events occurred in each prefecture. 121 (42.2%) prefectures experienced at least one anti-tax unrest; and 29 (10.1%) prefectures experienced four or more such events. To facilitate result interpretation, we also standardize anti-tax unrest ( $z$ -score) for robustness.

[Insert Table 1 about here]

#### 3.2 Patriotic civic engagement in Republican China

We measure (non-monetary) civic engagement during the Republican era (1912–1926) using three proxies, each capturing a distinct aspect of civilian contributions to the state.

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<sup>16</sup>China Proper refers to the core eighteen provinces of the Qing dynasty, excluding the frontier territories inhabited by non-Han Chinese populations, which were administrated by an alternative system.

<sup>17</sup>We went through each of the unrest events and classified them as follows: 121 anti-foreigner unrest, 133 anti-gentry unrest, 423 peasant riots, 66 banditry activities, 121 secret societies, and 333 strikes (of which 140 were unrelated to anti-tax unrest or anti-government unrest). For other forms of anti-government unrest, we identified 51 resist-tyranny unrest, 115 oppose-policies unrest, 37 uprisings, and 20 mutinies. The remaining 77 unrest events were with unknown reasons or forms, or concerned only with personal animosity. Note that an event could be identified with multiple reasons or forms. The above unrest incidents were used as placebo treated variables in Section 5.4.

First, we examine military participation, a key dimension of civic contribution under social contract theory (Caprettini and Voth, 2023; Qian and Tabellini, 2025). Using data from Chen (2006), we digitize enrollment records from the *Baoding Military Academy*, China’s leading military school, which operated from 1902 to 1923. With data for each cadet’s birthplace and enrollment year, we construct the cross-prefectural distribution of cadets before and after 1912 as a proxy for military participation across regimes and across regions.<sup>18</sup> Panel A of Figure B3 illustrates the spatial distribution of cadets.

Second, we track political associations because civilian engagement in political activities reflects state-building willingness (Rousseau, 1762). Under the Qing, civilian organizations were tightly restricted, but political participation increased after 1912 as citizens sought to exercise their rights and contribute to governance. Political associations were formed to express political opinions, support state-building efforts, and offer insights on local governance.

Third, we consider social associations, which provide public goods, social support, and aid to vulnerable groups such as women and teenagers. Using data from Zhang and Li (1999), we compile all (718) civilian organizations founded in the years 1912 through 1926 and identify 418 political associations and 195 social associations.<sup>19</sup> Panels B and C of Figure B3 illustrate the distributions of these associations across prefectures.

### 3.3 Principal components and standardization

As patriotism takes various forms, we employ principal component analysis (PCA) and standardization ( $z$ -score) to construct reduced-form proxies for the overall strength of patriotism across prefectures.

**Civic engagement 1912–26.** We construct two reduced-form variables to capture civic engagement between 1912–1926. (i) *PC civic engagement 1912–26* is the first principal component of the three previously defined civic engagement measures. This component alone explains 64.8% of the total variance. (ii)  $z$  *civic engagement 1912–26* is the  $z$ -score of the average  $z$ -scores of the three civic engagement measures. The double standardization ensures that the final measure has a mean of zero and a standard deviation of one, which facilitates result interpretation.

**Patriotism 1902–14.** Combining anti-tax unrest and three civic engagement measures, we construct two more variables to index the strength of civic capital that predicts patriotic engagement between 1902–1914, serving as alternative treatment variables in our DiD design which examines the role of patriotism in banking development. (i) *PC patriotism 1902–14*, is the

<sup>18</sup>The *Baoding Military Academy* drew students from both elite and grassroots backgrounds. Using Anhui as a case, Feng (2008) finds that over half of cadets came from the province’s poorest regions.

<sup>19</sup>After excluding *Shuntian* Fu (Beijing), the numbers are 436 associations, including 211 political associations and 162 social associations.

first principal component of anti-tax unrest (1902–1911) and the three civic engagement measures from 1912–1914. This component explains 56.6% of the total variance. (ii)  $z_{\text{patriotism } 1902-14}$  is the  $z$ -score of the average  $z$ -scores of anti-tax unrest and the three civic engagement measures before 1914.

### 3.4 Government bond subscriptions

Patriots could also demonstrate support for the state through financial contributions, primarily by subscribing to government bonds. We compile two datasets on bond subscription.

**The 1914 Patriotic Bonds.** We construct a novel dataset of reported prefecture-level subscriptions to the 1914 national bond campaign (described in Section 2.2) using the *Quanguo Baokan Suoyin* (National Newspapers and Periodical Index), the most comprehensive digital archive of Chinese newspapers and periodicals hosted by the Shanghai Library. We focus on this campaign because it represents the only nationally coordinated bond drive for which geographically disaggregated subscription records were systematically reported in the contemporary press; subsequent bond issuances lacked comparable coverage. We identify 269 newspaper articles systematically reporting bond subscription figures by locality from major Republican-era outlets, including *Shenbao* and *Shishi xinbao* (China Times), the largest Chinese newspapers at the time. For each article, we record the subscribing entity when reported, the subscription amount in silver dollars, and the locality name, which we map to prefectures using their 1911 boundaries.<sup>20</sup> The resulting dataset spans 105 prefectures across eleven provinces (with the spatial pattern shown in Figure B4), representing the full set of localities for which subscription figures were publicly reported in the contemporary press.<sup>21</sup>

At the prefecture level, we construct total subscription value (or collected value, in millions of silver dollars) and a per capita subscription measure, along with  $z$ -scores of each to facilitate coefficient interpretation. On average, prefectures subscribed 0.047 million silver dollars in total, or 0.019 million per 100,000 people.

**Provincial bonds (panel).** Since systematic subscription records comparable to the 1914 campaign are difficult to obtain for other bond issuances, we compile a second dataset at the national and provincial levels. Our sources include *Historical Materials on Government Bonds*

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<sup>20</sup>To avoid double counting, we treat repeated mentions of the same subscription figure across newspapers or dates as a single record; when reports provide cumulative totals reported as of a specific date, we retain the latest cumulative amount within the campaign period for that locality.

<sup>21</sup>Several features support the reliability of these newspaper-based figures. First, subscription announcements were designed to publicize campaign progress to a national audience and were frequently reproduced across independent outlets, creating scope for cross-validation. Second, reported subscriptions are typically dispersed across many small contributors, including military units, students, merchants, and ordinary citizens, rather than dominated by a few large lump-sum entries, consistent with grassroots mobilization rather than strategic fabrication. Third, for localities covered by multiple newspapers, reported amounts are highly consistent across sources, suggesting that common underlying subscription records rather than editorial discretion drive the figures.

in China (1894–1949) (Qian, 1984), *Domestic Bonds in China* (Survey Department of Southern Manchuria Railway Company, 1930), and *Provincial Bonds* (Ling, 1928). We also meticulously complement missing information using the “Fiscal Chronicles” from each provincial gazetteer.

From 1912 to 1926, a total of 139 bonds were issued, 29 by the central government and 110 by provincial governments. Figure 1 illustrates the time trends of successfully issued bonds, highlighting a noticeable increase in domestic bond issuances after 1914. Altogether, during this period, national bonds raised 500.4 million yuan, and provincial bonds raised 151 million yuan.

We also digitize key bond characteristics, including targeted and collected amounts, interest rates, maturity, issuance purpose, and whether they were guaranteed/collateralized. Based on issue descriptions, we classify bond purposes as military funding (52%), economic development (44%), and disaster relief (4%). Historical accounts indicate that military-related bonds were typically raised through compulsory allocation, whereas development bonds relied more on voluntary subscriptions.<sup>22</sup>

As subsequent national bond issuances lack geographically disaggregated subscription records, our regression analysis of bond financing focuses on provincial bonds, for which subscriptions were primarily purchased by local populations. To avoid potential biases from central government influence, we exclude the province of Zhili, the seat of the central government.

### 3.5 Modern banking development

Our main dependent variable, modern banking development, is measured by the annual number of modern Chinese banks (headquarters and branches) in each prefecture. This dataset was manually compiled from more than 2,000 gazetteers across county, prefectural, and provincial levels. To ensure completeness, we supplement missing entries using *Banking Yearbooks* (1934–1937) and relevant historical archives. See the supplementary materials of Lin et al. (2021) for the full list of data sources.

We focus on modern Chinese banks and exclude foreign banks that primarily operated in foreign exchange markets and were less influenced by Chinese domestic policies. Modern Chinese banks are classified into official and private banks based on their founding entities: Official banks were established by government bodies, while private banks were independently founded by the private sector. Joint ventures between government and private entities are categorized as

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<sup>22</sup>Several historical records document that military bonds were raised coercively. For example, one account states: “...during the Beiyang government period, the purposes for issuing Anhui provincial public bonds were basically to cover military and administrative expenses.... As for the methods of raising subscriptions...did not realize a genuinely public offering in the true sense....there were underwritten apportionments and compulsory distribution.” (Pan, 2014, p. 115). A similar pattern is documented for other provinces: “During periods of turmoil brought about by political instability, provincial warlord governments resorted to every means to raise military funds; practices such as forced loans, extortion, and advance levies were commonplace across the provinces” (Zhang and Li, 1989, p. 557).

official banks due to their direct political ties.

Table 1 presents the summary statistics. In the years 1912–1926, the 286 prefectures had an average of 1.92 modern Chinese banks. The standard deviation is large: 52.33% of prefecture-year observations had no banks, while *Songjiang* prefecture, where Shanghai is located, had 97 banks in 1926.

### 3.6 Instrumental variable: Massacres

To address endogeneity concerns, we use massacres during the Qing dynasty’s founding era (1644–1661) to instrument late Qing anti-tax unrest events. During the transition from Ming to Qing, the Qing army carried out brutal massacres to consolidate control; a lasting collective memory that increased the propensity for unrest against the Qing court continued during the late Qing period. These massacres followed the army’s conquest route and severely disrupted local economies, making it unlikely that the massacres directly promoted banking development through any overlooked economic factors. More details will be discussed in Section 5.2.

Drawing upon *The History of the Southern Ming Dynasty* (Gu, 1997), we identify 55 of 1,503 counties in our sample where at least one massacre occurred. After summing up to the prefectural level, 45 of 286 prefectures witnessed from one to three massacres.

### 3.7 Baseline controls

We control for baseline factors that may influence both anti-tax unrest and banking development; additional controls will be discussed in Section 5.3.

Population size serves as a key proxy for economic foundation, as both unrest and banking activity were likely to be more prominent in populous areas. Using population records in 1910 from Cao (2001), we control for the log of population density, i.e. the 1910 population divided by land area, interacted with the post-1914 dummy.

We also account for geographic factors that may shape the distribution of unrest and banking expansion: (1) prefecture land area (log), which affects the reach of financial services; (2) the prefecture’s distance to the coast (log), because coastal areas were historically more exposed to trade and external influences; (3) the river density (river length scaled by land area) within each prefecture (log), given the role of waterways in trade and economic activity; and (4) the altitude (log) and ruggedness of the terrain, which impact transportation and communication. Data are sourced from the 1911 basemap of CHGIS (2016). Lastly, we control for the (log) distance to the national capital, Beijing, as proximity to the political center may influence state relations and banking development. We report the summary statistics of all these variables in Table 1.

## 4 Social contract: From unrest to patriotic civic engagement

This section demonstrates that prefectures with stronger civic resistance to Qing authority, as captured by anti-tax unrest, were the same prefectures that channeled that civic energy into patriotic engagement once a more legitimate government was established in 1912; this is reflected in both non-monetary civic participation and financial contributions to the state.

### 4.1 Civic engagement

We measure civic engagement within the Republican era from three dimensions: military, political, and social participation. Each dimension reflects a distinct aspect of civilian contributions to the state. The analyses follow the cross-sectional regression model below:

$$Civic\_engagement_i = \alpha + \beta \times Anti\_tax\_unrest_i + \gamma \times \mathbf{X}_i + \epsilon_i \quad (1)$$

where  $Civic\_engagement_i$  refers to one of the three measures of civic engagement in prefecture  $i$  between 1912–1926. As a robustness check, we also use the first principal component of these three, or the  $z$ -score of the average  $z$ -scores of the three.  $Anti\_tax\_unrest_i$  denotes the number of unrest events related to anti-tax appeals in each prefecture during 1902–1911. If anti-tax unrest reflects the strength of social contract that embodies patriotism,  $\beta$  should be significantly positive.  $\mathbf{X}_i$  represents a vector of baseline controls that include the 1910 prefectural population density (log), land area (log), distance to the coast (log), river density (log), altitude (log), terrain ruggedness, and distance to Beijing (log).

The baseline regressions use heteroskedasticity-robust standard errors. To account for potential spatial correlation among the residuals, we also cluster the standard errors within a circle that covers all adjacent prefectures, using the method of [Colella, Lalive, Sakalli, and Thoenig \(2019\)](#). Given that the average “radius” of each prefecture is 68.3 kilometers in our sample, we use a distance of three times this radius (205 kilometers) to approximate all adjacent prefectures. This method calculates the bilateral spatial correlation within this 205-kilometer radius.

[Insert Table 2 about here]

Panel A of Table 2 examines whether late Qing anti-tax unrest was linked to higher levels of military participation in the Republican era. Social contract theory views military service as a key demonstration of loyalty to the state ([Caprettini and Voth, 2023](#); [Qian and Tabellini, 2025](#)). Using enrollment records from the *Baoding Military Academy* from 1902 through 1923 ([Chen, 2006](#)), we digitize cadet data and construct the cross-prefectural distribution of military participation before and after the Republic’s establishment in 1912.

Column 1 in Panel A of Table 2 regresses the number of cadets on the number of anti-tax unrest events and reveals a positive association between the strength of social contract and

military participation. Adding demographic and geographic controls (column 2) reduces the coefficient by 46%, but the effect remains significant. Quantitatively, each additional anti-tax unrest event corresponds to 3.51 more cadets, a 16% increase relative to the average (21.6 cadets). Columns 3 and 4 examine the unrest effects on the differences in cadet enrollment before and after 1912. The results indicate a significant rise of 22%, highlighting a stronger rise in patriotic engagement in regions with more unrest. All results remain robust to spatial correlation clustering. Appendix Table B1 reports the full set of coefficients for all control variables.

In the same vein, we examine whether regions with more instances of anti-tax unrest before 1912 saw greater formation of political and social associations after 1912. The results (Table 2, Panel B) reveal positive and significant associations. We find that each additional anti-tax unrest event corresponds to a 123% increase in political associations (column 2) and a 106% increase in social associations (column 4). These findings provide evidence that a strong social contract in the late Qing era was manifested in active patriotic engagements in the Republican era, contributing to the building of political and social institutions.

Panel C of Table 2 presents the findings using two alternative proxies for civic engagement: the first principal component of the three individual measures (columns 1-2) and the  $z$ -score of the average  $z$ -scores (columns 3-4). As shown in column 4, a one standard deviation increase in anti-tax unrest is associated with a 0.42 standard deviation increase in civic engagement.

A potential alternative is that the late Qing protesters turned to anarchism, reducing their role in Republican state-building. Appendix A.1 details the development of anarchism in China and provides empirical analyses. The results indicate a positive but statistically weak correlation between anti-tax unrest and anarchist associations. Given that anarchism did not emerge on a large scale in China until 1919 (Zhang and Li, 1999), we argue that the awareness of a social contract was actively channeled into patriotic support for the new government, rather than resulting in disengagement during the critical period examined in our study.

## 4.2 Government bond subscriptions

We now examine the impact of anti-tax unrest on government bond subscriptions, a key form of “monetary contribution” to the state and a primary channel through which anti-tax unrest influenced banking development.

As discussed in Section 2.2, the outbreak of WWI disrupted Western financial support to China, creating an urgent need for domestic financing. The 1914 Patriotic Bond was the first successful domestic government bond campaign and arguably the most representative of grassroots fiscal mobilization. We surveyed newspaper records and compiled a novel dataset of prefecture-level subscriptions for this bond.

We follow a regression model analogous to Equation 1, replacing the dependent variable

with the total collected value, per capita collected value, and their  $z$ -scores for the 1914 bond. Results are presented in Panel A of Table 3. Prefectures with more anti-tax unrest before 1912 subscribed significantly more to the 1914 bond. Each additional unrest event is associated with an increase of 0.057 million yuan in total collected value and 0.014 million yuan per 100,000 people (columns 1 and 3). Using standardized measures, a one-standard-deviation increase in anti-tax unrest corresponds to a 0.597 and 0.449 standard deviation increase in total and per capita subscription, respectively (columns 2 and 4).

[Insert Table 3 about here]

The WWI outbreak represented an exogenous fiscal shock to the Chinese government, which also allows us to implement a Difference-in-Differences (DiD) strategy to gauge the causal effect of the social contract on bond subscriptions over time. Since comparable locality-level bond data are unavailable for other years, we compile panel data at the provincial level for the DiD analysis. The regression model is specified as follows:

$$\begin{aligned} \text{CollectedValue}_{p,t} = & \alpha + \beta \times \text{Anti-tax\_unrest}_p \times \text{Post}_t + \gamma_1 \times \text{TargetValue}_{p,t} + \\ & + \gamma_2 \times \text{InterestRate}_{p,t} + \gamma_3 \times \text{Collateral}_{p,t} + \gamma_4 \times \text{Maturity}_{p,t} \quad (2) \\ & + \gamma_5 \times \mathbf{X}_p \times \text{Post}_t + \mu_p + \lambda_t + \varepsilon_{p,t}. \end{aligned}$$

where  $\text{CollectedValue}_{p,t}$  refers to the actual amount raised (in million yuan) in province  $p$  in year  $t$ .  $\text{Anti-tax\_unrest}_p$  denotes the incidences of unrest related to tax resistance in 1902–1911 in province  $p$ .  $\text{Post}_t$  is a dummy that equals one for years after the outbreak of WWI (1915–1926) and zero otherwise. We include control variables, which generally affect bond pricing and issuance efficiency:  $\text{TargetValue}_{p,t}$ ,  $\text{InterestRate}_{p,t}$ ,  $\text{Collateral}_{p,t}$  and  $\text{Maturity}_{p,t}$ ; these denote the total target amount (in million yuan), bond interest rate, whether the bonds were collateralized, and years of maturity, respectively.  $\text{InterestRate}_{p,t}$ ,  $\text{Collateral}_{p,t}$  and  $\text{Maturity}_{p,t}$  are averages of all bonds issued in province  $p$  in year  $t$ , weighted by target value. The vector  $\mathbf{X}_p$  refers to a set of baseline controls interacting with  $\text{Post}_t$ ; these include the 1910 provincial population density (log), land area (log), distance to the coast (log), river density (log), altitude (log), terrain ruggedness, and distance to Beijing (log). Province and year fixed effects are controlled for, and the standard errors are clustered at the province level.

The results are presented in Panel B of Table 3. Column 1 examines the overall impact of the anti-tax unrest on provincial bond subscriptions. Consistent with our hypothesis, regions with more anti-tax unrest in the late Qing period raised significantly more funds after 1914. On average, 17.88 instances of anti-tax unrest occurred in each province before 1912, and each additional instance corresponds to an increase of 0.03 million yuan in bond subscriptions—a 5.1% rise relative to the mean. Interpreted alternatively, a one-standard-deviation increase in anti-tax unrest is associated with a 21.2%-standard-deviation increase in bond collected value.

To probe mechanisms, columns 2-3 differentiate the effects across bond types. As outlined in Section 3.4, bonds issued for economic development are classified as voluntary, whereas bonds issued for military funding are classified as compulsory. The results show that a stronger social contract significantly increases the success of voluntary bonds, while compulsory bonds exhibit no comparable effect.<sup>23</sup> In Appendix Table B3, we use principal component and standardized measures of patriotism and of bond issuance for robustness. Overall, the results underscore that patriotic mobilization is effective when voluntary rather than coercive.

An alternative explanation is the demand-side effect: that government bonds were more attractive than other investment options, and regions with stronger social contracts had greater investment demand. In this case, once government bonds became available, their uptake would have increased more in regions with stronger social contracts. However, anecdotal evidence suggests otherwise. Government bonds offered no superior returns and carried high risk due to severe fiscal deficits.<sup>24</sup>

Provincial bonds in our sample had an interest rate ranging from 0% to 12% and averaging 5.1%, significantly lower than private loan rates. Sourcing from the *China Historical Interest Rate Database* (Chen, Peng, and Yuan, 2016),<sup>25</sup> we analyze the 2,141 private loans issued from 1912 through 1926.<sup>26</sup> The average annual rate was 41.93%; after excluding usury above 100%, the adjusted average remained at 18.12% across 2,046 loans. This substantial gap in returns suggests that financial incentives alone do not explain the higher number of bond subscriptions in patriotic regions. Using 2,046 loan records, we computed the average loan rate at the province-year level. To account for missing data, we interpolated values when both a preceding and a succeeding province-year had data. Including it as a control variable in Equation 2, the results indicate minimal impact of local loan rates on government bond subscriptions (Panel A

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<sup>23</sup>One nuanced hypothesis is that a stronger social contract might also raise compulsory allocations if governments lean harder on high-patriotism regions. Two facts address this: (1) these are province-level issues—placement was organized within provinces, could not easily be floated across provinces, and reflects fiscal needs; (2) in Appendix Table B2, anti-tax unrest predicts lower targets for all, voluntary, and compulsory bonds; a one-unit rise of unrest led to 6.1%, 11.3% and 5.9% lower face value relative to the mean, respectively. Combining results in Appendix Table B2 and Table 3, stronger social contracts predict a higher *success rate* of voluntary bonds. We do not directly use success rates as the dependent variable in the regressions because many province-years have no issuance, leaving the denominator undefined.

<sup>24</sup>In 1921, Zhou Ziqi, the Minister of Finance of the central government, stated in a report: “If we follow the current government debt regulations and pay the principal and interest on schedule, it would require as much as 39.7 million yuan. Given the current financial situation, it would be impossible to manage this” (Qian, 1984, p. 14-15).

<sup>25</sup>Chen et al. (2016) constructs the *China Historical Interest Rate Database*, comprising 23,489 records of private lending and borrowing spanning from 1563 to 1950. These records were compiled from diverse sources such as private archives, newspapers, periodicals, and surveys. Chen et al. (2022) also provide an extensive description of this dataset.

<sup>26</sup>Before 1926, China’s financial markets were underdeveloped, with few financial instruments. These markets included the Shanghai stock market, which emerged in the late 1860s but was primarily accessible to foreigners or Chinese within the Shanghai concession; a Beijing Stock Exchange that appeared after 1918; and a nascent corporate bond market that emerged after 1921 (Zhu, 2022). Thus, local borrowing and lending rates can reliably proxy the returns on investments in this period.

of Appendix Table B4). To test the robustness of our findings, we further excluded Zhejiang and Jiangsu provinces, given their populations' high awareness of investment. The results were robust to this exclusion (Panel B of Appendix Table B4).

Overall, findings in Section 4 suggest that pre-1912 anti-tax unrest reflects the strength of the social contract, which was manifested in patriotic efforts under the new government after 1912, including increased military participation, the establishment of political and social associations, and heightened government bond subscriptions. Among these, the bond subscription result is particularly telling. Purchasing government bonds at unfavorable terms (government bond rates were far below private lending rates) is not an act that generic collective action capacity alone can explain. A community's organizational strength might facilitate a coordinated bank run or a tax revolt, but it provides no motive to voluntarily accept financial losses for the state's benefit. The willingness to do so reflects a specifically patriotic disposition, a commitment to the state that transcends private financial interest. We therefore hypothesize that, following the 1914 fiscal shock, regions with a stronger social contract (or greater patriotism) would experience larger expansions in modern banking.

While one might be concerned that anti-tax unrest reflected political opportunism, with participants signaling loyalty to a potential new government for future political gains, in Appendix A.2, we further discuss and empirically disentangle anti-tax unrest from political opportunism.

## 5 Patriotism and banking development

### 5.1 Baseline results

In the previous section, we show that anti-tax efforts in the late Qing persisted across regimes and translated into civic engagement, particularly patriotic government bond subscriptions. As government bond underwriting drove early modern banks, regions with stronger social contracts should have experienced greater bank growth after the 1914 fiscal shock (as stated in Section 2.2). In a reduced-form expression, modern banks were fueled by patriotism. To formally test this, we employ a DiD approach:

$$Banks_{i,t} = \alpha + \beta \times Anti\_tax\_unrest_i \times Post_t + \gamma \times \mathbf{X}_i \times Post_t + \mu_i + \lambda_t + \varepsilon_{i,t}. \quad (3)$$

where  $Banks_{i,t}$  refers to the number of modern banks (including headquarters and branches) in prefecture  $i$  in the year  $t$  (1912–1926).  $Anti\_tax\_unrest_i$ , which denotes the total number of anti-tax unrest events in each prefecture  $i$  between 1902–1911, captures the social contract and embodies patriotic sentiment toward the Republican regime.  $Post$  is a dummy that equals one for years after the outbreak of WWI (1915–1926) and zero otherwise (1912–1914). The coefficient  $\beta$  captures the DiD effects of patriotic sentiment rooted in civic capital on bank establishment

after 1914. The vector  $\mathbf{X}$  denotes a set of baseline controls interacting with *Post*: these include the 1910 prefectural population density (log), land area (log), distance to the coast (log), river length density (log), altitude (log), terrain ruggedness, and distance to Beijing (log). We include prefectural fixed effects ( $\mu_i$ ) to control for time-invariant regional characteristics and year fixed effects ( $\lambda_t$ ) to absorb common annual shocks. We report both standard errors clustered at the prefecture level and standard errors adjusted for spatial autocorrelation, following [Colella et al. \(2019\)](#). The spatial cutoff is set to 205 km, approximately three times the average prefectural radius, to account for correlations across adjacent prefectures.

As a robustness check, we replace the treatment variable with three alternatives: (i) *PC patriotism 1902–14*, the first principal component of anti-tax unrest (1902–1911) and the three civic-engagement measures (1902–1914), (ii) *z-standardized anti-tax unrest*; and (iii) *z patriotism 1902–14*, defined as the *z*-score of the average of the *z*-scores of anti-tax unrest and the three civic-engagement measures before 1914. When the treatment is standardized, we also use the standardized number of banks as the dependent variable to facilitate result interpretation.

[Insert Table 4 about here]

The baseline results are presented in Table 4. Column 1 provides a benchmark by regressing the number of banks on *Anti-tax unrest*  $\times$  *Post*, controlling only for prefecture and year fixed effects. The significantly positive coefficient on the interaction term suggests that social contracts fostered banking development. In column 2, we add demographic and geographic controls. While the coefficient decreases by about 33%, the effect remains significant. Economically, each additional anti-tax unrest event before 1912 is associated with a 0.40 increase in the number of banks after 1914, a 21% rise relative to the mean (1.92 banks). Column 3 excludes *Zhejiang* and *Jiangsu* (where Shanghai is located) to address concerns that results might be driven by these economically advanced regions with dense pre-existing traditional banking networks. The result indicates that our findings are robust to this exclusion. In columns 4–6, we use principal-component and standardized measures of patriotism. Column 6 shows that a one-standard-deviation increase in the standardized patriotism index is associated with a 0.5-standard-deviation increase in the number of banks. To visualize the results, the geographical distributions of anti-tax unrest, patriotism *z*-score, and bank growth are depicted in Figure 2.

[Insert Figure 2 about here]

We assess the parallel trend assumption with dynamic DiD, which regresses the number of banks on interactions between anti-tax unrest and year dummies for 1912 through 1920 (1914 omitted as the reference year), while including all controls interacted with year dummies, as well as prefectural and year fixed effects. The estimated yearly coefficients with 95% confidence intervals are plotted in Figure 3. The coefficients are near-zero in 1912–1913 and turn

significantly positive after 1914. This confirms the parallel trend assumption and indicates a sharp shock from WWI. To address concerns about the short pre-trend period (only two years in the baseline sample), we extend the sample to 1908–1926 (Appendix Figure A1). Appendix A.3 further discusses the results’ sensitivity to the potential violation of the parallel trend assumption based on the honest DiD method proposed by Rambachan and Roth (2023).

[Insert Figure 3 about here]

To examine whether the impact of anti-tax unrest on banking development operates through patriotic sentiment, we regress the number of banks on *Anti-tax unrest*  $\times$  *Post* and *Civic engagement (1912–14)*  $\times$  *Post* simultaneously. Here, civic engagement is measured by military participation, political associations, and social associations between 1912 and 1914, as well as the principal component or *z*-score of all three. The results (Panel A of Appendix Table B5) show that adding patriotic civic engagement variables significantly reduces the coefficient for anti-tax unrest, even to the point of insignificance, indicating that patriotic sentiment is a key transmission channel. Panel B further adds 1914 Patriotic bond subscriptions  $\times$  *Post* to the horse race. Bond subscription emerges as the dominant predictor of banking development, suggesting that the patriotic mobilization effect on banking development operates primarily through the financial channel.

## 5.2 Instrumental variable

Despite the DiD strategy, endogeneity remains. First, economic conditions are difficult to fully control, raising the concern of omitted variables. It is possible that regions with stronger economic foundations had more exposure to modern ideologies, which fostered civic awareness and banking development after 1914. Second, modern banks emerged in China already in 1897. This raises the possibility of reverse causality, where early banking development introduced new ideas and influenced civic awareness.<sup>27</sup>

To tackle these concerns, we employ an IV approach, using the massacres taking place during the Qing regime’s founding era (1644–1661) as an instrument for anti-tax unrest in the late Qing. The Ming-Qing transition was marked by violent massacres carried out by the Qing army against the mass Han Chinese resisting Qing rule. These massacres were a key strategy for consolidating Qing control and resulted in substantial casualties. Scholars estimate that China’s population peaked at nearly 200 million in 1630 (Cao, 2001), but by 1680 had declined by 45–50 million because of these events. Notable massacres occurred in *Yangzhou*, *Taichang*, *Datong*,

<sup>27</sup>In Appendix Table B6, we present the regression results of banks predicted by anti-tax unrest, and anti-tax unrest predicted by banks between 1902 and 1911. There is no evidence of a positive correlation between anti-tax unrest and banks before 1911.

and *Ganzhou*. Based on Gu (1997), we identify one to three massacres as having occurred in each of 45 of the 286 prefectures. Figure 4 depicts the geographical distribution of massacres.

[Insert Figure 4 about here]

We argue that early Qing massacres satisfy the relevance condition for a valid instrument, as they could leave a profound and lasting imprint on collective memory (Fouka and Voth, 2023). Surviving records, such as the Record of *Ten Days in Yangzhou*, have preserved this historical trauma, which later fueled anti-Qing unrest as central authority weakened after the mid-nineteenth century (Li, 2023).<sup>28</sup> Thus, our first-stage hypothesis is that regions affected by the early Qing massacres were more likely to exhibit anti-tax unrest in the late Qing period.

This instrument likely satisfies the exclusion restriction. First, the massacres occurred over two centuries before the emergence of modern banks, eliminating the possibility of reverse causality. Second, the massacres were primarily driven by military, rather than economic, objectives and predominantly occurred along the Qing army’s conquest path. Third, their destructive impact on local economies renders it implausible that they directly promoted banking development two and a half centuries later, particularly after controlling for long-run economic conditions.

[Insert Table 5 about here]

The IV results are presented in Table 5. Panel A reports the 2SLS estimates, where *Anti-tax unrest*  $\times$  *Post* is instrumented by *Massacres*  $\times$  *Post*. The instrumented anti-tax unrest significantly promoted modern banks. Comparing the 2SLS estimate 0.63 to the OLS estimate 0.40 in Table 4 (column 2), the IV effect is 1.59 times larger, with no sign of a high coefficient inflation issue (Jiang, 2017). Panel B presents the first-stage results. As shown, historical massacres significantly predict anti-tax unrest, indicating that regions with greater historical trauma were more likely to resist government taxation. This finding extends Qian and Tabellini (2025) by providing evidence that discriminated groups may not only be less willing to support the state but may also be more likely to engage in efforts to overthrow it. We also discuss the potential issue of Local Average Treatment Effect (LATE) (Jiang, 2017) and complier characteristics in Appendix A.4.

Appendix Table B7 presents the reduced-form regressions. As expected, *Massacres*  $\times$  *Post* positively predicts the number of banks (column 1). When both *Massacres*  $\times$  *Post* and *Anti-tax unrest*  $\times$  *Post* are included (column 2), the coefficient of massacres turns insignificant, leaving

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<sup>28</sup>Even senior bureaucrats in the Qing government could sense the devastating effects of the massacres that occurred during the founding period. For example, Zhao Liewen (1832–1894), a disciple of Zeng Guofan, one of the most potent senior officials in the late nineteenth century, said, “Our (Qing) state could be subverted within 50 years...the massacres during the founding period were too gruesome. Even if the emperor has very good virtue, he could not save our state from its fate” (Zhao, 2013).

anti-tax unrest as the key driver of banking development. This suggests that historical massacres affected banking growth by mobilizing anti-tax unrest.

In Appendix Table B8, we use massacres conducted by non-Qing forces, primarily those led by historical peasant revolt leaders Li Zicheng (1606–1645) and Zhang Xianzhong (1606–1647) and their followers between 1644 and 1661 as a placebo instrument. Appendix Figure B5 maps the geographical distribution of these massacres and provides a brief historical background. The results indicate that massacres by non-Qing actors do not significantly predict anti-tax unrest in the late Qing period. This suggests that such unrest was not driven by a general breakdown of order, but specifically reflected a rupture in the social contract with the state responsible for prior harm.

### 5.3 Additional controls

This section outlines robustness checks incorporating additional covariates that could affect banking development. We begin by discussing the rationale and data, followed by presenting the results for each factor.

First, the weakening of Western powers in China after 1914 likely altered trade patterns and spurred local industrial growth. Specifically, World War I disrupted imports into China, boosted Chinese exports, and thereby stimulated domestic industries, generating increased demand for external financing (Rawski, 1989). Therefore, we additionally control for the prefecture-year number of industrial firms, using data from Du (2014).

Second, Treaty Ports were centers of foreign presence in China. They connected China to global trade, technology, and culture, likely fostering modern ideologies and civic awareness, which may have influenced banking development (Ma and Rubin, 2019). By 1926, fifty-four treaty ports spanned forty-two prefectures, hosting 50.4% (436 of 866) of modern Chinese banks. We thus control for the number of treaty ports per prefecture using data from Yan (2012).

Third, to directly capture the impact of trade, we include customs duty income with records from *Historical Materials of Chinese Maritime Customs: 1859–1948* (Chinese Second Historical Archive, 2001).

Fourth, civic awareness likely evolved alongside the presence of educated elites. To distinguish whether anti-tax unrest influenced banking primarily through patriotism or simply reflected human capital concentration, we control for the interaction between (log) civil examination quotas and the post-1914 dummy. These quotas, set by the central government and largely unchanged since the nineteenth century, serve as a historical measure of regional human capital (Lin et al., 2025).

Fifth, Confucian clans, formed through kinship networks, have been shown to delay the emergence of modern banks in China. These clans established internal markets for resource pooling and risk sharing within families, thereby reducing the demand for external financial

instruments during times of financial need (Chen et al., 2022). To measure the strength of these clans, we use the prefectural-level number of genealogy books as a proxy, drawing on data from Shanghai Library (2009).

Sixth, advancements in communication and transportation technology, including telegraph, railway networks, and postal routes, could also facilitate bank expansion. To capture such progress, we control for the lagged number of telegraph stations, railway stations, and postal offices in each prefecture (Lin et al., 2021).

Seventh, civil wars and natural disasters could hinder economic development and affect the government's likelihood of issuing bonds. Using data from Shizhengchu (1967) and Chinese Academy of Meteorological Sciences (1981), we count the number of civil wars and natural disasters (floods and droughts) per prefecture in the previous year.

[Insert Figure 5 about here]

Figure 5 shows the coefficients, along with 95% confidence intervals, of *Anti-tax unrest*  $\times$  *Post* while controlling for each of the above-mentioned additional controls, or all of them simultaneously. It also plots the IV-DiD estimates. The figure shows that the estimated effects of anti-tax unrest on banks change little under all the circumstances. Appendix Figure B6 replicates the test using standardized patriotism as the treatment variable and standardized bank as the dependent variable. The results remain robust.

In an additional robustness test, we replicate previous regressions at the county-level to account for variations in banks and anti-tax unrest within prefectures. Appendix A.5 discusses the models and results in detail. The results remain robust.

## 5.4 Falsification tests

In this section, we present falsification tests to validate that anti-tax unrest reflects the social contract rather than alternative explanations or other influencing factors.

First, anti-tax unrest could simply reflect high tax burdens. To address this concern, we construct two direct proxies for tax burden: (1) per capita land tax in 1820, derived from the latest available prefectural tax records (Liang, 2008), and (2) the number of commercial tax (*likin*) bureaus around 1909 (Luo, 1936). We replicate Equation 3 but replace anti-tax unrest with these two measures. As shown in Figure 6 (rows 1-2), tax levels had a marginal impact on banking development after 1914.<sup>29</sup>

<sup>29</sup>Appendix Figure B10 (Panels A and B) illustrates the spatial distribution of land tax and commercial tax bureaus. Appendix Table B9, Panel A shows a strong positive correlation between tax burden and the frequency of anti-tax unrest events. However, in a horse-race regression (Panel B) where both anti-tax unrest events and tax burden are included as predictors of banking development, the tax burden coefficients become insignificant. These results suggest that anti-tax unrest reflects not only attitudes toward tax levels but also fundamental attitudes toward government.

[Insert Figure 6 about here]

Next, we distinguish between constructive patriotism and xenophobia (blind patriotism) (Schatz et al., 1999). We highlight that constructive patriotism reflects a nuanced understanding of the social contract and fosters civic engagement. We use two proxies for radical xenophobic activity.<sup>30</sup> First, based on the *Boxer Protocol*, we construct a dummy indicating whether a prefecture experienced episodes of violent attacks on missionaries and expatriates between 1899 and 1901 during the Boxer Rebellion. Second, we count incidents of anti-foreigner unrest during the period 1902–1911 based on Zhang and Ding (1982); these include incidents such as “attacking foreigners”, “attacking missionaries”, or “destroying churches”. We identify 121 such events. Panel C of Appendix Figure B10 shows the spatial distributions of both proxies. Since their distributions do not strongly overlap, we create a combined dummy variable to capture either event. Regression results (Figure 6, row 3) show no significant impact of xenophobic sentiment on bank growth after 1914, suggesting that radical nationalism did not drive modern financial development.

Finally, we distinguish between anti-tax unrest and general discontent or disorder. Using data from Zhang and Ding (1982), we count: (1) incidents of anti-gentry unrest targeting local elites or landowners (133 incidents), (2) riots by peasants triggered by land disputes or natural disasters (423), (3) banditry or criminal activities like robbery (66), (4) secret societies, or oath-bound, clandestine popular associations (121), and (5) strikes (333, of which 140 were unrelated to anti-tax unrest or anti-government unrest). Note that an event could be identified with multiple reasons or forms. Their influences on banks are shown in rows 4–8 of Figure 6. Notably, strikes were concentrated in prosperous industrial cities, most notably Shanghai. Therefore, when we exclude *Zhejiang* and *Jiangsu* provinces, the strike effect on banks becomes statistically insignificant.<sup>31</sup>

The remaining unrest types in Zhang and Ding (1982) were mainly anti-government actions other than tax resistance: resistance to tyranny (collective protests against abusive local officials) (51), opposition to specific late-Qing policies (115), generic uprisings (37), and military mutinies (20).<sup>32</sup> These episodes were often triggered by particular corrupt officials or discrete policy shocks and are thus more idiosyncratic, less likely to proxy a broad “social contract” channel than anti-tax unrest. Figure B11 plots their placebo effects on the number of modern banks. “Resistance to tyranny” shows a positive association, but it becomes statistically insignificant once *Zhejiang* and *Jiangsu* are excluded.

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<sup>30</sup>Patriotism did not necessarily lead to xenophobia. In contrast, during the Self-strengthening Movement (1860–1894), enlightened patriots highlighted assimilating Western technology with the slogan “*Shi yi changji yi zi qiang*” (learning the foreigners’ advanced technology to strengthen our country) (Yan, 2012).

<sup>31</sup>In contrast, the effect of anti-tax unrest on banks is largely unchanged when these provinces are excluded (column 3 of Table 4).

<sup>32</sup>The remaining 77 incidents of unrest were with unknown reasons or forms, or concerned only with personal animosity.

In short, the above falsification tests support that anti-tax unrest is the most representative proxy for patriotic sentiment rooted in social-contract awareness. A natural concern is that anti-tax unrest simply proxies for a region’s capacity for collective action. However, banditry and secret societies required substantial organizational infrastructure and coordination, arguably more than tax protests, which were often spontaneous responses to specific fiscal impositions. Yet neither predicts banking development (Figure 6, rows 5-6). Similarly, peasant riots, which involved large-scale mobilization, show no significant effect. The common feature of these insignificant placebo variables is that they targeted private grievances or non-state actors. Only unrest directed at the state’s fiscal authority, the defining feature of social-contract engagement, predicts subsequent banking growth. This pattern is inconsistent with a generic collective action interpretation and instead supports our patriotism hypothesis.

## 6 Channels: bank profitability and public confidence

In this section, we investigate the channels through which patriotic sentiment fostered banking growth. We document evidence on two mechanisms. First, modern banks increased their profitability by underwriting and trading government bonds, strengthening their financial foundations. Second, citizens’ interactions with modern banks through bond subscriptions cultivated familiarity with these institutions, which gradually built trust in modern banking as a legitimate and reliable form of financial intermediation.

### 6.1 Exposure to patriotism and bank performance

In this section, we use bank-level balance sheet data to examine whether banks that expanded into more patriotic regions (relative to those that expanded into less patriotic ones) held more government bonds and achieved stronger financial performance.

For the explanatory variable, we construct a panel measure of each bank-year’s additional exposure to patriotic sentiment rooted in civic capital, proxied by the weighted sum of the number of anti-tax unrest events in prefectures where a bank opened new branches in that year, with the weight being the number of new branches opened in each prefecture.<sup>33</sup> To account for bank size, this measure is scaled by the bank’s total number of branches.

If banks expand strategically into patriotic regions to attract government bond investors, increased exposure to these areas should enhance bond holdings and profitability. We assemble financial statements for the years 1922–1926 from nineteen major Chinese banks, sourced from

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<sup>33</sup>For instance, consider a bank that established five new branches within the year—two in Prefecture A and three in Prefecture B. The bank’s additional exposure to patriotic sentiment in that year, as measured by anti-tax unrest, is calculated as the sum of the unrest episodes in Prefecture A multiplied by two and those in Prefecture B multiplied by three.

[Economic Research Department \(1933\)](#). We measure bank performance using return on assets (ROA), defined as net profit to total assets ( $\times 100$ ). Bond holdings are less straightforward, as government bonds could appear on balance sheets as either reserves for banknotes or marketable securities ([Ho and Li, 2014](#)).<sup>34</sup> Based on historical narratives, we assume that 30% of banknote issuance was backed by government bonds and 75% of marketable securities were government bonds.<sup>35</sup> Additionally, we consider bond holdings and profit growth, defined as the difference between current-year and previous-year bond holdings (profits), scaled by the previous year's total assets ( $\times 100$ ). Bank-level controls include total assets (log), cash-to-asset ratio ( $\times 100$ ), and leverage (debt-to-asset ratio  $\times 100$ ).

We regress banks' bond holdings and profitability on their lagged exposure to patriotic sentiment, controlling for bank characteristics and including bank and year fixed effects. [Table B12](#) reports summary statistics, and [Table 6](#) reports the results. We find that increased exposure to patriotic regions significantly boosts both banks' bond holdings and performance. These findings provide direct evidence that operating in more patriotic regions enhanced bank performance.

## 6.2 Official versus private banks

A second, longer-term channel operates through the gradual buildup of public familiarity and trust in modern banking. As citizens repeatedly interacted with modern banks through bond subscriptions and transactions, they accumulated direct experience with these institutions, developing familiarity and, over time, financial literacy ([Hilt et al., 2022](#)). This accumulation of experience gradually built public trust in modern banks as a legitimate and reliable form of financial intermediation.

Because trust in the banking sector is inherently difficult to measure directly, we rely on two indirect tests to provide suggestive evidence. The first test examines whether trust in official banks spilled over to private banks, reflecting a broader buildup of public confidence in modern banking as an institution.

As discussed in [Section 2.1](#), government bond underwriting was initially dominated by state-backed official banks; private banks entered later with limited involvement in the underwriter role. Thus, the public's first engagement with modern banking was through official banks. As

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<sup>34</sup>Marketable securities, as recorded in bank balance sheets, refer to investments in financial instruments that can be easily traded in the market. They are typically liquid assets held by banks for managing short-term needs or for investment purposes; these include stocks, bonds, and other financial instruments that can be readily converted into cash. Based on a survey of 28 banks from 1921 to 1934, [Lan \(2005\)](#) found that more than three-quarters of these marketable securities consisted of government bonds.

<sup>35</sup>Beginning in 1928, banks were officially required to back the issuance of banknotes with 40% government bonds as reserves ([Braggion, Manconi, and Zhu, 2020](#)). Before 1928, the ratio was not stipulated. With some variation across banks, this ratio was about 30% in the largest official banks in China, including the Bank of China and the Bank of Communications ([Qian, 1984](#)).

these banks demonstrated operational reliability, the public’s confidence gradually extended to private banks offering similar services. We posit that official bank growth in the immediate aftermath of 1914 primarily reflects the public’s engagement with state-backed financial instruments, while the subsequent expansion of private banks is more consistent with a broader buildup of confidence in modern banking as an institution.

To test this, we analyze the dynamic growth of official and private banks separately using the event-study dynamic DiD model, following the method described in Figure 3 but replacing the dependent variable with the number of official or private banks per prefecture-year. The findings, visualized in Figure 7, reveal that official and private banks responded in distinct ways to anti-tax unrest and the 1914 shock. Official banks saw a sharp jump in 1914, aligning with anecdotal evidence of their direct response to the fiscal shock. In contrast, private banks grew gradually, reflecting their slower bond market entry and the progressive buildup of public trust in modern banking. While official banks plateaued after 1914, private banks continued gaining momentum, eventually surpassing the official banks. This suggests that patriotic sentiment first bolstered trust in official banks before extending to private ones.

[Insert Figure 7 about here]

The ultimate, larger effects of anti-tax unrest on private banks are confirmed by Appendix Table B10, which estimates the average anti-tax effects for official and private banks separately. The results show that, on average from 1914 to 1926, each additional incident of unrest increased the number of official banks by 13.66% (Panel A, column 3) and private banks by 23.55% (Panel B, column 3) when evaluated at the mean. This aligns with the intuition that, for the banking sector and for others, once the private sector gains momentum, it expands rapidly and dominates the official sector.

To complement the identification of trust spillover, we conduct a less formal test to explore whether the effects of anti-tax unrest on private banks were channeled through official banks. Specifically, we assess the impact of anti-tax unrest on each bank type while controlling for the lagged presence of the other. Appendix Table B11 shows that anti-tax unrest significantly influenced official banks, even when we account for lagged private banks (column 1). However, when controlling for lagged official banks, the effect of anti-tax unrest on private banks becomes insignificant (column 2). This suggests that patriotism directly drove official bank growth, while private bank expansion was indirectly influenced by official banks.

### 6.3 Banking crisis in 1921

In the second test, we exploit the 1921 banking crisis as a revealed-preference test of public confidence. Bank runs, by their nature, reflect depositors’ mistrust in financial institutions; regions where branches survived are thus regions where public confidence proved resilient.

China experienced nationwide banking panics and widespread mistrust in the financial sector in 1921, triggered by two distinct crises. The first was a crisis of confidence in the central government, driven by a growing fiscal deficit and diplomatic failure at the Washington Naval Conference (1921–1922) (Wu, 2019). This led to a large-scale bank run targeting the two *de facto* central banks, the Bank of China and the Bank of Communications. At the same time, a second financial crisis unfolded in Shanghai, fueled by the collapse of unregulated stock and commodity exchanges. Given that many bank headquarters were based in Shanghai, concerns about their solvency quickly spread, leading to runs on their branches in other provinces and cities (Zhu, 1998). Additional details about these crises, along with historical anecdotes highlighting the role of patriotism in mitigating panic, are provided in [Appendix A.6](#).

Empirically, we construct a bank-prefecture-level panel dataset. On the banking side, our sample includes the Bank of China, the Bank of Communications, and all other banks that were either headquartered in Shanghai or operated at least one branch there at the end of 1920. We assign a unique bank ID to each bank headquarters and its branches. This sample includes 484 bank entities corresponding to 73 unique bank IDs.<sup>36</sup> These bank entities span 49 prefectures.

As we restrict the sample to bank-prefecture pairs where a bank operated at least one branch in the prefecture in 1920, the regression sample is an unbalanced panel of 73 bank groups across 49 prefectures. During the 1921 crisis, 35 banks (7 headquarters and 28 branches) closed across 27 prefectures. Our regression model is specified as follows:

$$\# \text{ bank closure}_{i,b} = \alpha + \beta \times \text{Anti-tax\_unrest}_i + \gamma_1 \times \mathbf{X}_i + \gamma_2 \times \text{HQ-B distance (log)}_b + \mu_b + \lambda_t + \epsilon_i \quad (4)$$

where  $\# \text{ bank closure}_{i,b}$  denotes the number of branch closures in 1921 for bank  $b$  in prefecture  $i$ .<sup>37</sup> Prefecture-level controls include baseline controls as specified in Equation 1, as well as conditional controls as specified in Section 5.3. The regression further controls for the log distance between the bank headquarters ( $HQ$ ) and the branch ( $B$ ). Prefecture *Shuntian* is excluded. Standard errors are clustered at the bank ID level.

[Insert Table 7 about here]

The results reported in Table 7 indicate that anti-tax unrest is negatively associated with the likelihood of a bank entity closing during the 1921 banking crisis. Because our model includes bank fixed effects, this relationship rules out, at least partially, the effects from bank-level fundamentals. Thus, variation in branch-level resilience likely captures differences in depositor trust toward the banking sector more broadly. Specifically, the finding that banks are more resilient

<sup>36</sup>The 484 bank entities consist of 71 headquarters and 413 branches. Two banks had headquarters located outside the sample provinces (e.g., *Liaoning*) and are therefore only represented by their branches.

<sup>37</sup>A bank can have several branches in a prefecture as they might have multiple branches in different counties within a prefecture, or have more than one branch in large counties.

in regions with stronger patriotic sentiment suggests that patriotism fosters institutional trust in banks. This trust helps sustain depositor confidence during times of financial distress, thereby mitigating the risk of bank runs when fundamental shocks occur.

## 7 Generalization

While systematic cross-country evidence is beyond the scope of this paper, historical narratives from other countries suggest that the patriotism-banking nexus is not unique to China. To demonstrate the wider applicability of our findings, we provide historical examples showing that this pattern extends beyond China.

**Great Britain.** The Bank of England (BOE), the first modern bank in Great Britain, was founded in 1694 amid ongoing military conflicts and shortly after the Glorious Revolution of 1688. It was founded as a private bank but served as the government's banker, with the primary mission of managing public debt and financing war efforts (North and Weingast, 1989). During the Nine Years' War (1688–1697), monetary contributions were promoted as a patriotic duty (Murphy, 2006). Motivated by this sentiment, a third of the first government war loan underwritten by the BOE was subscribed on the first day, another third within two days, and the entire loan was fully subscribed within ten days (North and Weingast, 1989). The BOE's success in raising capital laid the foundation for modern capital markets in England. Throughout the eighteenth and nineteenth centuries, modern private banks emerged, gradually replacing traditional institutions like local goldsmiths (North and Weingast, 1989).<sup>38</sup>

**United States.** The foundations of American modern banking were closely linked to patriotic efforts during the Revolutionary War (1775–1783). The nation's first congressionally chartered bank, the Bank of North America (1782), was established under the leadership of Robert Morris to manage the financial logistics of the war. Morris, who even pledged his personal credit to secure loans and supplies, came to be regarded as the “patriotic financier” (Oberholtzer, 1903).<sup>39</sup>

Another example is the First Bank of the United States, which was established by the Treasury of the United States in 1791 to stabilize the national economy after the Revolutionary War. To promote investment as a patriotic act, the Treasury deliberately scheduled the initial stock sale for July 4, 1791, coinciding with Philadelphia's fifteenth Independence Day celebration. The 20,000 publicly available shares sold out within hours, reflecting strong public support for

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<sup>38</sup>Similar to China, various countries had traditional financial institutions before the establishment of modern banks, such as the goldsmiths in Great Britain. These traditional institutions were generally small in scale and relied on local, soft information to operate. Public trust in these banks was based on reputation and the owners' unlimited liability.

<sup>39</sup>His legacy is memorialized until today by a statue standing beside the First Bank of the United States in Philadelphia (see Figure B12).

the state (Cowen, 2000).

**Japan.** Japan’s first modern bank, the First National Bank of Japan, was founded in 1873 after the Meiji Restoration (1868). In 1876, the government issued Kinroku bonds to convert samurai stipends into interest-bearing securities, easing resistance to reform and funding modernization. To support this transition, the National Banking Act allowed government banks to use Kinroku bonds as capital and to issue convertible notes (Yamamura, 1967). This policy spurred rapid banking expansion, with the number of banks rising from six in 1876 to 153 within three years, creating a nationwide banking network (Tang and Basco, 2023).

Patriotism remains relevant in reinforcing public and financial trust in modern societies. While government backing has been proven to help mitigate financial instability, such as bank runs (Diamond and Dybvig, 1983), trust in governments is fragile and often declines during crises (Algan et al., 2017; Doerr et al., 2022). Acting as the ultimate safeguard, patriotism can sustain public support even when trust in government is low, helping to stabilize the state and to reinforce financial and economic stability.

## 8 Conclusion

This paper examines the role of patriotism in the emergence of modern banks, focusing on early twentieth-century China. The outbreak of World War I in 1914 created an exogenous fiscal shock, prompting the newly formed Republican government to issue domestic bonds despite its limited credibility. We argue that patriotism reinforced public trust in the government, enabling banks, key underwriters of these bonds, to expand.

Inspired by social contract theory, we use anti-tax unrest under the preceding late Qing regime as a proxy for the strength of social contract, which embodied patriotic sentiment and was manifested in patriotic civic engagement efforts after institutional change. Regions with greater pre-1912 anti-tax unrest exhibited stronger state-building efforts during the Republican era. Using a DiD approach, we find that there were more subscriptions to government bonds in patriotic regions, fostering modern banking expansion after 1914. This result holds under the DiD-IV approach, additional socioeconomic controls, and falsification tests.

In terms of channels, we first find that banks operating in more patriotic regions subsequently held more government bonds and achieved higher profitability, consistent with the direct profitability channel. We also provide suggestive evidence on the trust channel. Dynamic DiD analyses reveal that while anti-tax unrest spurred the growth of official banks immediately after 1914, the impact on private banks emerged only after 1920, consistent with public confidence gradually extending from state-backed to private financial institutions. Further, with bank fixed effects controlled for, the 1921 bank-run regressions show that branches in more patriotic prefectures were significantly more resilient, consistent with patriotic engagement having

bolstered depositor confidence and mitigated run risk when shocks hit.

Finally, the pattern we document in China is not unique. Across vastly different political and institutional contexts, patriotism repeatedly served as a substitute for formal institutional credibility in the formative stages of modern banking.

Overall, this study underscores patriotism as a crucial driver of trust in government and financial institutions, one that shapes the interactions among civilians, the state, and banking development.

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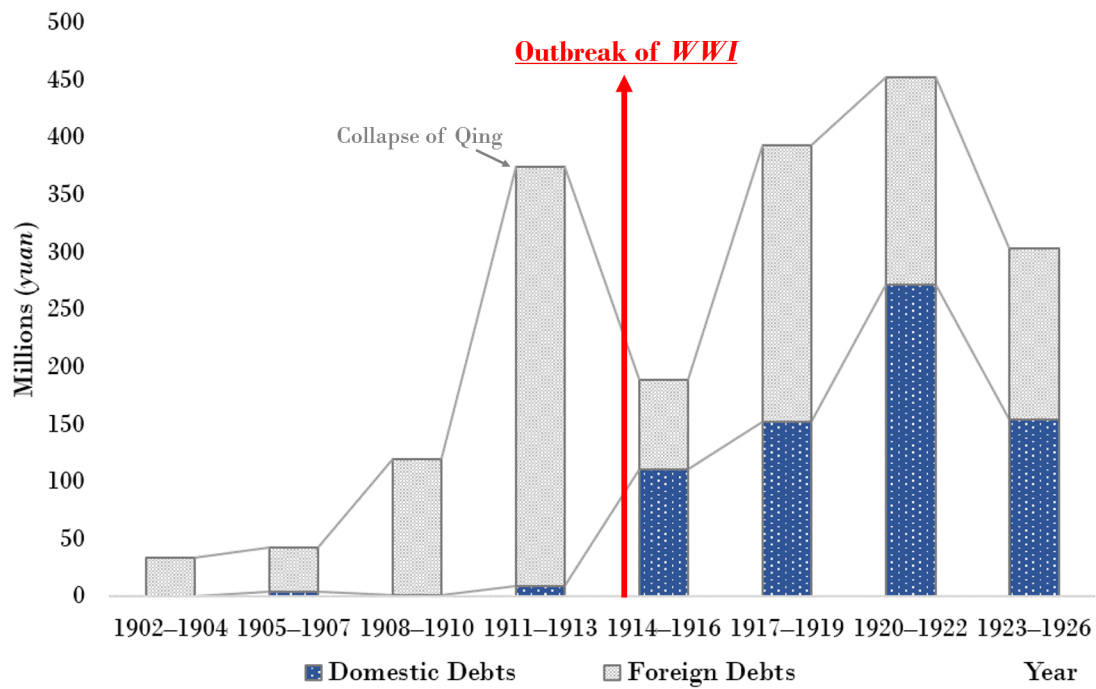
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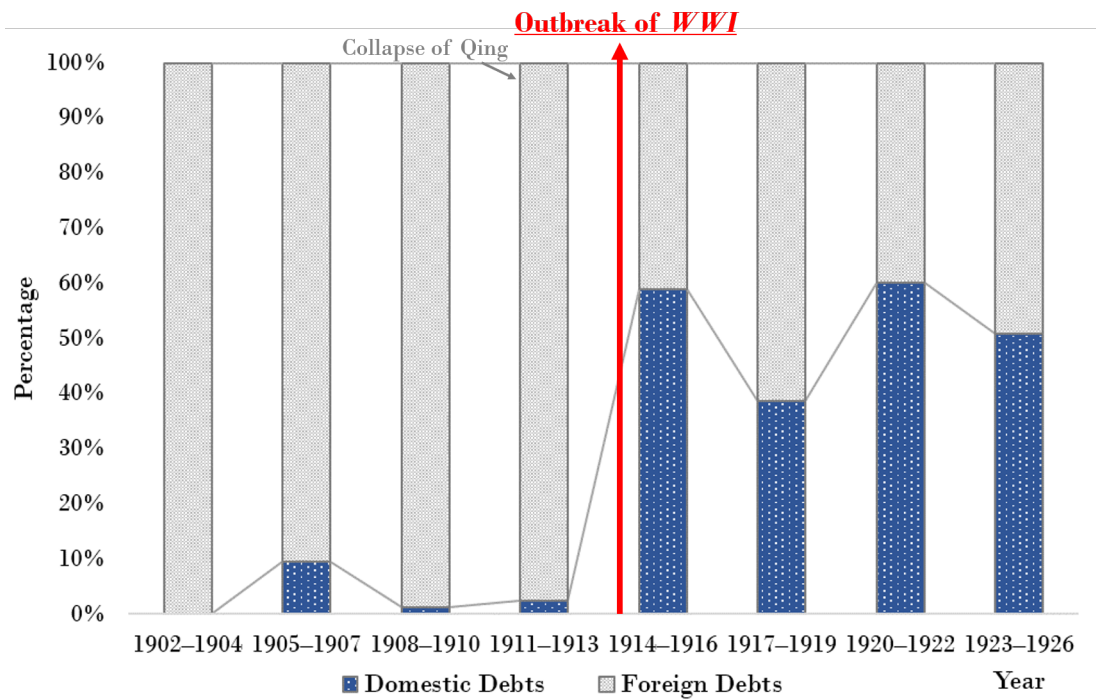
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Panel A: The collected amount of domestic and foreign debts



Panel B: The percentage share of domestic and foreign debts

Figure 1. Domestic and foreign debts (1902–1926)

Note: The figures depict the collected amount (in million *Yuan*) and percentage share of domestic and foreign debts in China from 1902 to 1926. Domestic debts refer to Chinese government bonds issued by the central government and 18 provincial governments. Foreign debts refer to debts borrowed by the Chinese government from foreign banks and foreign governments. Data for foreign debts are from Xu (1962), and the data for domestic debts are from Qian (1984), Survey Department of Southern Manchuria Railway Company (1930), Ling (1928), and Fiscal Chronicles from provincial gazetteers.

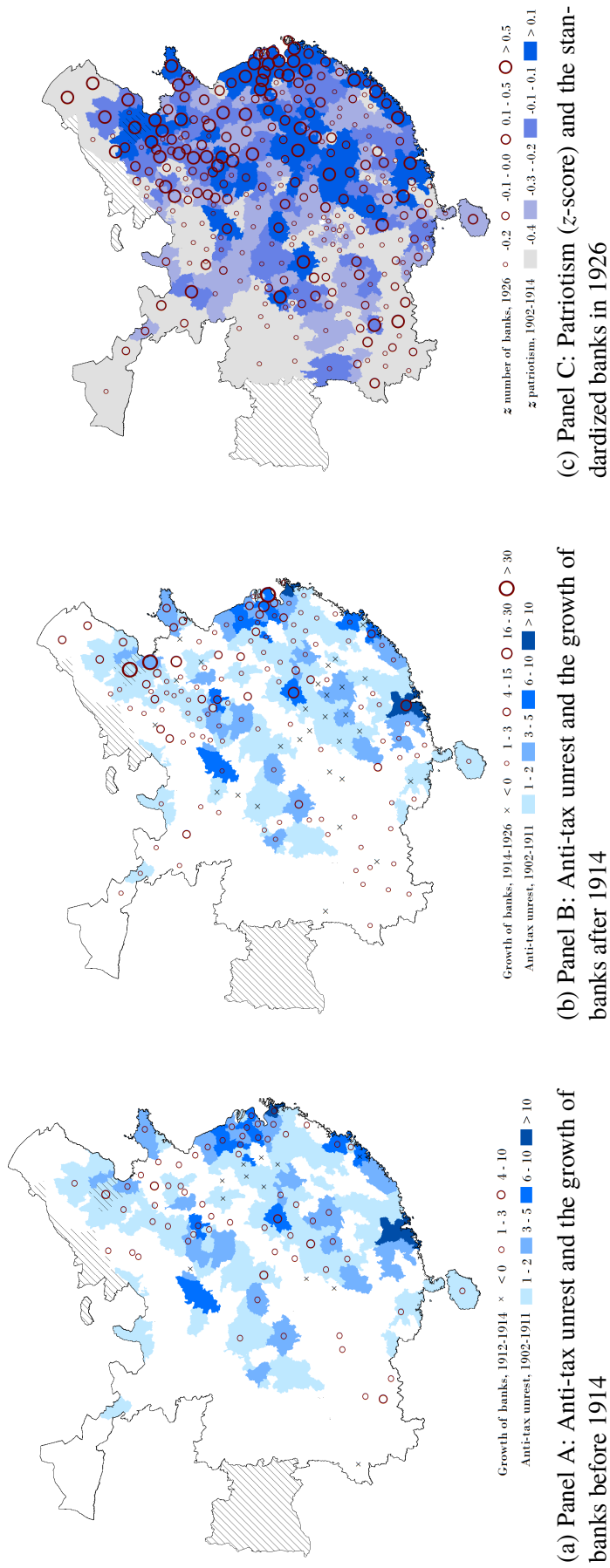


Figure 2. Distributions of anti-tax unrest and growth of modern banks

Note: These maps illustrate the distribution of anti-tax unrest and the growth of modern banks. Anti-tax unrest refers to the total number of unrest events involving tax resistance occurring in 1902–1911 in each prefecture. Growth of Chinese banks refers to the difference in the number of banks between 1912 and 1914 (Panel A), and between 1914 and 1926 (Panel B). Panel C maps the standardized measures of patriotism (the z-score of the average z-scores of anti-tax unrest as well as the three civic engagement measures before 1914) and of banks in 1926. The data are at the prefectural level. The map covers 298 prefectures from 18 provinces in China proper; prefectures outside the regression sample are shaded.

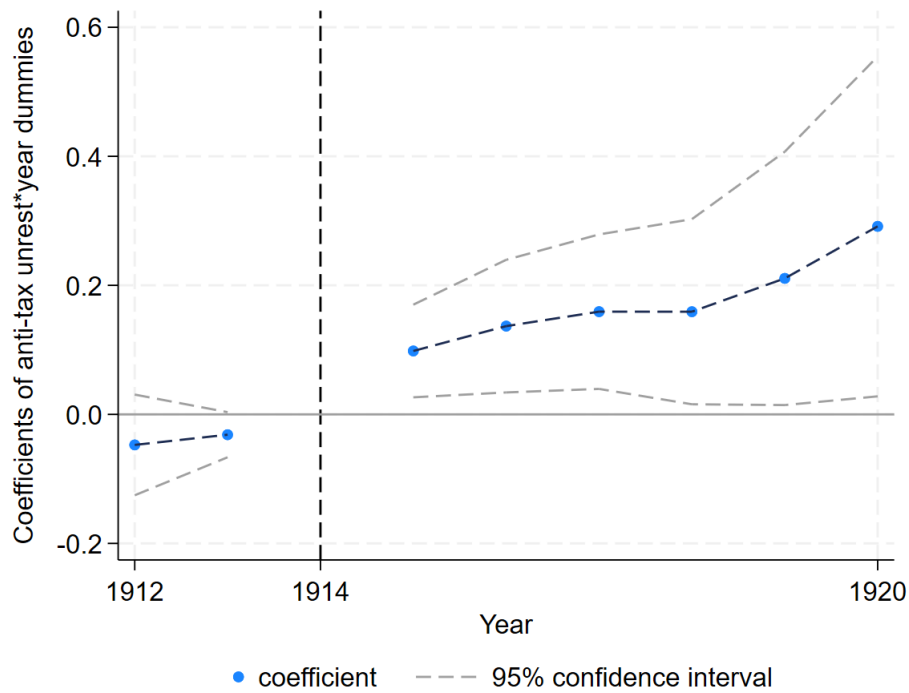


Figure 3. **Parallel trends: The effect of patriotism on modern banks**

Notes: This figure shows the pre- and post-1914 trend in the effect of anti-tax unrest on the number of modern Chinese banks. The coefficients (with 95% confidence intervals) are obtained from regressing the number of banks on the interaction terms between the anti-tax unrest and the year dummies for the years 1912–1920, conditional on the prefectural and year fixed effects and the interactive effects between the year dummies and the baseline prefectural factors (logarithm of prefectural population density in 1910, the logarithm of prefectural land area, the logarithm of the prefecture’s distance to the coast, the logarithm of the river density within each prefecture, the logarithm of the prefectural altitude, the ruggedness of the terrain, and the logarithm of the prefecture’s distance to capital (Beijing)). The year 1914 is set to be the reference year.

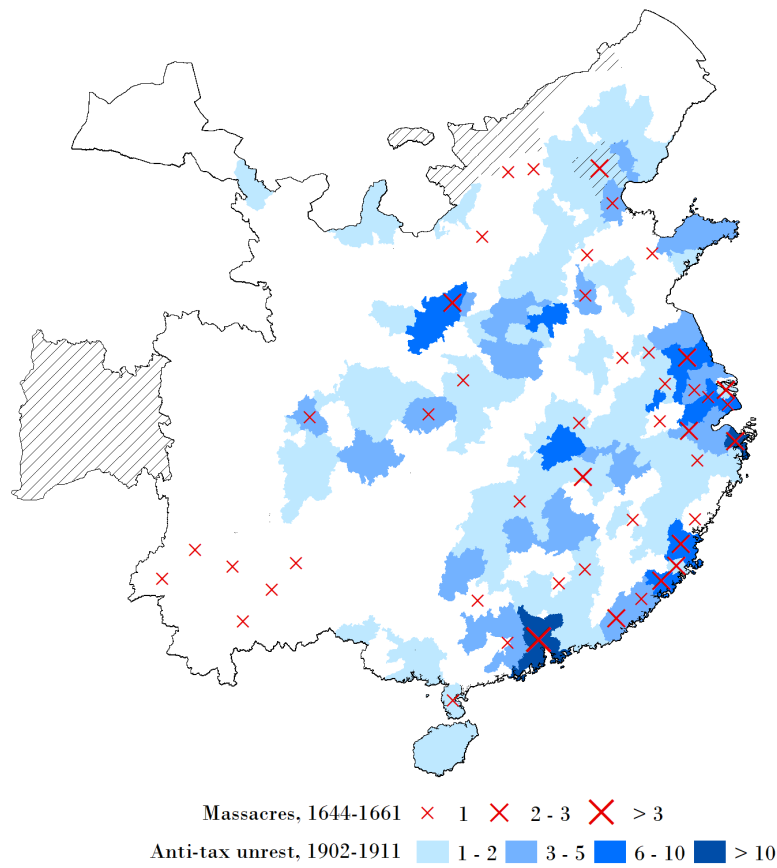


Figure 4. **Distributions of anti-tax unrest and historical massacres**

Notes: This figure depicts the spatial distribution of anti-tax unrest and historical massacres taking place from 1644 through 1661. The data are at the prefectural level. The map covers 298 prefectures from 18 provinces in China proper; prefectures outside the regression sample are shaded.

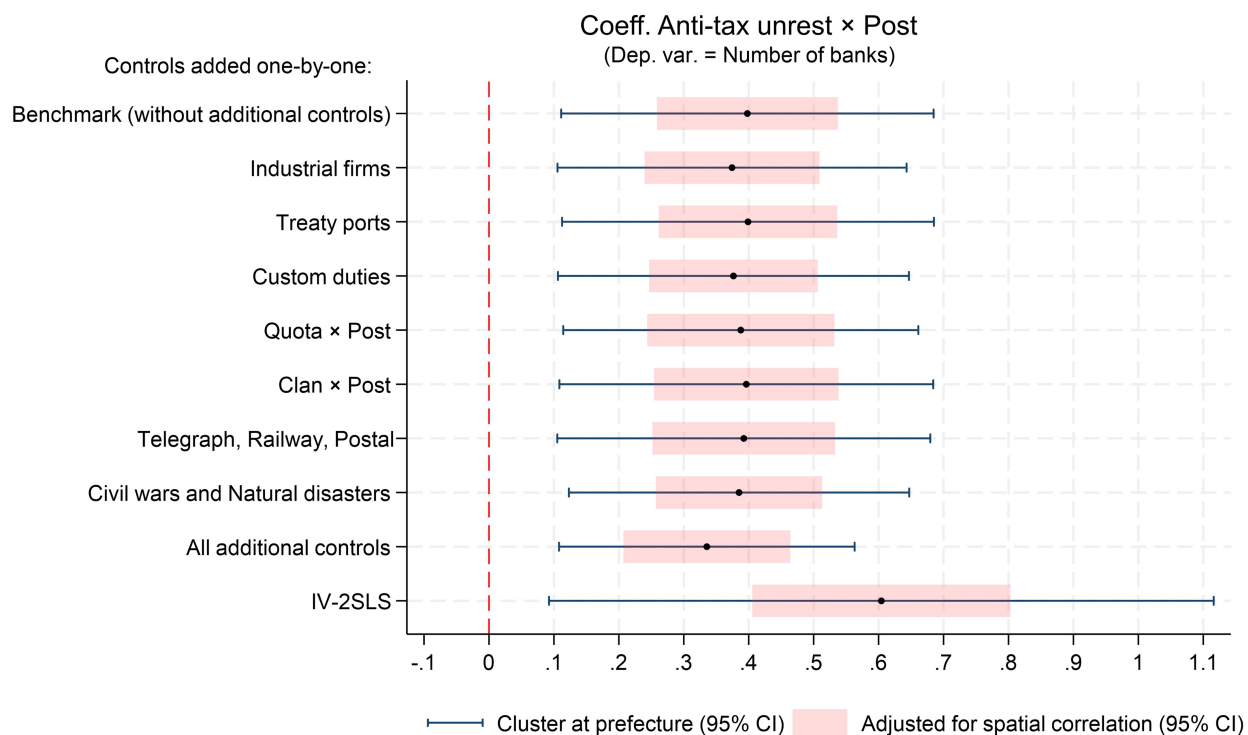


Figure 5. **Additional controls**

Note: This figure examines the effects of anti-tax unrest on modern banks with additional controls. In all models, the dependent variable is the number of banks in each prefecture-year, and the figure plots the coefficient of the main explanatory variable, Anti-tax unrest × Post. Model 1 replicates column 2 of Table 4, serving as a benchmark. Additional controls are added one-by-one, including: the logarithm of the number of newly established industrial firms (model 2), the number of treaty ports (model 3), the logarithm of the custom duty income in these treaty ports (model 4), the interaction term between log civil examination quota and post-1914 dummy (model 5), the interaction term between log clan and post-1914 dummy (model 6), the number of telegraph stations, railway stations and postal offices (model 7), the occurrence of local civil wars and the occurrence of natural disasters (flood and drought) (model 8). All time-variant additional controls are lagged for one year. In model 9, all the above-mentioned controls are included simultaneously. Column 10 employs a DiD-IV approach where Anti-tax unrest × Post is instrumented by Massacres × Post. Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures, covering the period 1912–1926. Standard errors are either clustered at the prefecture level or calculated using the method of [Colella et al. \(2019\)](#) with a distance parameter of 205 kilometers (3 times prefectural radius). Appendix Figure B6 plots the coefficients using models with the standardized number of banks and the standardized measure of patriotism.

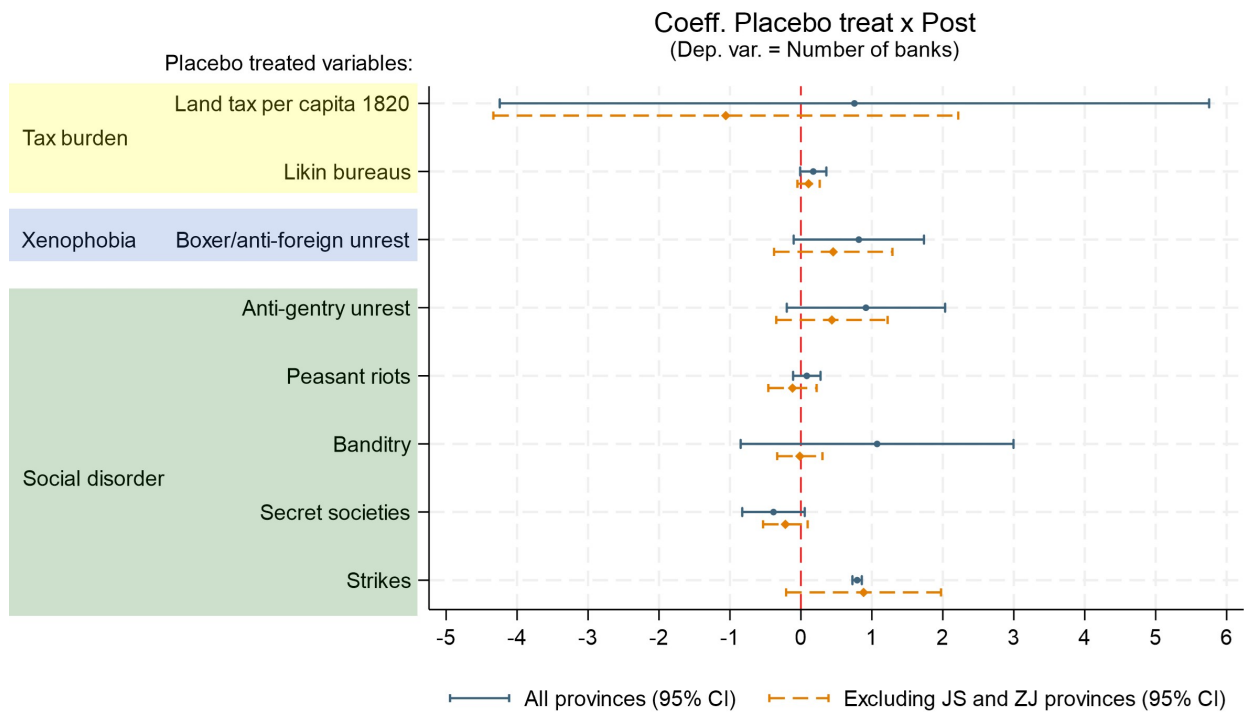


Figure 6. **Falsification tests**

Note: This figure displays the coefficients of placebo explanatory variables from falsification tests. In all models, the dependent variable is the number of modern Chinese banks in each prefecture for each year, and the placebo explanatory variable is structured as treat  $\times$  post. The placebo “treat” variables fall into three categories. In models 1-2, tax burden is measured by the logarithm of per capita land tax in 1820 (model 1) and the number of commercial tax bureaus (model 2). Model 3 examines xenophobia, which is proxied by a dummy that indicates the occurrence of either the Boxer movement or anti-foreigner unrest. Models 4-8 examine general social disorder which is proxied by five measures: the number of anti-gentry unrest events in 1902–1911 (model 4), the number of peasant unrest events in 1902–1911 (model 5), the number of banditry activities in 1902–1911 (model 6), the number of secret societies (model 7), and the number of strikes (model 8). Detailed variable definitions are provided in Section 5.4. All controls and model specifications are identical to those in column 2 of Table 4. Coefficients for the placebo treatment variables are reported for the full sample and for a subsample that excludes *Zhejiang* and *Jiangsu* provinces.

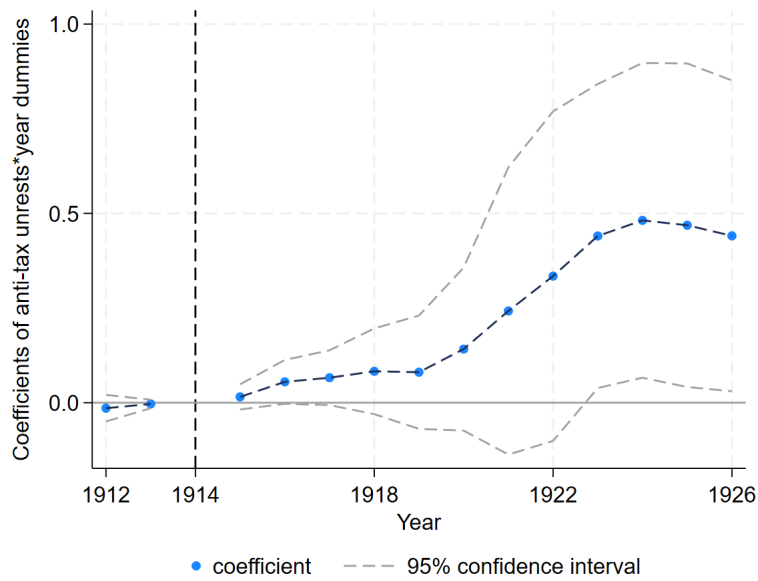
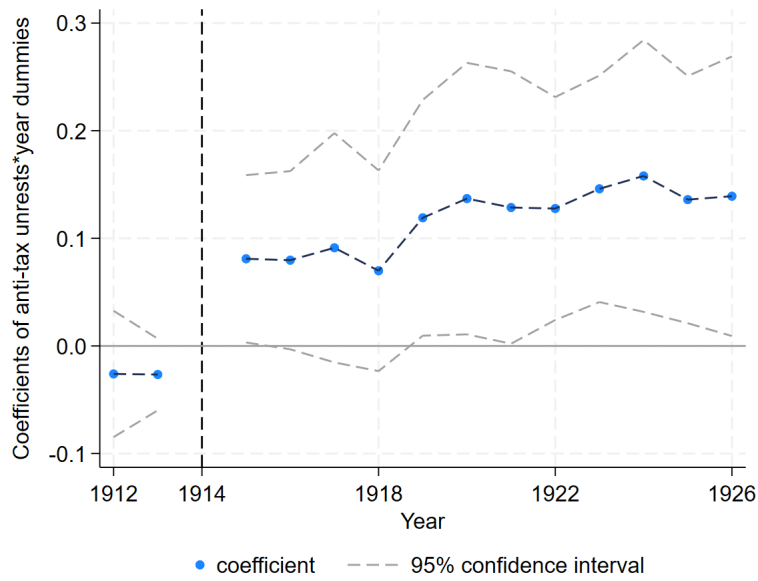


Figure 7. **The effect of patriotism on official and private banks**

Notes: The figures show the dynamic effects of anti-tax unrest on the number of official banks (Panel A) and private banks (Panel B). The coefficients (with 95% confidence intervals) are obtained from regressing the number of official (private) banks on the interaction terms between the anti-tax unrest and the year dummies for the years 1912–1926, conditional on the prefectural and year fixed effects and the interactive effects between the year dummies and the baseline prefectural factors (logarithm of prefectural population in 1910, the logarithm of prefectural land area, the logarithm of the prefecture's distance to the coast, the logarithm of the river density within each prefecture, the logarithm of the prefectural altitude, the ruggedness of the terrain, and the logarithm of the prefecture's distance to capital (Beijing)). The year 1914 is set to be the reference year.

Table 1. Summary Statistics

|                                                                                              | Obs.  | Mean    | S.D.    | Min    | Max      |
|----------------------------------------------------------------------------------------------|-------|---------|---------|--------|----------|
| <b>A. Prefecture-level variables</b>                                                         |       |         |         |        |          |
| Anti-tax unrest                                                                              | 286   | 1.143   | 2.179   | 0      | 21       |
| Cadets in Baoding Academy after 1912                                                         | 286   | 21.612  | 45.610  | 0      | 387      |
| Cadets in Baoding Academy diff. (after-before 1912)                                          | 286   | 13.608  | 28.394  | -15    | 239      |
| Political associations (1912–1926)                                                           | 286   | 0.738   | 5.156   | 0      | 77       |
| Social associations (1912–1926)                                                              | 286   | 0.566   | 3.180   | 0      | 35       |
| PC civic engagement (1912–1926)                                                              | 286   | 0       | 1.394   | -0.371 | 17.050   |
| z civic engagement (1912–1926)                                                               | 286   | 0       | 1       | -0.336 | 10.844   |
| PC patriotism (1902–1914)                                                                    | 286   | 0       | 1.505   | -0.562 | 18.099   |
| z patriotism (1902–1914)                                                                     | 286   | 0       | 1       | -0.440 | 10.565   |
| Collected value for 1914 patriotic bonds (in million yuan)                                   | 286   | 0.047   | 0.209   | 0      | 2.38     |
| z Collected value for 1914 patriotic bonds                                                   | 286   | 0       | 1       | -0.228 | 11.184   |
| Collected value for 1914 patriotic bonds per capita<br>(in million yuan, per 100,000 people) | 286   | 0.019   | 0.067   | 0      | 0.856    |
| z Collected value for 1914 patriotic bonds per capita                                        | 286   | 0       | 1       | -0.289 | 12.505   |
| Chinese banks                                                                                | 4,290 | 1.915   | 5.538   | 0      | 101      |
| Official banks                                                                               | 4,290 | 1.171   | 2.160   | 0      | 19       |
| Private banks                                                                                | 4,290 | 0.692   | 3.585   | 0      | 78       |
| <i>Baseline controls</i>                                                                     |       |         |         |        |          |
| Population in 1910 (in 100,000 people)                                                       | 286   | 13.995  | 13.397  | 0.330  | 75.770   |
| Land area (in 1,000 sq.km.)                                                                  | 286   | 14.663  | 14.205  | 0.262  | 131.190  |
| Population density (1,000 people/sq.km.)                                                     | 286   | 0.134   | 0.159   | 0.000  | 1.878    |
| Distance to the coast (km)                                                                   | 286   | 500.337 | 382.237 | 0.381  | 2181.953 |
| River length (km)                                                                            | 286   | 3.378   | 5.790   | 0.024  | 38.037   |
| River density (m/sq.km.)                                                                     | 286   | 0.824   | 4.322   | 0.001  | 58.341   |
| Altitude (in 100 m)                                                                          | 286   | 7.484   | 8.102   | 0.040  | 42.778   |
| Ruggedness                                                                                   | 286   | 1.004   | 0.006   | 1.000  | 1.040    |
| Distance to Beijing (km)                                                                     | 286   | 1.230   | 0.589   | 0.099  | 2.499    |
| <i>Instrumental variable</i>                                                                 |       |         |         |        |          |
| Massacres                                                                                    | 286   | 0.213   | 0.580   | 0      | 5        |
| <i>Additional controls (lagged)</i>                                                          |       |         |         |        |          |
| Modern industrial firms                                                                      | 4,290 | 0.379   | 2.141   | 0      | 51       |
| Treaty ports                                                                                 | 4,290 | 0.157   | 0.449   | 0      | 4        |
| Custom duties (in million yuan)                                                              | 4,290 | 0.116   | 0.942   | 0      | 24.360   |
| Civil examination quota                                                                      | 286   | 103.164 | 76.009  | 0      | 354      |
| Clan (before 1911)                                                                           | 286   | 20.811  | 79.174  | 0      | 718      |
| Telegraph stations                                                                           | 4,290 | 1.999   | 1.754   | 0      | 11       |
| Railway stations                                                                             | 4,290 | 0.624   | 1.442   | 0      | 8        |
| Postal offices                                                                               | 4,290 | 4.701   | 2.847   | 0      | 16       |
| Civil wars                                                                                   | 4,290 | 0.035   | 0.250   | 0      | 5        |
| Natural disasters                                                                            | 4,290 | 0.167   | 0.373   | 0      | 1        |
| <i>Placebo treats</i>                                                                        |       |         |         |        |          |
| Land tax per capita in 1820                                                                  | 286   | 1.082   | 0.102   | 1      | 1.959    |
| Commercial tax bureaus                                                                       | 286   | 2.654   | 3.482   | 0      | 20       |
| Boxer or anti-foreigner unrest                                                               | 286   | 0.336   | 0.473   | 0      | 1        |
| Anti-gentry unrest                                                                           | 286   | 0.476   | 1.358   | 0      | 10       |
| Peasant riots                                                                                | 286   | 1.608   | 3.548   | 0      | 29       |
| Banditry                                                                                     | 286   | 0.269   | 0.907   | 0      | 7        |
| Secret societies                                                                             | 286   | 0.458   | 0.938   | 0      | 7        |
| Strikes                                                                                      | 286   | 0.490   | 3.279   | 0      | 53       |
| Resist tyranny                                                                               | 286   | 0.178   | 0.670   | 0      | 6        |
| Oppose policies                                                                              | 286   | 0.413   | 1.088   | 0      | 8        |
| Uprisings                                                                                    | 286   | 0.126   | 0.548   | 0      | 7        |
| Mutiny                                                                                       | 286   | 0.063   | 0.271   | 0      | 2        |
| All resist government unrest                                                                 | 286   | 0.780   | 1.738   | 0      | 10       |
| <b>B. Province level variables</b>                                                           |       |         |         |        |          |
| Anti-tax unrest events                                                                       | 17    | 17.882  | 13.242  | 0.000  | 44.000   |
| Collected value (in million yuan)                                                            | 255   | 0.544   | 1.751   | 0.000  | 17.750   |
| Collected value for voluntary bonds (in million yuan)                                        | 255   | 0.226   | 1.095   | 0.000  | 14.968   |
| Collected value for compulsory bonds (in million yuan)                                       | 255   | 0.276   | 1.334   | 0.000  | 17.750   |
| Target value (in million yuan)                                                               | 255   | 0.879   | 2.334   | 0.000  | 18.250   |
| Target value for voluntary bonds (in million yuan)                                           | 255   | 0.382   | 1.545   | 0.000  | 15.000   |
| Target value for compulsory bonds (in million yuan)                                          | 255   | 0.425   | 1.704   | 0.000  | 18.250   |

**Table 2. Anti-tax unrest and civic engagement**

This table examines the effects of anti-tax unrest on civic engagement activities. Panel A focuses on military participation. The dependent variable in column 1-2 is the number of cadets at the Baoding Military Academy from 1912 through 1923. The prefectural distribution of cadets is identified based on the students' birthplace. In column 3-4, the dependent variable is the difference in the number of cadets post- and pre-1912. Panel B focuses on political and social participation. The dependent variable in column 1 and 2 is the number of associations promoting political ideologies and the dependent variable in column 3 and 4 is the number of associations aimed at social welfare, such as charities and support groups for vulnerable demographics like women and youth. In Panel C, the dependent variables are the principal component of the number of cadets, the number of political associations, and the number of social associations between 1912–1926 (column 1-2), and the z-score of the average z-scores of the three (column 3-4). The main explanatory variable for all panels is the number of anti-tax unrest events (or its z-score in panel C) in each prefecture from 1902 through 1911. Control variables include the log of prefectural population density in 1910, the log of prefectural land area, the log of the prefecture's distance to the coast, the log of the river density within each prefecture, the log of the prefectural altitude, the ruggedness of the terrain, and the log of the prefecture's distance to capital (Beijing). Appendix Table B1 reports the full set of coefficients for all controls. Prefecture fixed effects are included. The analysis encompasses 286 prefectures from 18 provinces within China proper. Robust standard errors are reported in parentheses. Standard errors that account for spatial correlation using [Colella et al. \(2019\)](#) with a distance parameter of 205 kilometers (3 times prefectural radius) are reported in square brackets. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

| Panel A. Military participation                                 | # cadets at the Baoding Military Academy |                                   |                                   |                                   |
|-----------------------------------------------------------------|------------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|                                                                 | After 1912                               |                                   | Diff.(After 1912-Before 1912)     |                                   |
|                                                                 | 1                                        | 2                                 | 3                                 | 4                                 |
| Anti-tax unrest                                                 | 6.509<br>(1.415)***<br>[1.795]***        | 3.513<br>(1.433)**<br>[1.536]**   | 4.598<br>(1.064)***<br>[1.382]*** | 3.053<br>(1.006)***<br>[1.176]*** |
| R-squared                                                       | 0.094                                    | 0.256                             | 0.121                             | 0.210                             |
| Mean of dependent variable                                      | 21.612                                   |                                   | 13.608                            |                                   |
| Panel B. Political and social participation                     | # political associations                 |                                   | # social associations             |                                   |
|                                                                 | 1                                        | 2                                 | 3                                 | 4                                 |
| Anti-tax unrest                                                 | 0.939<br>(0.356)***<br>[0.292]***        | 0.906<br>(0.349)***<br>[0.320]*** | 0.621<br>(0.222)***<br>[0.215]*** | 0.599<br>(0.237)**<br>[0.236]**   |
| R-squared                                                       | 0.155                                    | 0.149                             | 0.178                             | 0.176                             |
| Mean of dependent variable                                      | 0.738                                    |                                   | 0.566                             |                                   |
| Panel C. Principal component (PC) and standardization (z-score) | PC civic engagement 1912–26              |                                   | z civic engagement 1912–26        |                                   |
|                                                                 | 1                                        | 2                                 | 3                                 | 4                                 |
| Anti-tax unrest                                                 | 0.299<br>(0.092)***<br>[0.083]***        | 0.268<br>(0.098)***<br>[0.094]*** |                                   |                                   |
| z anti-tax unrest                                               |                                          |                                   | 0.480<br>(0.130)***<br>[0.120]*** | 0.417<br>(0.144)***<br>[0.140]*** |
| R-squared                                                       | 0.216                                    | 0.230                             | 0.227                             | 0.265                             |
| Controls                                                        |                                          | Y                                 |                                   | Y                                 |
| Observations                                                    | 286                                      | 286                               | 286                               | 286                               |

Table 3. **Anti-tax unrest and government bond subscription**

This table examines the effects of anti-tax unrest on government bond issuance. Panel A presents cross-sectional estimates of the effect of anti-tax unrest on prefecture-level subscriptions to the 1914 Patriotic bond. The dependent variables are the total collected value (in millions of yuan), collected value per capita (in millions of yuan per 100,000 people), and their respective  $z$ -scores. The explanatory variables and controls are identical to Panel C of Table 2. Panel B presents panel estimates of the effect of anti-tax unrest on provincial bond issuance. The dependent variable, collected value (in million yuan), refers to the amount of funds raised by provincial government through bond issuance in each year. The main explanatory variable, *Anti-tax unrest*  $\times$  *Post*, is the number of anti-tax unrest events in each province during 1902–1911 interacting with the post-1914 dummy. Other control variables, target value, interest rate, collateral, and maturity, denote the total target amount (in million yuan), bond interest rate, whether the bonds are collateralized, and years of maturity, respectively. The latter three variables are the average of all bonds issued in a province in a given year, weighted by target value. Column 1 includes all bonds issued by provincial governments. Column 2 focuses on voluntary bonds. Column 3 focuses on compulsory bonds. The regressions include the interaction between post-1914 dummy and each of the baseline controls (identical to column 2 of Table 2, except at the provincial level). Province and year fixed effects are included. The analysis encompasses 17 provinces, covering the period 1912–1926. Standard errors are clustered at the province level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

Panel A. The 1914 Patriotic bond (cross-section)

|                            | Collected value (in million yuan) |                    | Collected value per capita<br>(in million yuan, per 100,000 people) |                     |
|----------------------------|-----------------------------------|--------------------|---------------------------------------------------------------------|---------------------|
|                            | Value                             | $z$ value          | Value                                                               | $z$ value           |
|                            | 1                                 | 2                  | 3                                                                   | 4                   |
| Anti-tax unrest            | 0.057**<br>(0.022)                |                    | 0.014***<br>(0.003)                                                 |                     |
| $z$ anti-tax unrest        |                                   | 0.597**<br>(0.230) |                                                                     | 0.449***<br>(0.095) |
| Controls                   | Y                                 | Y                  | Y                                                                   | Y                   |
| R-squared                  | 0.366                             | 0.366              | 0.247                                                               | 0.247               |
| Observations               | 286                               | 286                | 286                                                                 | 286                 |
| Mean of dependent variable | 0.048                             |                    | 0.019                                                               |                     |

Panel B. Provincial bonds (panel)

|                                | Collected value (in million yuan) |                     |                     |
|--------------------------------|-----------------------------------|---------------------|---------------------|
|                                | All bonds                         | Voluntary bonds     | Compulsory bonds    |
|                                | 1                                 | 2                   | 3                   |
| Anti-tax unrest $\times$ Post  | 0.028**<br>(0.010)                | 0.020**<br>(0.008)  | 0.008<br>(0.008)    |
| Target value (in million yuan) | 0.678***<br>(0.131)               | 0.646***<br>(0.192) | 0.718***<br>(0.191) |
| Interest rate (weighted avg)   | 5.043<br>(4.472)                  | 2.032<br>(7.383)    | 8.372*<br>(4.567)   |
| Collateral (weighted avg)      | -0.553*<br>(0.288)                | -0.861*<br>(0.480)  | -0.484*<br>(0.231)  |
| Maturity (weighted avg)        | -0.048<br>(0.042)                 | -0.017<br>(0.076)   | -0.084<br>(0.063)   |
| Controls $\times$ Post         | Y                                 | Y                   | Y                   |
| Province and Year FE           | Y                                 | Y                   | Y                   |
| R-squared                      | 0.766                             | 0.729               | 0.782               |
| Observations                   | 255                               | 255                 | 255                 |
| Mean of dependent variable     | 0.544                             | 0.226               | 0.276               |

Table 4. **Patriotism and modern banking development**

This table explores the relationship between patriotism and the development of modern banking in China. In Columns 1-4, the dependent variable is the number of modern banks, including both headquarters and branches, in each prefecture for each year. Columns 5-6 use the standardized (z-score) number of banks as the dependent variable. The key explanatory variable is structured as  $\text{treat} \times \text{post}$ , where “treat” is (i) the number of anti-tax unrest events occurring in each prefecture in 1902–1911 (column 1-3), (ii) patriotism, or the principal component of anti-tax unrest in 1902-1911 and the three civic engagement measures (see Table 2 for definitions) in 1912-1914 (column 4), (iii) the standardized anti-tax unrest (column 5), and (iv) the standardized patriotism (column 6). *Post* is a dummy variable that equals one for years after 1914 and zero for years up to and including 1914. Control variables include the interaction of post dummy and the log of prefectural population density in 1910, the log of prefectural land area, the log of the prefecture’s distance to the coast, the log of the river density within each prefecture, the log of the prefectural altitude, the ruggedness of the terrain, and the log of the prefecture’s distance to capital (Beijing). Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures from 18 provinces within China proper, covering the period from 1912, the establishment year of Republican China, through 1926, just before the Northern Expedition. Standard errors clustered at the prefecture level are reported in parentheses. Standard errors that account for spatial correlation using the method of [Colella et al. \(2019\)](#) with a distance parameter of 205 kilometers (3 times prefectural radius) are reported in square brackets. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                                     | Number of banks                   |                                   |                                   |                                   | z number of banks                 |                                   |
|-------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|                                     | 1                                 | 2                                 | 3                                 | 4                                 | 5                                 | 6                                 |
| Anti-tax unrest $\times$ Post       | 0.596<br>(0.186)***<br>[0.063]*** | 0.398<br>(0.146)***<br>[0.051]*** | 0.300<br>(0.110)***<br>[0.045]*** |                                   |                                   |                                   |
| PC patriotism 1902–14 $\times$ Post |                                   |                                   |                                   | 1.789<br>(0.398)***<br>[0.333]*** |                                   |                                   |
| z anti-tax unrest $\times$ Post     |                                   |                                   |                                   |                                   | 0.171<br>(0.063)***<br>[0.031]*** |                                   |
| z patriotism 1902–14 $\times$ Post  |                                   |                                   |                                   |                                   |                                   | 0.502<br>(0.155)***<br>[0.103]*** |
| Pop. density in 1910 $\times$ Post  |                                   | -0.010<br>(0.254)                 | 0.141<br>(0.239)                  | -0.230<br>(0.218)                 | -0.002<br>(0.050)                 | -0.065<br>(0.052)                 |
| Land area $\times$ Post             |                                   | 0.548**<br>(0.238)                | 0.606***<br>(0.217)               | 0.143<br>(0.248)                  | 0.108**<br>(0.047)                | 0.013<br>(0.055)                  |
| Distance to coast $\times$ Post     |                                   | -0.184<br>(0.130)                 | -0.281**<br>(0.112)               | -0.190<br>(0.123)                 | -0.036<br>(0.026)                 | -0.034<br>(0.027)                 |
| River density $\times$ Post         |                                   | 0.533<br>(0.363)                  | 0.361<br>(0.231)                  | 0.369<br>(0.243)                  | 0.105<br>(0.072)                  | 0.084<br>(0.054)                  |
| Altitude $\times$ Post              |                                   | -0.742<br>(0.455)                 | -0.319<br>(0.402)                 | -0.346<br>(0.227)                 | -0.146<br>(0.090)                 | -0.079*<br>(0.047)                |
| Ruggedness $\times$ Post            |                                   | 74.153**<br>(36.373)              | 52.558<br>(36.598)                | 47.054*<br>(26.164)               | 14.621**<br>(7.172)               | 9.683*<br>(5.223)                 |
| Distance to Beijing $\times$ Post   |                                   | -1.273**<br>(0.493)               | -1.435***<br>(0.465)              | -1.298***<br>(0.437)              | -0.251**<br>(0.097)               | -0.247***<br>(0.086)              |
| Prefecture and Year FE              | Y                                 | Y                                 | Y                                 | Y                                 | Y                                 | Y                                 |
| R-squared                           | 0.745                             | 0.755                             | 0.774                             | 0.786                             | 0.755                             | 0.781                             |
| Observations                        | 4290                              | 4290                              | 3930                              | 4290                              | 4290                              | 4290                              |
| Mean of dep. var.                   | 1.915                             | 1.915                             | 1.533                             | 1.915                             | 1.915                             | 1.915                             |
| Sample                              | Full                              | Full                              | Excl. JS &<br>ZJ prov.            | Full                              | Full                              | Full                              |

Table 5. **Instrumental variable (IV)**

This table examines the effects of anti-tax unrest on modern banks using the IV-2SLS approach. The dependent variable is the number of modern banks in each prefecture for each year. The explanatory variable, *Anti-tax unrest*  $\times$  *Post*, is instrumented by *Massacres*  $\times$  *Post*, where massacres refer to the number of massacres that each prefecture had experienced during 1644 and 1661. Panel A reports the 2SLS estimation results. Column 1 reports the result without any controls. Column 2 reports the result with baseline controls identical to column 2 of Table 4. Column 3 excludes Jiangsu and Zhejiang provinces. Column 4 uses standardized banks and patriotism measures (see section 3.3 for definition). Panel B reports the first-stage regression accordingly. Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures, covering the period 1912–1926. Standard errors clustered at the prefecture level are reported in parentheses. Standard errors that account for spatial correlation using the method of Colella et al. (2019) with a distance parameter of 205 kilometers (3 times prefectural radius) are reported in square brackets. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

| Panel A: 2SLS                             | Number of banks                   |                                   |                                  | <i>z</i> number of banks                  |
|-------------------------------------------|-----------------------------------|-----------------------------------|----------------------------------|-------------------------------------------|
|                                           | 1                                 | 2                                 | 3                                | 4                                         |
| Anti-tax unrest $\times$ Post             | 0.749<br>(0.279)***<br>[0.130]*** | 0.631<br>(0.270)**<br>[0.112]***  | 0.476<br>(0.221)**<br>[0.099]*** |                                           |
| <i>z</i> patriotism 1902-14 $\times$ Post |                                   |                                   |                                  | 0.322<br>(0.110)***<br>[0.055]***         |
| Kleibergen Paap F-statistic               | 16.182                            | 10.194                            | 5.229                            | 9.798                                     |
| Kleibergen Paap F-statistic, spatial      | 79.219                            | 51.945                            | 24.337                           | 49.626                                    |
| Panel B: First Stage                      | Anti-tax unrest $\times$ Post     |                                   |                                  | <i>z</i> patriotism 1902–14 $\times$ Post |
|                                           | 1                                 | 2                                 | 3                                | 4                                         |
| Massacres $\times$ Post                   | 2.149<br>(0.534)***<br>[0.240]*** | 1.823<br>(0.571)***<br>[0.251]*** | 1.731<br>(0.757)**<br>[0.348]*** |                                           |
| <i>z</i> massacre $\times$ Post           |                                   |                                   |                                  | 0.409<br>(0.131)***<br>[0.058]***         |
| R-squared                                 | 0.863                             | 0.882                             | 0.870                            | 0.857                                     |
| Controls $\times$ Post                    |                                   | Y                                 | Y                                | Y                                         |
| Prefecture and Year FE                    | Y                                 | Y                                 | Y                                | Y                                         |
| Observations                              | 4,290                             | 4,290                             | 3,930                            | 4,290                                     |
| Sample                                    | Full                              | Full                              | Excl. JS & ZJ<br>prov.           | Full                                      |

Table 6. The effects of anti-tax unrest on bank performance

This table examines the effects of anti-tax unrest on bank performance. Based on the financial statements of 19 major Chinese modern banks from 1922 through 1926, we construct bank-year level financial measures. The dependent variable in column 1 and 2, Bonds/assets, refers to the amount of government bonds held by the bank scaled by total assets ( $\times 100$ ). The amount of government bonds holdings in the bank is the sum of 30% of banknotes and 75% of marketable securities. The dependent variable in column 3 is the bond-holding difference between the current and previous years, scaled by previous year's total assets. The dependent variable in columns 4 and 5 is ROA, defined as the net profit to total assets ratio ( $\times 100$ ). The dependent variable in column 6 is the profit difference between the current and previous years, scaled by previous year's total assets. The main explanatory variable, *Additional exposure to unrest*, is constructed by calculating the weighted sum of anti-tax unrest events in the prefectures where a bank has established new branches within a specific year, with the weighting based on the number of new branches in each prefecture. This weighted sum of unrest is then scaled by the total number of branches in the bank. Columns 2–3 and 5–6 control for bank-level covariates, including the bank's total assets (log), cash-to-asset ratio ( $\times 100$ ), and leverage or debt-to-asset ratio ( $\times 100$ ). The explanatory variables are all lagged for one year. Bank and year fixed effects are included. Standard errors are clustered at the bank level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                                        | Government bond holding |                          |                    | Profitability              |                     |                     |
|----------------------------------------|-------------------------|--------------------------|--------------------|----------------------------|---------------------|---------------------|
|                                        | Bonds/assets            | (Growth of bonds)/assets | ROA                | (Growth of profits)/assets |                     |                     |
|                                        | (1)                     | (2)                      | (3)                | (4)                        | (5)                 | (6)                 |
| Additional exposure to unrest (lagged) | 0.566**<br>(0.232)      | 0.768***<br>(0.265)      | 1.402**<br>(0.318) | 0.307***<br>(0.060)        | 0.268***<br>(0.064) | 0.443***<br>(0.094) |
| Assets (log, lagged)                   |                         | 0.801<br>(2.533)         | -4.088<br>(3.992)  |                            | 0.906<br>(0.626)    | -0.984<br>(1.139)   |
| Leverage (lagged)                      |                         | 0.091<br>(0.067)         | 0.186<br>(0.114)   |                            | -0.028<br>(0.018)   | 0.048*<br>(0.026)   |
| Cash-to-asset (lagged)                 |                         | -0.023<br>(0.186)        | 0.182<br>(0.194)   |                            | 0.012<br>(0.046)    | 0.157***<br>(0.052) |
| R-squared                              | 0.760                   | 0.767                    | 0.082              | 0.836                      | 0.837               | 0.171               |
| Observations                           | 95                      | 95                       | 95                 | 95                         | 95                  | 95                  |
| Mean of dependent variable             | 9.152                   | 9.152                    | 1.690              | 2.021                      | 2.021               | 0.019               |

Table 7. **Banking crisis in 1921**

This table examines the effect of patriotism on bank closures during the 1921 run. The dependent variable is the number of bank entities (head offices or branches) that closed in 1921 in each bank–prefecture cell. The explanatory variable is the number of anti-tax unrest events in 1902–1911 by prefecture. In column 3, anti-tax unrest is instrumented by the number of massacres a prefecture experienced in 1644–1661. Prefectural controls are identical to those in Table 2; additional controls are described in Section 5.3. For time-varying controls, we use their 1920 (lagged) values. Columns 2 and 3 also include the log Headquarter–branch distance. All specifications include bank ID fixed effects. The sample comprises 73 bank IDs and 484 entities across 49 prefectures. Standard errors are clustered at the bank-ID level, and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                             | Number of bank closure |                      |                      |
|-----------------------------|------------------------|----------------------|----------------------|
|                             | OLS                    |                      | IV                   |
|                             | 1                      | 2                    | 3                    |
| Anti-tax unrest             | -0.005***<br>(0.001)   | -0.005***<br>(0.001) | -0.021***<br>(0.006) |
| R-squared                   | 0.424                  | 0.424                |                      |
| Kleibergen–Paap F-statistic |                        |                      | 32.214               |
| Prefectural controls        | Y                      | Y                    | Y                    |
| Additional controls         | Y                      | Y                    | Y                    |
| HQ-B distance               |                        | Y                    | Y                    |
| Bank FE                     | Y                      | Y                    | Y                    |
| Observations                | 307                    | 302                  | 302                  |
| Mean of dep. var.           | 0.100                  | 0.100                | 0.100                |

**Appendices for**  
**Bonds of Love: Patriotism and the Rise of Modern**  
**Banks**

## Appendix A Additional results

### Appendix A.1 Anti-tax unrest and anarchism

Anarchism emerged in late Qing China amid civilian unrest and intellectual ferment. Seeking solutions to national decline and foreign subjugation, some Chinese thinkers turned to anarchism. The *Chunfeng Hui* (Spring Wind Society), formed in Tokyo around 1907, was among the earliest attempts by Chinese students to promote anarchist ideology. However, anarchism remained marginal in China until the May Fourth Movement in 1919. Given its limited presence in China, it is unlikely to confound our results.

To formally test to what extent anti-tax protesters during the late Qing (from 1902 to 1911) adopted anarchism instead of engaging in state-building in the Republican era (from 1912 onward), we compiled data on anarchist associations during the years 1912 through 1926 using the data from [Zhang and Li \(1999\)](#). The data confirm that anarchism remained marginal before 1919, with only 12 of 33 documented groups founded from 1912 through 1919. Post-1919, anarchist ideas spread but remained less influential than those of other political movements. Empirically, we regress anarchist association counts on anti-tax unrest events, using both OLS and Poisson models to account for geographic sparsity and potential zero inflation.

The results reported in Appendix Table [A1](#) show a weak correlation between anti-tax unrest and anarchist associations. This, along with the fact that anarchism only emerged after 1919, suggests anti-tax unrest was unlikely driven by anarchism. Even if such a correlation existed, it would attenuate the association between anti-tax unrest and state formation.

**Table A1. Anti-tax unrest and anarchism**

This table examines whether anti-tax unrest events were associated with anarchism. The dependent variable is the number of anarchistic associations established in each prefecture from 1912 through 1926. The explanatory variable is the number of anti-tax unrest events in each prefecture from 1902 through 1911. Control variables are identical to Table 2. The analysis encompasses 286 prefectures from 18 provinces within China proper. Standard errors are robust and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                            | Number of anarchistic associations |                   |
|----------------------------|------------------------------------|-------------------|
|                            | 1<br>OLS                           | 2<br>OLS          |
| Anti-tax unrest            | 0.168*<br>(0.089)                  | 0.170*<br>(0.100) |
| Controls                   |                                    | Y                 |
| R-squared                  | 0.216                              | 0.208             |
| Observations               | 286                                | 286               |
| Mean of dependent variable | 0.101                              | 0.101             |

## Appendix A.2 Anti-tax unrest and political opportunism

One concern is that anti-tax unrest reflected political opportunism, with participants signaling loyalty to a potential new government for future political gains. Distinguishing patriotism from opportunism is challenging. We resort to both historical narratives and empirical evidence to shed light on their differences.

First, most anti-tax unrest events were grassroots movements led by ordinary citizens without explicit political slogans or objectives. Second, our sample includes events of unrest from as early as a decade before the Qing collapse, a period when no clear alternative government had emerged. Late Qing China saw multiple failed revolutionary movements and lacked a unified opposition to the Qing court (Cameron, 1963). Given the low likelihood of success and the absence of a viable new authority, it is unlikely that early incidents of anti-tax unrest were primarily motivated by political opportunism.

We empirically test political opportunism using “merely symbolic” alignment with the Xinhai Revolution as a proxy. The Xinhai Revolution, triggered by the Wuhan Uprising on October 10, 1911, led to the collapse of the Qing court. Following the uprising, many regions—provinces, prefectures, and counties—formally declared their separation from the Qing. A significant share of these declarations came from Qing officials and local elites and contributed to a largely bloodless transition.

To illustrate the ease of alignment, on November 5, 1911, Jiangsu Governor Cheng Dequan, responding to requests from Suzhou gentry and merchants, declared Jiangsu’s “independence”. He converted the governor’s office into a military governor’s headquarters, replaced the flags at the main entrance, and had a few roof tiles removed from the main hall with bamboo poles to symbolically “destroy for the revolution”. With these simple acts, the “revolution” was considered complete (Li, 2013).

This lack of widespread violence was partly driven by opportunistic officials who abandoned the Qing court and aligned with the revolution in response to shifting power dynamics.

As an example of opportunistic alignment, we reference Liu Xianshi, who served as the magistrate of Xingyi county in Guizhou province during the Qing dynasty and had previously aided the Qing government in suppressing local peasant and secret society uprisings. After the outbreak of the Xinhai Revolution, Liu initially intended to suppress the uprising. However, upon learning that Guizhou’s provincial capital had fallen to revolutionary forces, he recognized that supporting the revolution would be more advantageous and chose to betray the Qing court, aligning strategically with the new order. As a political outcome, he later rose to the position of Guizhou provincial governor under the early Republican government (Worthing, 2007).

Using the *Historical Atlas of the Xinhai Revolution* (Xinhai Revolution and Wuchang Uprising Memorial Museum, 1991), we manually identified 401 local alignments within our sample region. Excluding military conquests and failed alignments, we classify 293 alignments as

“symbolic”. While acknowledging most participants in the Xinhai Revolution were genuinely patriotic, we suggest symbolic aligners were more opportunistically motivated than those who initiated violent conquests.

In Appendix Tables A2 and A3, we replicate equations 1 and 2, replacing the main independent variable with *Symbolic Xinhai alignment* in Appendix Table A2, and with *Symbolic Xinhai alignment*  $\times$  *Post* in Appendix Table A3. Additionally, we include both *Symbolic Xinhai alignment* and *Anti-tax unrest* to conduct horse-race regressions, testing whether the social contract (anti-tax unrest) or political opportunism better predicts civic engagement and bond subscription. Results show political opportunism fails to predict either outcome, while anti-tax unrest coefficients remain significant with magnitudes comparable to those in Tables 2 and 3.

Table A2. Political opportunism and civic engagement

This table examines the effects of political opportunism on civic engagement activities. Political opportunism is measured by *Symbolic Xinhai alignment*—that is, within each prefecture, the ratio of counties that symbolically declared abandonment of the Qing court during the Xinhai Revolution. Panel A focuses on military participation. Panel B focuses on political and social participation. Panel C focuses on the principal component and standardized civic engagement measures. The definitions of dependent variables and model specifications follow Table 2. Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures from 18 provinces within China proper. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

| Panel A. Military participation                                 | # cadets at the Baoding Military Academy |                     |                          |                     |
|-----------------------------------------------------------------|------------------------------------------|---------------------|--------------------------|---------------------|
|                                                                 | After 1912                               |                     | Before 1912              |                     |
|                                                                 | 1                                        | 2                   | 3                        | 4                   |
| Symbolic Xinhai alignment                                       | -7.365<br>(6.561)                        | -12.398*<br>(6.779) | -3.207<br>(4.414)        | -7.498<br>(4.722)   |
| Anti-tax unrest                                                 |                                          | 3.744**<br>(1.489)  |                          | 3.193***<br>(1.056) |
| R-squared                                                       | 0.235                                    | 0.257               | 0.167                    | 0.211               |
| Mean of dependent variable                                      | 21.612                                   |                     | 13.608                   |                     |
| Panel B. Political and social participation                     | # political associations                 |                     | # social associations    |                     |
|                                                                 | 1                                        | 2                   | 3                        | 4                   |
|                                                                 |                                          |                     |                          |                     |
| Symbolic Xinhai alignment                                       | 3.051<br>(2.345)                         | 1.880<br>(1.948)    | 1.613<br>(1.134)         | 0.829<br>(0.883)    |
| Anti-tax unrest                                                 |                                          | 0.871***<br>(0.330) |                          | 0.583**<br>(0.233)  |
| R-squared                                                       | 0.050                                    | 0.153               | 0.054                    | 0.176               |
| Mean of dependent variable                                      | 0.738                                    |                     | 0.566                    |                     |
| Panel C. Principal component (PC) and standardization (z-score) | PC civic engagement 12–26                |                     | z civic engagement 12–26 |                     |
|                                                                 | 1                                        | 2                   | 3                        | 4                   |
|                                                                 |                                          |                     |                          |                     |
| Symbolic Xinhai alignment                                       | 0.679<br>(0.545)                         | 0.326<br>(0.434)    |                          |                     |
| Anti-tax unrest                                                 |                                          | 0.262***<br>(0.095) |                          |                     |
| z symbolic Xinhai alignment                                     |                                          |                     | 0.097<br>(0.086)         | 0.036<br>(0.068)    |
| z anti-tax unrest                                               |                                          |                     |                          | 0.400***<br>(0.141) |
| R-squared                                                       | 0.100                                    | 0.230               | 0.140                    | 0.264               |
| Controls                                                        | Y                                        | Y                   | Y                        | Y                   |
| Observations                                                    | 286                                      | 286                 | 286                      | 286                 |

Table A3. **Political opportunism and government bond subscription**

This table examines the effects of political opportunism on government bond issuance. Political opportunism is measured by *Symbolic Xinhai alignment*—that is, within each prefecture, the ratio of counties that symbolically declared abandonment of the Qing court during the Xinhai Revolution. Panel A focuses on military participation. Panel B focuses on political and social participation. Panel C focuses on the principal component and standardized civic engagement measures. The definitions of dependent variables and model specifications follow Table 3. Province and year fixed effects are included. The analysis encompasses 17 provinces, covering the period from 1912 through 1926. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                                         | Collected value (in million yuan) |                    |                   |
|-----------------------------------------|-----------------------------------|--------------------|-------------------|
|                                         | All bonds                         | Voluntary bonds    | Compulsory bonds  |
|                                         | 1                                 | 2                  | 3                 |
| Panel A. Political opportunism          |                                   |                    |                   |
| Symbolic Xinhai alignment $\times$ Post | 0.176<br>(0.955)                  | 0.955<br>(0.665)   | -0.238<br>(0.478) |
| R-squared                               | 0.766                             | 0.729              | 0.782             |
| Panel B. Horse race                     |                                   |                    |                   |
| Symbolic Xinhai alignment $\times$ Post | -0.961<br>(1.177)                 | 0.339<br>(0.601)   | -0.605<br>(0.798) |
| Anti-tax unrest $\times$ Post           | 0.034**<br>(0.014)                | 0.019**<br>(0.008) | 0.011<br>(0.011)  |
| R-squared                               | 0.765                             | 0.728              | 0.782             |
| Bond controls                           | Y                                 | Y                  | Y                 |
| Controls $\times$ Post                  | Y                                 | Y                  | Y                 |
| Province and Year FE                    | Y                                 | Y                  | Y                 |
| Observations                            | 255                               | 255                | 255               |
| Mean of dependent variable              | 0.392                             | 0.234              | 0.13              |

### Appendix A.3 The Parallel Trends Assumption

We employ the honest DiD method proposed by [Rambachan and Roth \(2023\)](#) to examine the extent to which our baseline DiD estimation is robust against potential violations of the parallel trends assumption.

Specifically, [Rambachan and Roth \(2023\)](#) define  $\delta_t$  as the difference in trends between the treated and control groups that would have occurred in the absence of the treatment effect in a given period  $t$ .  $\Delta$  is the set of  $\delta_t$  for all  $t$ . The authors formulate  $\Delta$  as a function of a parameter  $M$  that bounds how different the post-treatment differential trend may be from the pre-treatment differential trend.<sup>40</sup> A larger value of  $M$  indicates a greater allowance for violations of the parallel trends assumption, pushing the estimated treatment effects toward statistical insignificance. For any given  $M$ , the honest DiD method estimates a coefficient and its confidence interval. The break-even  $M$  is of particular interest as it represents the minimum level of assumption violation required to render the estimated treatment effects statistically insignificant. [Rambachan and Roth \(2023\)](#) propose two types of parallel trend violations:

**1. Relative Magnitudes Bounds (RM).** In the RM approach, the violation of parallel trends in the post-treatment period is assumed to be  $\bar{M}$  times the largest violation of parallel trends in the pre-treatment period.  $\Delta$  is defined as:

$$\Delta^{\text{RM}}(\bar{M}) = \{\delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq \bar{M} \cdot \max_{s < 0} |\delta_{s+1} - \delta_s|\} \quad (5)$$

**2. Smoothness Restrictions (SD).** The SD (“second differences” or “second differential”) assumes that the differential trends evolve smoothly over time, and  $M$  bounds the extent to which the slope of the difference in trends can vary across consecutive periods.  $\Delta$  is defined as:

$$\Delta^{\text{SD}}(M) = \{\delta : \forall t \geq 0, |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq M\} \quad (6)$$

### Applications to our settings

In our baseline model, which estimates the effects of anti-tax unrest on modern banking (equation 3), there are only two periods of pre-treatment trends (1912–1913). To lengthen the pre-trend window, thereby improving the credibility of the honest DiD estimation, we extend the sample back to 1908 and use a sample period of 1908–1926. Appendix Figure A1 illustrates the event study estimations using this window.

Based on this 1908–1926 sample and focusing on the coefficient of *anti-tax unrest* × *year*<sub>1915</sub>, or the first year post-shock, we implement the honest DiD method to assess the robustness of

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<sup>40</sup>The conventional DiD parallel-trends assumption is the special case where post-treatment differential trend is identical to pre-treatment differential trend in the absence of the treatment effect.

the results against varying levels of trend violations. Panels A and B of Appendix Figure [A2](#) plot the results using RM and SD methods, respectively.

For the RM approach, our result is robust up to  $\bar{M} \approx 0.35$ . In other words, the effects of anti-tax unrest on banks in 1915 would become insignificant only if a violation in the post-treatment period (i.e., a change in the trend) were more than 35% as large as the maximum violation observed in the pre-treatment period.

For the SD approach, our results are robust to linear violations of the trend ( $M = 0$ ), and remain statistically significant until  $M$  reaches approximately 0.02.

The above analysis signifies that our core finding—that social contract strength fostered bank development—would only be challenged by the presence of unobserved confounding factors or shocks capable of inducing a sharp and non-linear change in the differential trend of bank growth.

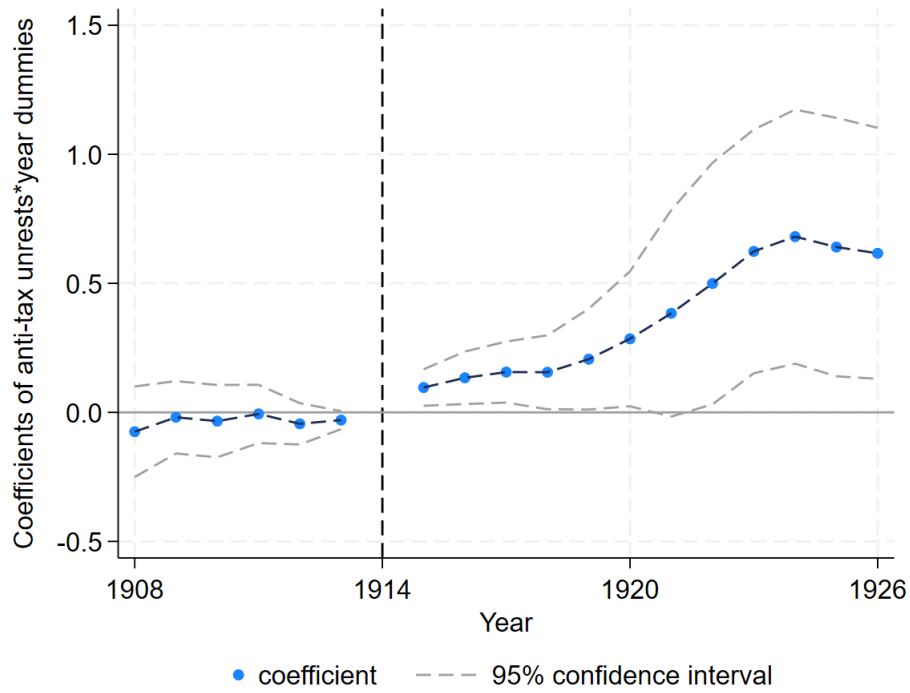
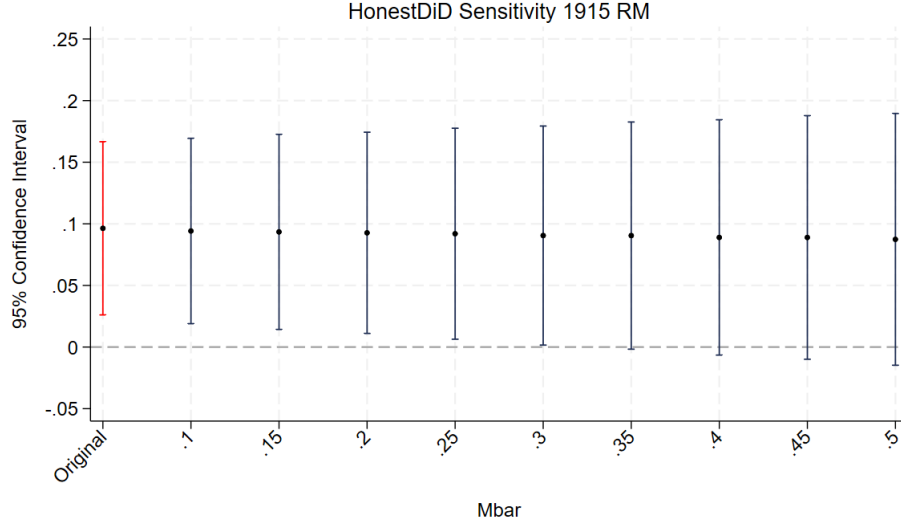
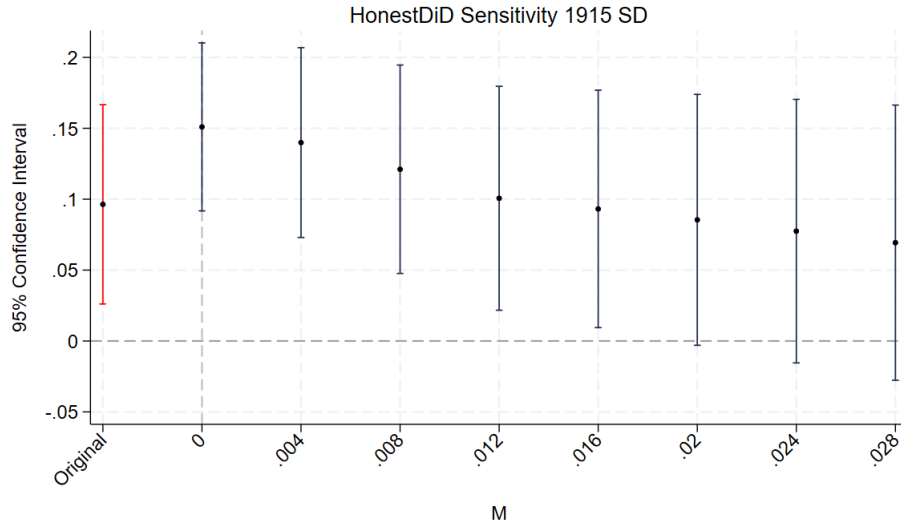


Figure A1. **Extended parallel trends**

Notes: This figure presents an extended version of the pre- and post-1914 trend in the effect of anti-tax unrest on the number of Chinese modern banks. The time span is prolonged to be 1908 through 1926. The coefficients (with 95% confidence intervals) are obtained from regressing the number of banks on the interaction terms between the anti-tax unrest and the year dummies, conditional on the prefectural and year fixed effects and the interactive effects between the year dummies and the baseline prefectural factors. The year 1914 is set to be the reference year.



Panel A. Honest DiD under Relative Magnitudes Bounds (RM)



Panel B. Honest DiD under Smoothness Restrictions (SD)

Figure A2. **The Parallel Trends Assumption**

This figure examines the sensitivity of our results to the potential violations of the parallel trend assumption. The analysis is based on the model in Appendix Figure A1 and focuses on the coefficient of year 1915. The x-axis corresponds to the sets of  $M$ , while the y-axis plots the corresponding coefficient as well as the 95% confidence interval given each value  $M$ . Panel A employs the RM method and Panel B employs the SD method. Detailed explanations of the methods are presented in Section Appendix A.3.

## Appendix A.4 LATE and complier characteristics

Instrumental Variable (IV) approach estimates a Local Average Treatment Effect (LATE), rather than an Average Treatment Effect (ATE) (Jiang, 2017). It is therefore essential to characterize the subpopulation for which this effect is estimated—the compliers. Following the framework of Imbens and Rubin (1997) and assuming no defiers, we classify prefectures into three groups based on their response to the instrument (experiencing a massacre,  $Z$ ) and their subsequent action (engaging in anti-tax unrest,  $D$ ). These groups are defined as follows:

Compliers are prefectures whose behavior follows the instrument; they exhibit unrest if and only if a massacre occurs ( $[Z = 1, D = 1]$  and  $[Z = 0, D = 0]$ ). This group constitutes the majority of our sample (61.89%), and the LATE represents the causal effect for these prefectures. Always-takers are prefectures that experience unrest regardless of the instrument ( $[Z = 0, D = 1]$ ). They represent a substantial portion of the sample (32.17%), suggesting that other significant factors also drive unrest. Never-takers are prefectures that do not experience unrest even if a massacre occurs ( $[Z = 1, D = 0]$ ). This is the smallest group (5.94%).

As shown in Table A4, a comparative analysis of these groups' characteristics reveals a distinct profile. Most notably, always-takers are, on average, located in significantly more developed regions. Compared with compliers, they are closer to Beijing (by 269 km,  $p < 0.001$ ) and the coast (by 119 km,  $p < 0.01$ ), and situated at substantially lower altitudes (by 496 m,  $p < 0.001$ ) with less rugged terrain ( $p < 0.001$ ). This suggests that in more developed and centrally located areas, unrest may be driven by factors other than our instrument, such as a greater awareness of the “social contract”, as we hypothesize. In contrast, the observable differences between compliers and never-takers are not statistically significant, which may be due to the small sample size of the never-taker group ( $N=17$ ). Ultimately, this characterization clarifies the scope of our findings. The LATE estimated by our IV model represents the causal effect of anti-tax unrest on provoking banking development, but this effect is concentrated within a specific subpopulation: the compliers, which are predominantly the prefectures that are relatively less developed, more remote, and located in inland, mountainous areas.

Table A4. **Complier characteristics**

This table reports the characteristics of compliers, never-takers and always-takers in our IV design. Detailed discussions are presented in Appendix Section [Appendix A.4](#)

|                                              | Full sample | Compliers<br>(C) |          | Never-takers<br>(N) |      | N – C    |           | Always-<br>takers (A) |      | A – C       |          |
|----------------------------------------------|-------------|------------------|----------|---------------------|------|----------|-----------|-----------------------|------|-------------|----------|
|                                              |             | Mean             | Mean     | Mean                | Mean | Diff.    | s.e.      | Mean                  | Mean | Diff.       | s.e.     |
| # prefectures                                | 286         |                  | 177      | 17                  |      |          |           | 92                    |      |             |          |
| % prefectures                                | 100%        |                  | 61.89%   | 5.94%               |      |          |           | 32.17%                |      |             |          |
| Pop. density (1,000 people/km <sup>2</sup> ) | 0.134       |                  | 0.125    | 0.101               |      | -0.025   | (0.046)   | 0.158                 |      | 0.033       | (0.021)  |
| Land area (thousand km <sup>2</sup> )        | 14.663      |                  | 15.407   | 15.679              |      | 0.273    | (4.140)   | 13.045                |      | -2.362      | (1.859)  |
| Distance to the coast (km)                   | 500.337     |                  | 544.784  | 439.336             |      | -105.448 | (101.022) | 426.097               |      | -118.687**  | (49.422) |
| River density (km/km <sup>2</sup> )          | 0.824       |                  | 1.162    | 0.410               |      | -0.751   | (1.328)   | 0.250                 |      | -0.912      | (0.571)  |
| Altitude (hundreds of m)                     | 7.484       |                  | 9.012    | 10.138              |      | 1.126    | (2.287)   | 4.052                 |      | -4.960***   | (0.999)  |
| Ruggedness                                   | 1.004       |                  | 1.005    | 1.005               |      | 0.000    | (0.002)   | 1.003                 |      | -0.002***   | (0.001)  |
| Distance to Beijing (km)                     | 1230.234    |                  | 1314.314 | 1355.221            |      | 40.907   | (150.422) | 1045.376              |      | -268.938*** | (72.372) |

## **Appendix A.5 Anti-tax unrest and modern banks: County-level evidence**

We conduct a county-level robustness test to account for variations in banks and anti-tax unrest within prefectures. Our sample includes 1,503 counties across 286 prefectures, averaging five counties per prefecture. County-level data is not used as the baseline due to severe zero inflation, as banks and unrest were concentrated in prefectural capitals. From 1912 through 1926, 77.5% of county-level observations recorded zero banks, compared to 33.6% at the prefecture level.

At the county-year level, we apply the same DiD and IV-DiD strategies as in Section 5, except that the measures are at the county level. Appendix Table A5 presents the results. We start with a baseline model (column 1), add demographic and geographic controls (column 2), and further incorporate additional controls (column 3). The results remain robust; each additional anti-tax unrest event (1902–1911) leads to 0.423 more banks after 1914 (column 2), translating into a 0.182 ( $=0.423/2.330$ ) standard deviation increase, similar to the prefecture-level estimate (0.183  $= 0.398/2.179$ ) reported in Table 4, column 2. Columns 4–6 show IV results, instrumenting anti-tax unrest events with early Qing county-level massacres. The effects remain consistent, though statistical significance weakens.

Table A5. County-level evidence

This table examines the effects of anti-tax unrest on modern banks at the county level. The dependent variable is the number of modern Chinese banks in each county for each year. The explanatory variable, *Anti-tax unrest*  $\times$  *Post* is the number of anti-tax unrest events in each county from 1902 through 1911 interacting with the post-1914 dummy. Column 1–3 report the regression results with OLS. Column 4–6 report results using IV-2SLS approach, where the *Anti-tax unrest*  $\times$  *Post* is instrumented by *Massacres*  $\times$  *Post*. *Massacres* refer to the number of massacres that occurred in each county from 1644 through 1661. Columns 1 and 4 report the result without any controls. Columns 2 and 5 report the result with baseline controls identical to those in Table 4, column 2. Columns 3 and 6 further include additional controls identical to those presented in Section 5.3. County and year fixed effects are included. The analysis encompasses 1,503 counties, covering the period from 1912 through 1926. Standard errors are clustered at the county level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                               | Number of banks    |                    |                    |                   |                   |                  |
|-------------------------------|--------------------|--------------------|--------------------|-------------------|-------------------|------------------|
|                               | OLS                |                    |                    | IV-2SLS           |                   |                  |
|                               | 1                  | 2                  | 3                  | 4                 | 5                 | 6                |
| Anti-tax unrest $\times$ Post | 0.485**<br>(0.216) | 0.423**<br>(0.195) | 0.374**<br>(0.172) | 1.083*<br>(0.573) | 1.094*<br>(0.642) | 1.002<br>(0.710) |
| Controls $\times$ Post        |                    | Y                  | Y                  |                   | Y                 | Y                |
| Additional controls           |                    |                    | Y                  |                   |                   | Y                |
| Year and County FE            | Y                  | Y                  | Y                  | Y                 | Y                 | Y                |
| R-squared                     | 0.742              | 0.743              | 0.745              |                   |                   |                  |
| Kleibergen–Paap F-stat.       |                    |                    |                    | 12.859            | 10.910            | 10.542           |
| Observations                  | 22,545             | 22,545             | 22,545             | 22,545            | 22,545            | 22,545           |
| Mean of dep. var.             | 0.364              | 0.364              | 0.364              | 0.364             | 0.364             | 0.364            |

## Appendix A.6 The 1921 banking crises and the role of patriotism

China experienced nationwide banking panics and widespread mistrust of the financial sector in 1921, triggered by two distinct crises.

The first was a large-scale bank run that began in Beijing in November 1921. Citizens there withdrew massive amounts of deposits from the Bank of China and the Bank of Communications, which were not only the two largest banks at that time but also functioned as *de facto* central banks. This panic quickly spread to other cities where these banks operated branches, including Jinan, Hankou, and Shanghai (Wu, 2019).

This Beijing-originated bank run had two underlying triggers. First, in the early 1920s, the central government's fiscal deficit had grown substantially due to rising military expenditures from ongoing warlord conflicts. This led to a default on foreign indemnity payments in October 1921 (Reinhart and Rogoff, 2008) and forced the government to consolidate previously issued national bonds due to repayment difficulties (Yan, 2015). Such a fiscal crisis eroded public confidence in the government's credibility and solvency. A second trigger was the government's diplomatic failure at the Washington Naval Conference (1921–1922). News and even rumors about this failure, widely reported through telegraphs and newspapers, intensified public fears of losing state sovereignty and facing complete colonization by foreign powers like Japan (Wu, 2019). These combined factors led citizens to lose confidence in the two state-owned banks.

Simultaneously, another banking crisis occurred in Shanghai, driven by the collapse of unregulated stock and commodity exchanges. These had become highly speculative and profitable businesses in the early 1920s (Zhu, 1998). Following the establishment of Shanghai's first commodity exchange in 1920, 136 exchanges and 12 trust companies were created within a year, with the trust companies serving as investment agents for these exchanges. However, in August 1921, the central government began enforcing stricter regulations, banning unregistered exchanges. This regulatory crackdown led to the widespread failure of both exchanges and trust companies. By April 1922, only six exchanges and two trust companies remained operational (Zhu, 1998). This failed speculation had spillover effects on the banking sector. Between 1921 and 1923, six private banks in Shanghai that had invested in these exchanges went bankrupt (Yang, 1930), leading to widespread fears about the safety of deposits throughout Shanghai and its neighboring cities' banking system.

Under such widespread panics, and with the government unable to alleviate the crises, regions with historically patriotic civilians who had founded the Republican regime demonstrated greater resilience. For instance, members of chambers in Shanghai and Tianjin, who had also been significant buyers of government bonds since 1914, along with patriotic scholars like Liang Qichao (1873–1929), delivered public speeches and published articles in major newspapers (L'impartial (Dagong Bao), 1921; The China Press, 1921). These efforts called for citizens to maintain confidence in government banks, to disregard rumors, and to cease withdrawing their

deposits as a patriotic act supporting the state. All these initiatives helped the banks in these areas, particularly the official ones, resume operations even after enduring bank runs.

In contrast, in some prefectures without a historical presence of patriotic figures, the banking panics were more prolonged. A notable example is Zhoukou in Henan province, the hometown of Yuan Shikai (1859–1916). Zhoukou, home to branches of both the Bank of China and the Bank of Communications, was also affected by the banking panics that originated in Beijing. According to *Banker's Weekly* on December 7, 1921 ([Banker's Weekly, 1921](#)), China's leading finance-specialized magazine established in 1917, while the Bank of China branch in Zhoukou partially stabilized the market, the Bank of Communications branch, which primarily focused on remittance services, ultimately closed. This closure was driven by the collapse of the remittance business, as no one stepped forward to dispel the rumors or rebuild public confidence in the banks.

## **Appendix B   Figures and tables**

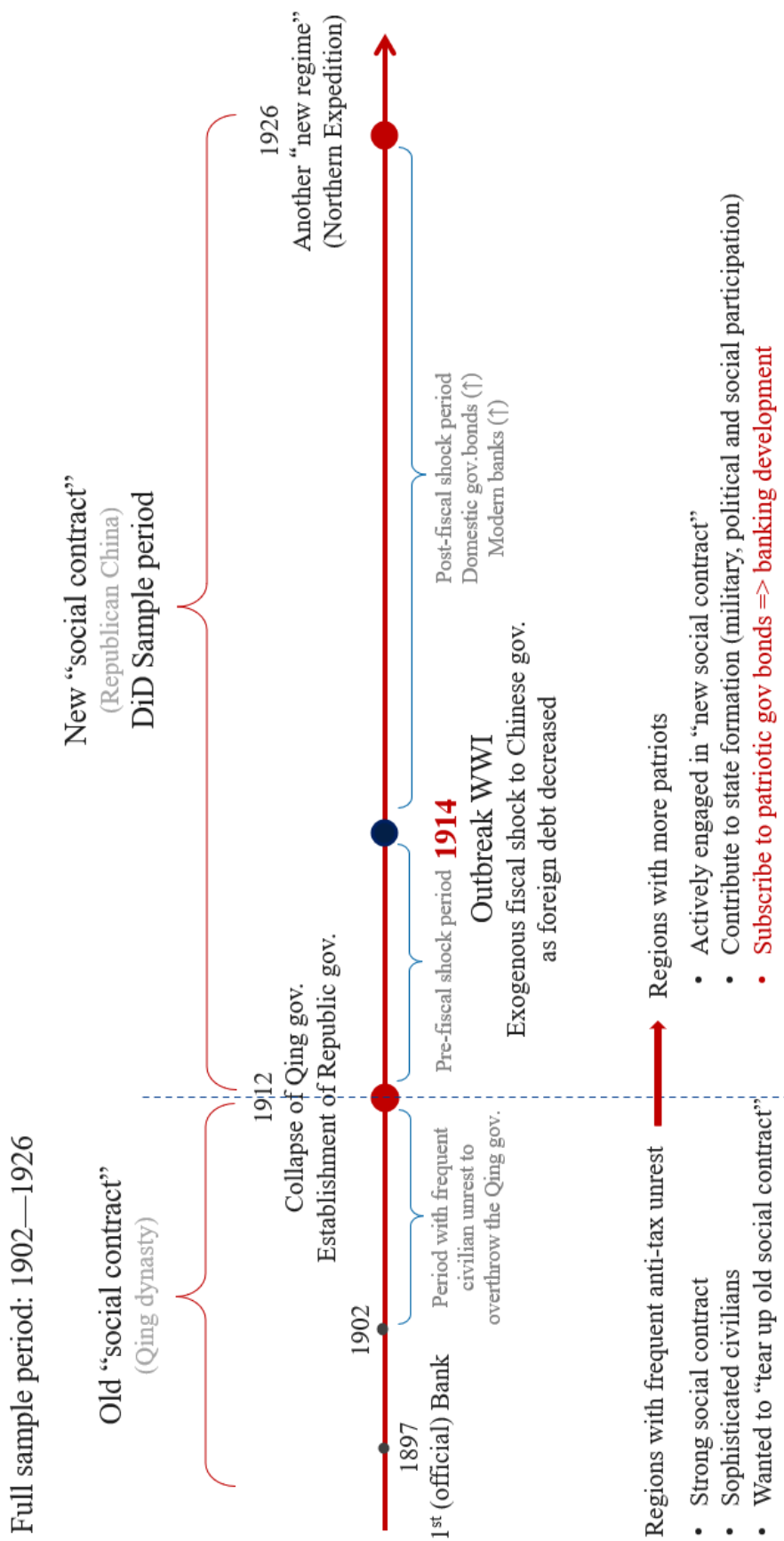


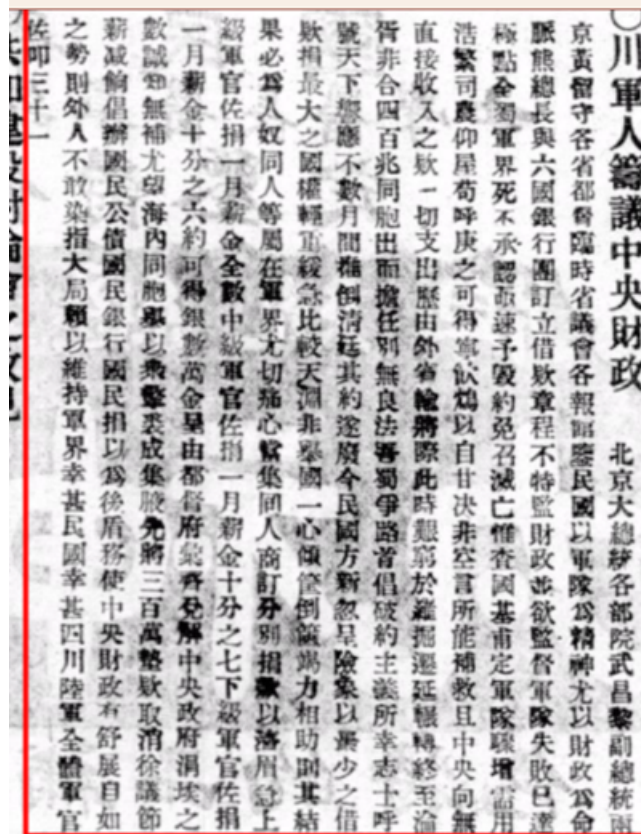
Figure B1. **Timeline**



Panel A: The Qing government's violent crackdown on the Sichuan Railway Protection Movement  
 Source: *Minli Huabao* [Minli Pictorial], 1911. *Xinnian Guangan Hua* [New Year's Impressions Illustration], p.1



Panel B: Monument to remember the martyrs killed in the Railway Protection Movement  
 Source: [https://en.wikipedia.org/wiki/Railway\\_Protection\\_Movement](https://en.wikipedia.org/wiki/Railway_Protection_Movement)  
 Location: People's Park, Chengdu, Sichuan province



Panel C: Chuanjunren Chouyi Zhongyang Caizheng [Sichuan Military Discusses Central Finance]  
Source: Minli Bao (Minli Daily), June 3, 1912.

Figure B1. Sichuan Railway Protection Movement

Authors' translation of [Sichuan Military Discusses Central Finance] (emphasis added):

*To President, Ministries, and Courts in Beijing; Vice President Li in Wuchang; Governor-General Huang in the Two Capitals; Governors and Temporary Provincial Assemblies of All Provinces; Newspapers and Media Outlets:*

*The Republic relies on its military for strength and on its finances for survival. Minister Xiong has negotiated loan agreements with the consortium of six foreign banks, not only placing financial oversight in foreign hands but also subjecting our military to foreign control—this is an utter failure. The military community in Sichuan firmly refuses to recognize this agreement and demands its immediate annulment to prevent national ruin.*

*However, we recognize that the foundations of the Republic are still fragile. The sudden expansion of the military requires vast resources, and the Treasury is already depleted. Though securing funds is urgent, resorting to foreign loans is akin to drinking poison to quench thirst—mere words cannot resolve this crisis. The central government lacks independent revenue and has always relied on provincial contributions for expenses. In this time of hardship, such indirect and delayed funding will inevitably lead to collapse. There is no solution except for all four hundred million compatriots to shoulder this burden together.*

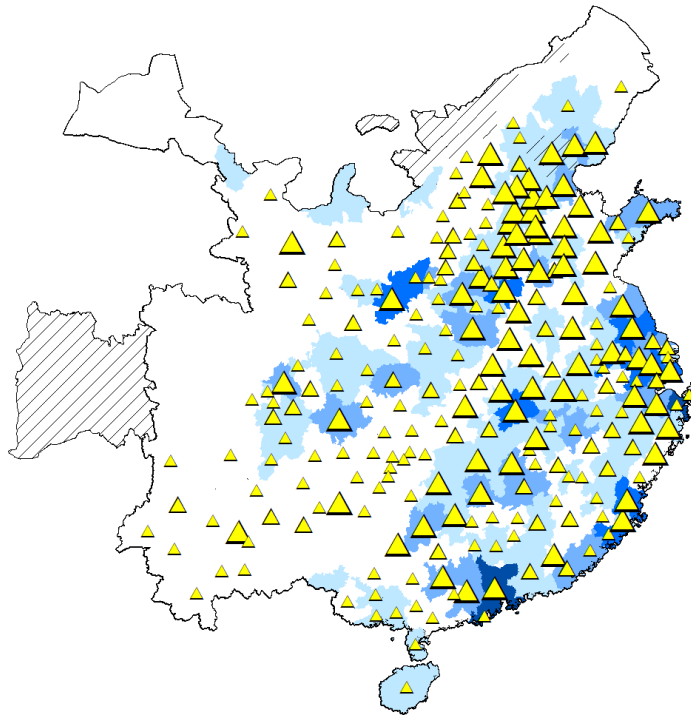
*Sichuan took the lead in resisting the Qing government's foreign railway loans, and thanks to the efforts of patriots nationwide, the imperial court was overthrown within months, rendering the treaty void. Now, as our newly founded Republic faces another crisis, the loss of sovereignty in exchange for a minimal loan is an even greater threat. **If the entire nation does not unite and contribute to financial self-sufficiency**, we will inevitably become subjugated once again.*

*As military officers, we feel this responsibility deeply. Thus, we have collectively decided to contribute a portion of our salaries to address the immediate crisis: Senior officers will donate one month's full salary. Mid-level officers will donate 70% of one month's salary. Junior officers will donate 60% of one month's salary.*

*These contributions are expected to raise tens of thousands of silver taels, which will be submitted to the Governor's Office for remittance to the central government. Though this amount is merely a drop in the ocean, we urge our fellow countrymen to unite in this effort. Together, we can eliminate the existing three-million-tael loan, and subsequently, **discuss further cost-saving measures, including salary reductions and promoting national bonds, a national bank, and voluntary citizen donations.** Only by stabilizing the central government's finances can we ensure national sovereignty and deter foreign interference.*

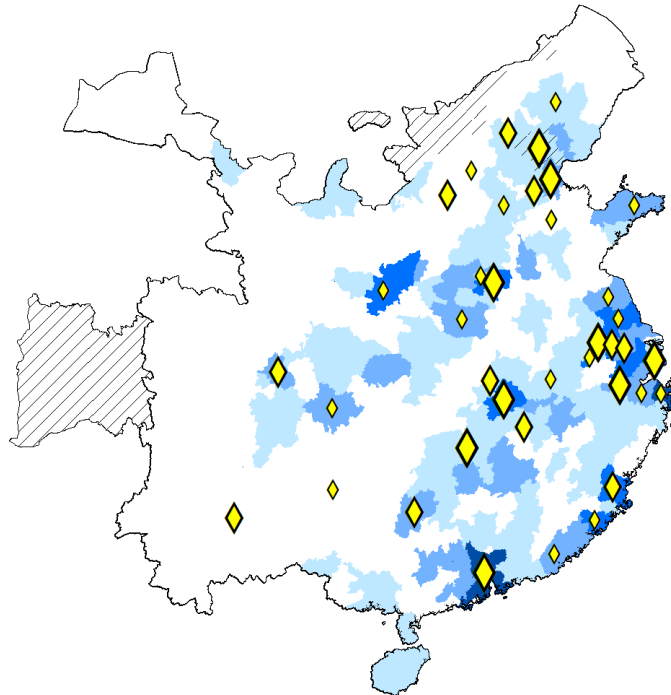
*This is vital for both the military and the Republic's survival.*

*Signed by all officers of the Sichuan Army*



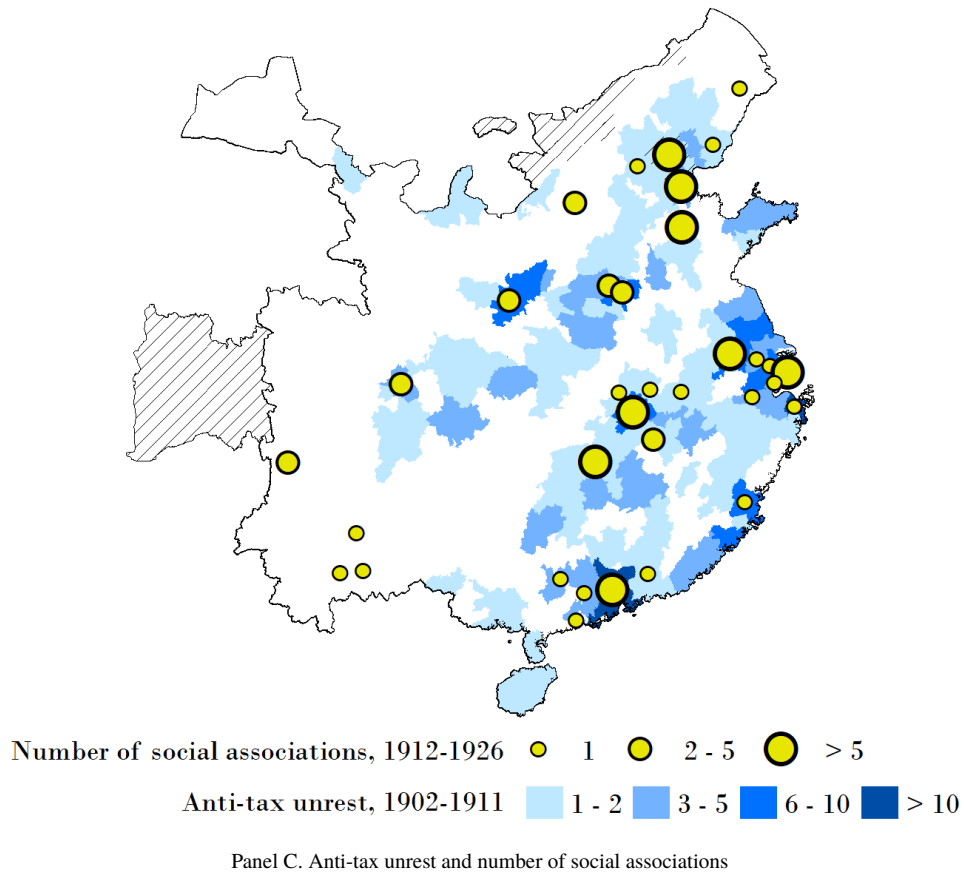
Cadets in Baoding Academy, 1912-1923    ▲ 1 - 10    ▲ 11 - 20    ▲ > 20  
 Anti-tax unrest, 1902-1911    1 - 2    3 - 5    6 - 10    > 10

Panel A. Anti-tax unrest and number of cadets at the Baoding Military Academy



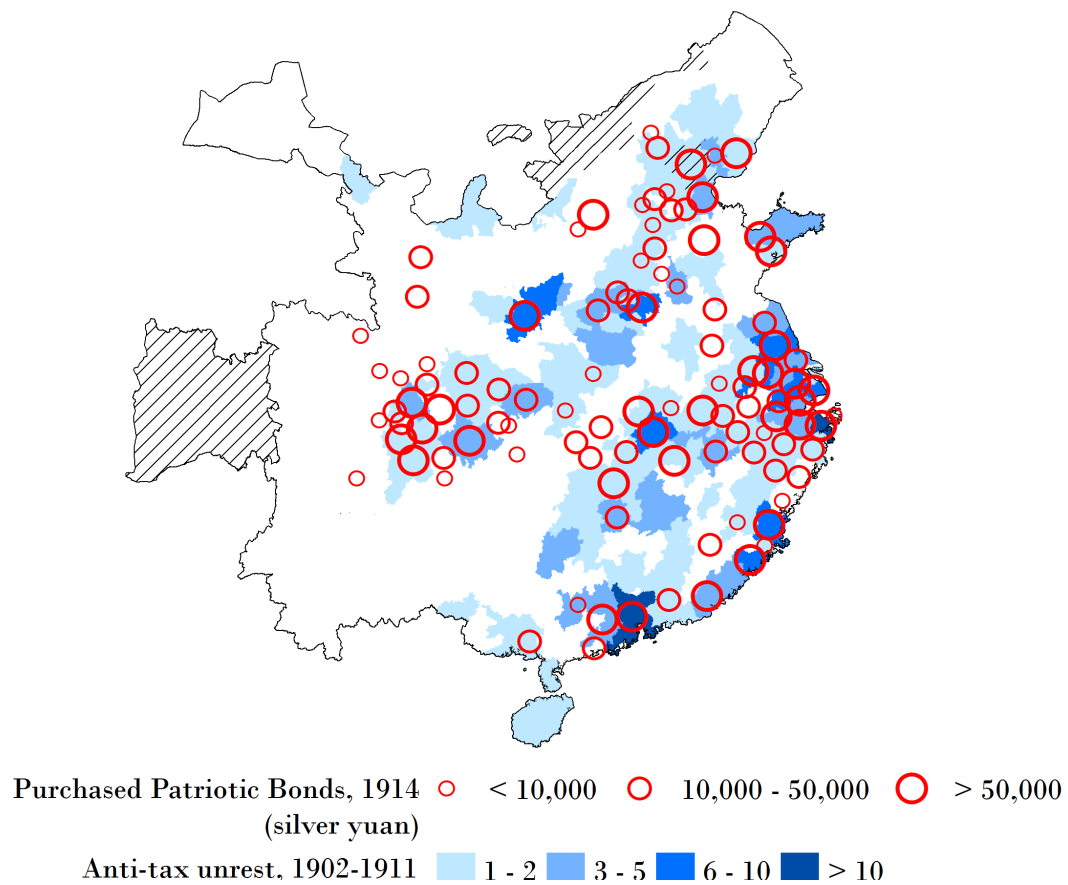
Number of political associations, 1912-1926    ◆ 1    ◆ 2 - 5    ◆ > 5  
 Anti-tax unrest, 1902-1911    1 - 2    3 - 5    6 - 10    > 10

Panel B. Anti-tax unrest and number of political associations



**Figure B3. Distributions of anti-tax unrest and civic engagement measures**

Notes: This figure maps the prefectural-level correlation between anti-tax unrest (1902–1911) and subsequent civic engagement. The panels display three proxies for Republican-era civic engagement: military participation (number of Baoding Military Academy cadets, 1912–1923; Panel A), political participation (number of political associations, 1912–1926; Panel B), and social participation (number of social associations, 1912–1926; Panel C). Prefectures outside the regression sample are shaded.



**Figure B4. Distribution of anti-Tax unrest and 1914 Patriotic Bond subscriptions**

Notes: This figure displays the spatial distribution of anti-tax unrest and prefecture-level subscriptions to the 1914 Patriotic Bond. Subscription data are drawn from the *Quanguo Baokan Suoyin* (National Newspapers and Periodical Index), the most comprehensive digital archive of Chinese newspapers and periodicals hosted by the Shanghai Library.

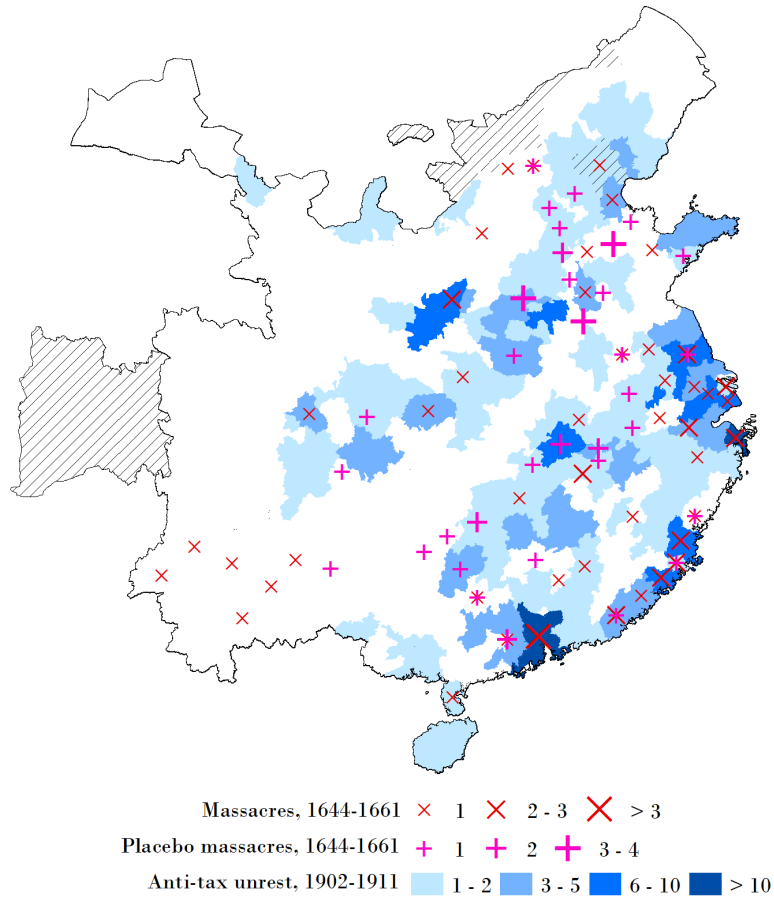
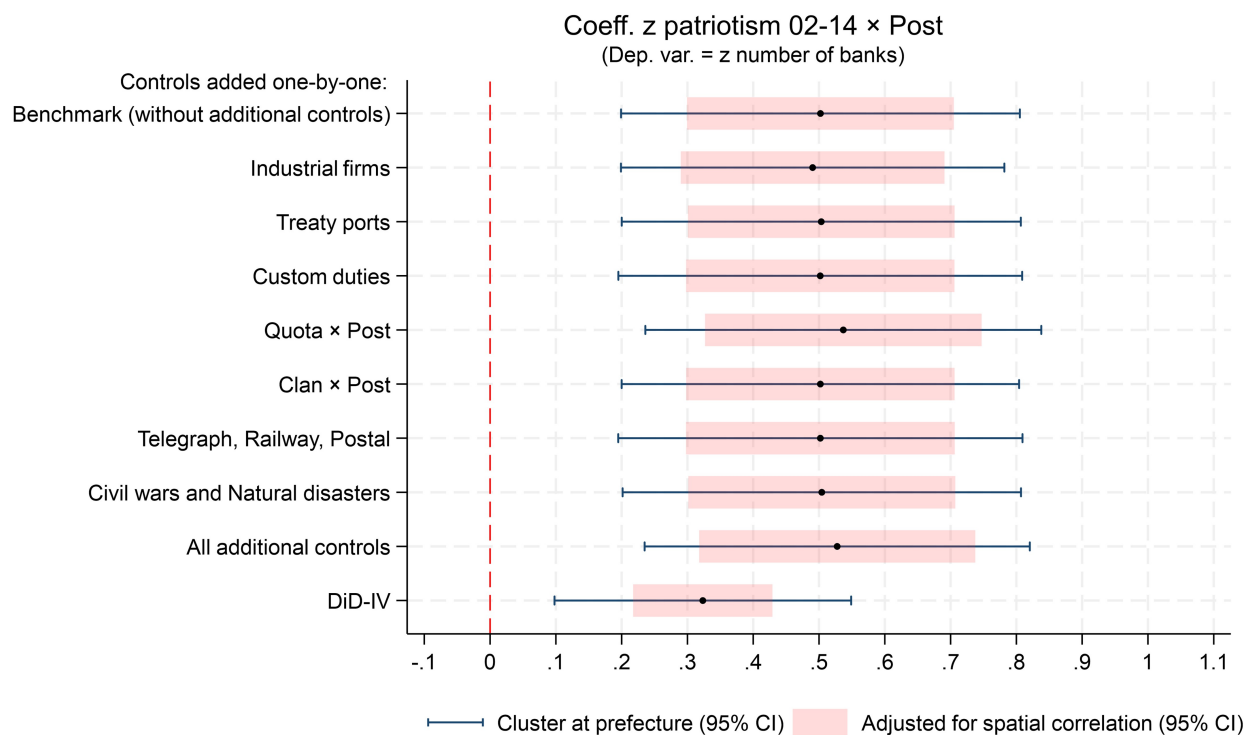


Figure B5. **Placebo IV: non-Qing massacres**

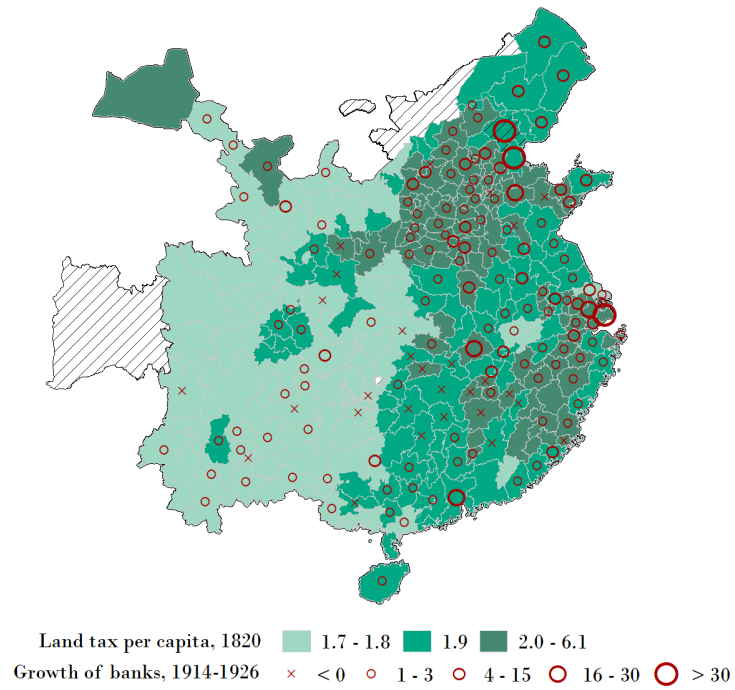
Notes: This figure depicts the spatial distribution of anti-tax unrest, historical massacres by the Qing armies taking place from 1644 through 1661 (IV), and the massacres by non-Qing forces (mostly by Li Zicheng and Zhang Xianzhong) between 1644 and 1661. The data are at the prefectural level. The map covers 298 prefectures from 18 provinces in China proper; prefectures outside the regression sample are shaded.

**A brief historical background.** After the fall of Beijing in 1644, historical peasant revolt leaders like Li Zicheng and Zhang Xianzhong carried out violent campaigns marked by brutal killings and retaliatory slaughters. These acts often stemmed from political distress, local resistance, or an attempt to consolidate power amid crumbling authority. Li's retreating forces and Zhang's reign in Sichuan became notorious for their ruthless treatment of civilians, sometimes bordering on indiscriminate violence. However, unlike the Qing massacres which were largely state-sanctioned and aimed at suppressing Ming loyalists and asserting imperial rule, the violence under Li and Zhang was more erratic, factional, and driven by fear of betrayal or rebellion from within. Their actions, while devastating, reflected the chaotic and fragmented nature of post-Ming resistance rather than the calculated brutality of a conquering regime.

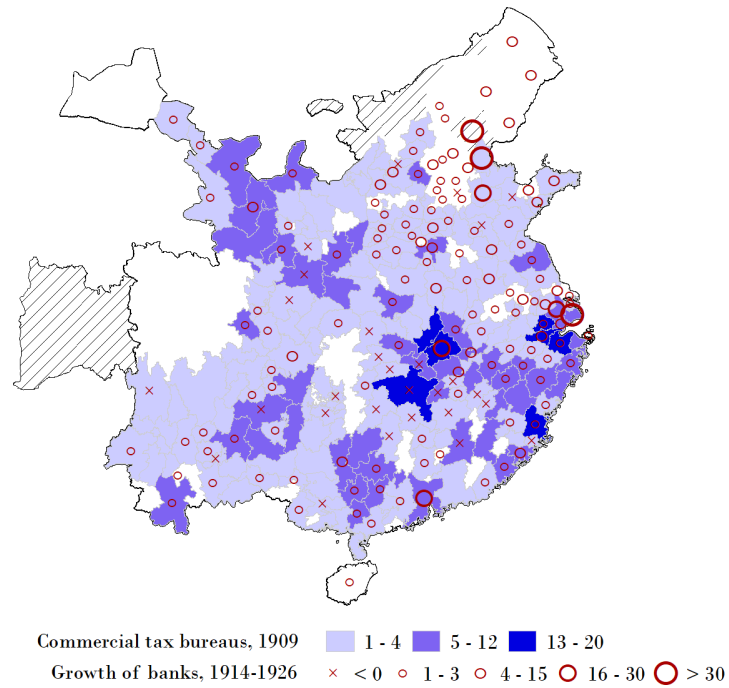


**Figure B6. Additional controls (z-score)**

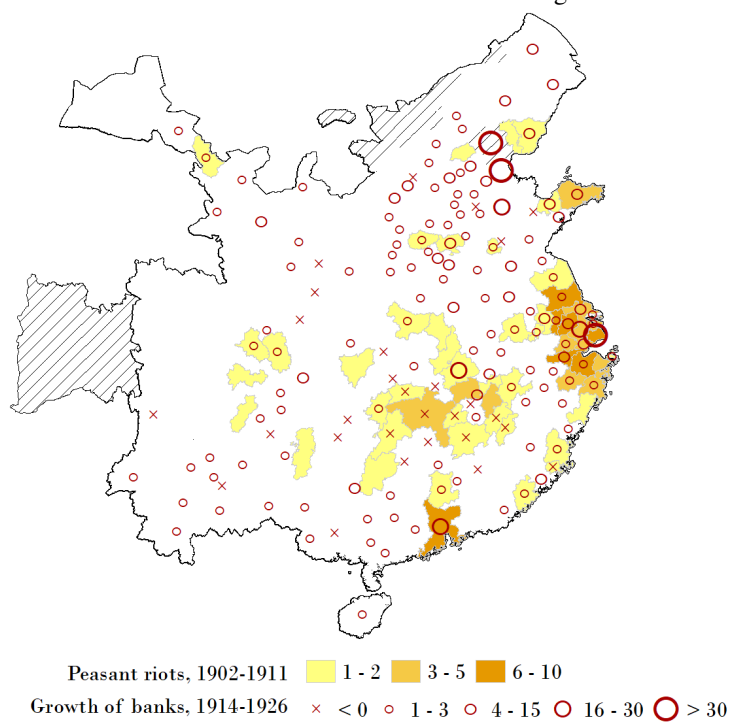
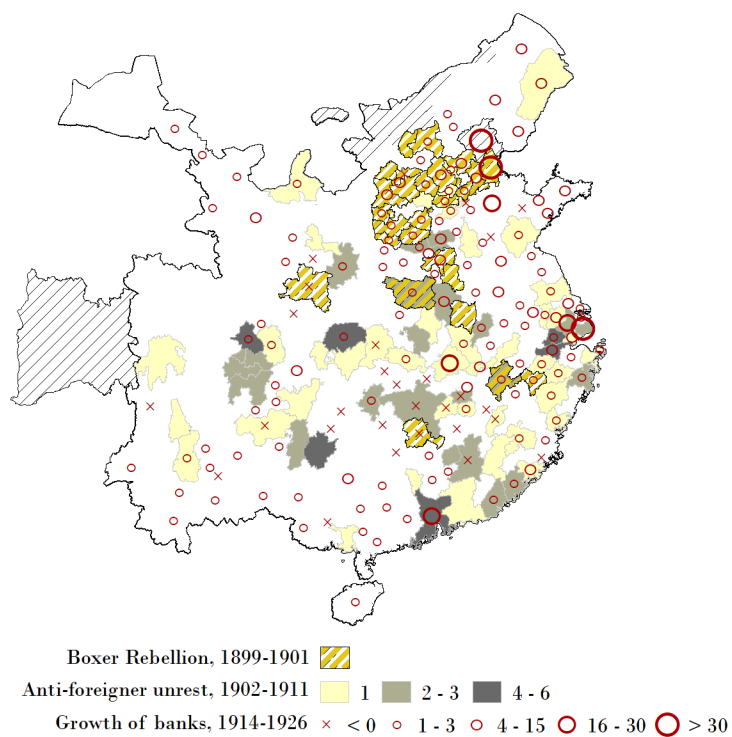
Note: This figure is identical to Figure 5 with two exceptions. First, the dependent variable, Number of banks, is standardized. Second, the main independent variable is  $z$  patriotism 02-14  $\times$  Post, where  $z$  patriotism 02-14 is the z-score of the average z-scores of anti-tax unrest and the three civic engagement measures before 1914.

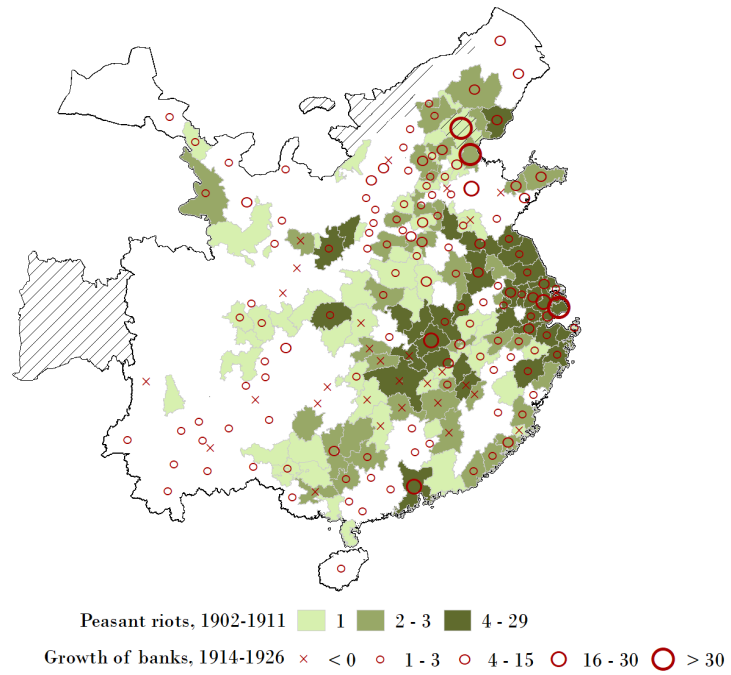


Panel A. Land tax per capita in 1820

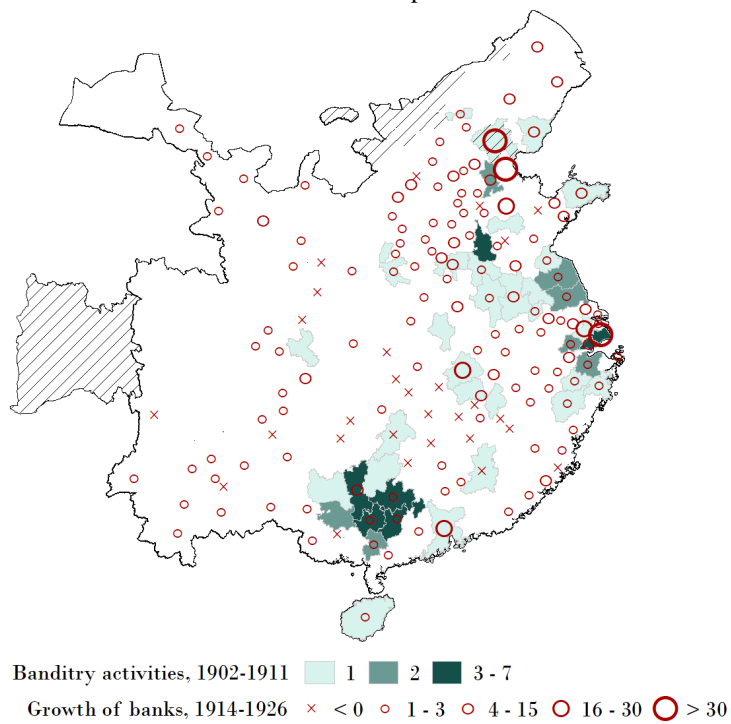


Panel B. Number of commercial tax bureaus

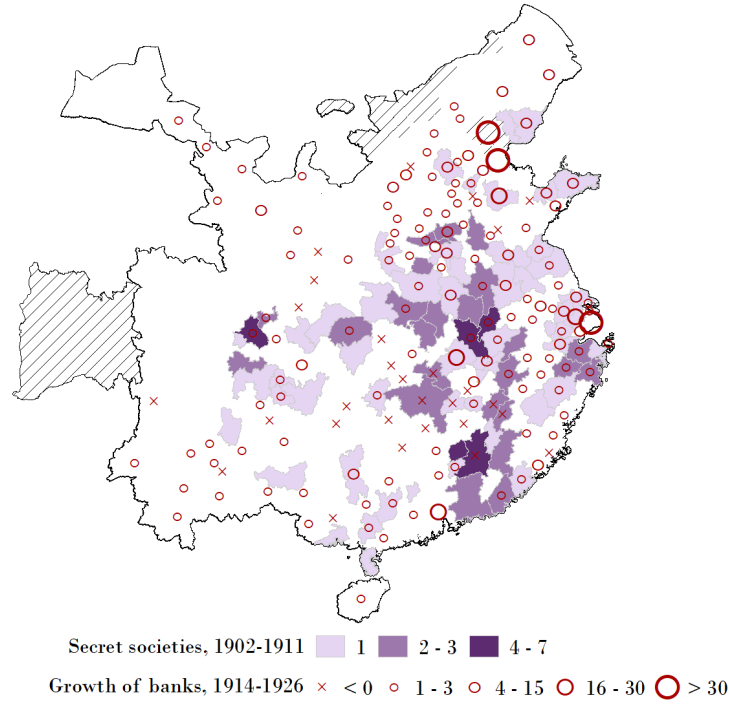




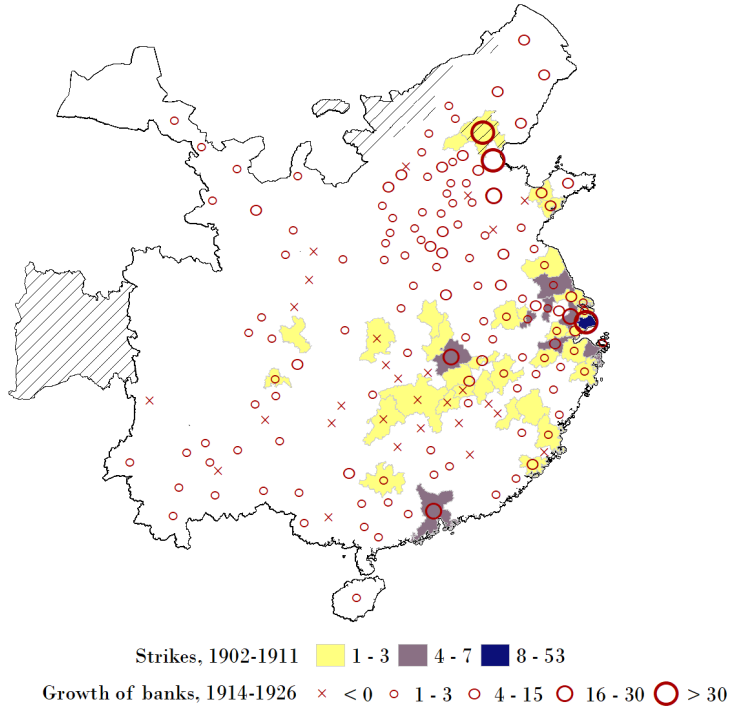
Panel E. Number of peasant riots



Panel F. Number of banditry activities



Panel G. Number of secret societies



Panel H. Number of strikes

Figure B10. The distribution of placebo treat variables

Notes: These figures map the growth of modern banks and of the placebo “treat” variables as discussed in Section 5.4. The growth of modern banks refers to the difference in the number of banks between 1926 and 1914. The placebo “treat” variables include: the level of land tax per capita in 1820 (Panel A), the number of commercial tax bureaus around 1909 (Panel B), the number of anti-foreigner unrest events 1902–1911 and the occurrence of the Boxer Rebellion (Panel C), the number of anti-gentry unrest events 1902–1911 (Panel D), the number of peasant riots 1902–1911 (Panel E), the number of banditry activities from 1902–1911 (Panel F), the number of secret societies from 1902–1911 (Panel G), and the number of strikes from 1902–1911 (Panel H).

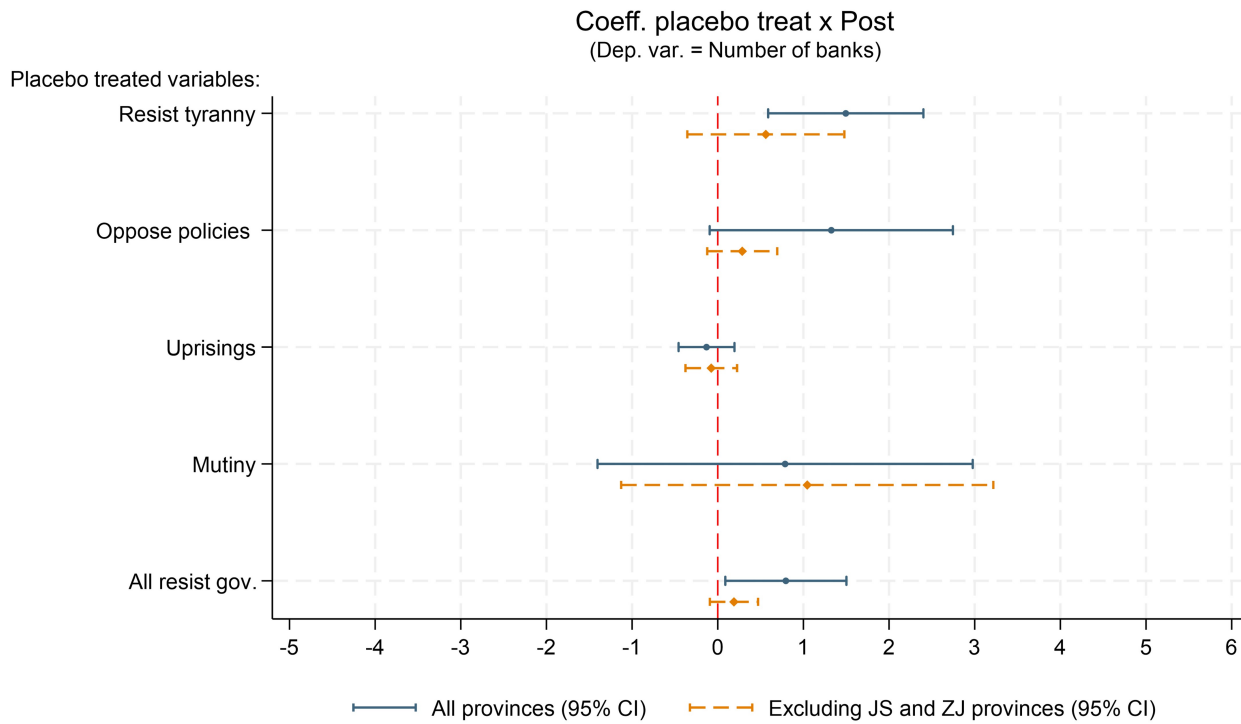


Figure B11. **Falsification tests: alternative anti-government unrest**

Note: This figure displays the coefficients of alternative anti-government unrest as placebo treatment variables. In all models, the dependent variable is the number of modern Chinese banks in each prefecture for each year, and the placebo explanatory variable is structured as treat  $\times$  post. The placebo “treat” variables include the incidence of (1) resistance to tyranny (collective protests against abusive local officials) (51 events), (2) opposition to specific late-Qing policies (115 events), (3) generic uprisings (37 events), (4) military mutinies (20 events), and (5) the sum of the above between 1902 and 1911 (221 events). Note that an event can take multiple forms. When aggregating, we count one for each event. All controls and model specifications are identical to those in column 2 of Table 4. Coefficients for the placebo treatment variables are reported for the full sample and for a subsample that excludes Zhejiang and Jiangsu provinces.



**Figure B12. Robert Morris Statue**

Note: This is an author-taken photograph of the Robert Morris statue located beside the First Bank of the United States in Philadelphia. The statue commemorates Robert Morris (1734–1806), a Founding Father who served as a patriot, statesman, and financier of the American Revolution. Morris played a pivotal role in financing the Revolutionary War and was instrumental in establishing the nation’s first congressionally chartered national bank, Bank of North America.

Table B1. Anti-tax unrest and civic engagement (full table)

This table examines the effects of anti-tax unrest on civic engagement activities. The regression models are identical to Table 2 except that all coefficients of the controls are explicitly presented here. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

Panel A. Military participation

|                      | Number of cadets at the Baoding Military Academy |                       |                               |                      |
|----------------------|--------------------------------------------------|-----------------------|-------------------------------|----------------------|
|                      | After 1912                                       |                       | Diff.(After 1912-Before 1912) |                      |
|                      | 1                                                | 2                     | 3                             | 4                    |
| Anti-tax unrest      | 6.509***<br>(1.415)                              | 3.513**<br>(1.433)    | 4.598***<br>(1.064)           | 3.053***<br>(1.006)  |
| Pop. density in 1910 |                                                  | 10.281***<br>(2.578)  |                               | 7.311***<br>(1.690)  |
| Land area            |                                                  | 13.903***<br>(2.860)  |                               | 8.411***<br>(2.091)  |
| Distance to coast    |                                                  | 1.825<br>(2.544)      |                               | 0.854<br>(1.980)     |
| River density        |                                                  | 3.118<br>(2.998)      |                               | 0.814<br>(2.050)     |
| Altitude             |                                                  | -5.098<br>(3.698)     |                               | -0.757<br>(2.187)    |
| Ruggedness           |                                                  | 650.445*<br>(360.052) |                               | 164.407<br>(170.144) |
| Distance to Beijing  |                                                  | -17.763**<br>(7.322)  |                               | -6.178*<br>(3.238)   |
| R-squared            | 0.094                                            | 0.256                 | 0.121                         | 0.210                |
| Observations         | 286                                              | 286                   | 286                           | 286                  |

Panel B. Political and social participation

|                      | Number of political associations |                     | Number of social associations |                    |
|----------------------|----------------------------------|---------------------|-------------------------------|--------------------|
|                      | 1                                | 2                   | 3                             | 4                  |
| Anti-tax unrest      | 0.939***<br>(0.356)              | 0.906***<br>(0.349) | 0.621***<br>(0.222)           | 0.599**<br>(0.237) |
| Pop. density in 1910 |                                  | -0.308<br>(0.322)   |                               | -0.107<br>(0.191)  |
| Land area            |                                  | 0.125<br>(0.219)    |                               | 0.310<br>(0.194)   |
| Distance to coast    |                                  | 0.174<br>(0.147)    |                               | 0.188<br>(0.138)   |
| River density        |                                  | 0.871*<br>(0.520)   |                               | 0.638**<br>(0.280) |
| Altitude             |                                  | -0.618<br>(0.698)   |                               | -0.377<br>(0.335)  |
| Ruggedness           |                                  | 28.795<br>(41.580)  |                               | 21.498<br>(23.148) |
| Distance to Beijing  |                                  | 0.200<br>(0.289)    |                               | -0.030<br>(0.228)  |
| R-squared            | 0.155                            | 0.149               | 0.178                         | 0.176              |
| Observations         | 286                              | 286                 | 286                           | 286                |

Table B1. Anti-tax unrest and civic engagement (full table) (cont.)

C. Principal component (PC) and standardization (z-score)

|                      | PC civic engagement 12–26 |                     | z civic engagement 12–26 |                     |
|----------------------|---------------------------|---------------------|--------------------------|---------------------|
|                      | 1                         | 2                   | 3                        | 4                   |
| Anti-tax unrest      | 0.299***<br>(0.092)       | 0.268***<br>(0.098) |                          |                     |
| z anti-tax unrest    |                           |                     | 0.480***<br>(0.130)      | 0.407***<br>(0.144) |
| Pop. density in 1910 |                           | 0.013<br>(0.083)    |                          | 0.056<br>(0.058)    |
| Land area            |                           | 0.183**<br>(0.078)  |                          | 0.181***<br>(0.060) |
| Distance to coast    |                           | 0.076<br>(0.051)    |                          | 0.056<br>(0.040)    |
| River density        |                           | 0.270**<br>(0.125)  |                          | 0.185**<br>(0.084)  |
| Altitude             |                           | -0.196<br>(0.162)   |                          | -0.148<br>(0.108)   |
| Ruggedness           |                           | 12.969<br>(10.442)  |                          | 11.258<br>(7.390)   |
| Distance to Beijing  |                           | -0.110<br>(0.099)   |                          | -0.152*<br>(0.088)  |
| R-squared            | 0.216                     | 0.230               | 0.227                    | 0.265               |
| Observations         | 286                       | 286                 | 286                      | 286                 |

Table B2. Anti-tax unrest and government bond face value

This table examines the effects of anti-tax unrest on government bond target (face) value. The dependent variable, target value (in million yuan), refers to the total face value of provincial government bonds issued in each year. The main explanatory variable, *Anti-tax unrest*  $\times$  *Post*, is the number of anti-tax unrest events in each province during 1902–1911 interacting with the post-1914 dummy. Other controls follow Table 3. Standard errors are clustered at the province level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                               | Target value (in million yuan) |                      |                      |
|-------------------------------|--------------------------------|----------------------|----------------------|
|                               | All bonds                      | Voluntary bonds      | Compulsory bonds     |
|                               | 1                              | 2                    | 3                    |
| Anti-tax unrest $\times$ Post | -0.054***<br>(0.009)           | -0.043***<br>(0.012) | -0.025***<br>(0.008) |
| Interest rate (weighted avg)  | 11.143<br>(12.278)             | 32.523*<br>(17.534)  | 10.010<br>(12.470)   |
| Collateral (weighted avg)     | 0.510<br>(0.592)               | -0.086<br>(1.208)    | 0.407<br>(0.703)     |
| Maturity (weighted avg)       | 0.278**<br>(0.126)             | 0.188<br>(0.133)     | 0.268*<br>(0.149)    |
| Controls $\times$ Post        | Y                              | Y                    | Y                    |
| Province and Year FE          | Y                              | Y                    | Y                    |
| R-squared                     | 0.271                          | 0.370                | 0.176                |
| Observations                  | 255                            | 255                  | 255                  |
| Mean of dependent variable    | 0.879                          | 0.382                | 0.425                |

Table B3. **Anti-tax unrest and government bond subscription, robustness**

This table examines the effects of anti-tax unrest on government bond issuance. In panel A, the dependent variable, collected value (in million yuan), refers to the amount of funds raised by provincial government through bond issuance in each year. The main explanatory variable, *PC patriotism 02–14 × Post*, is the patriotism measured by first principal component of anti-tax unrest and the three civic engagement measures between 1912–1914 interacting with the post-1914 dummy. Other controls follow Table 3. In panel B, the dependent variable is the standardized collected value, and the explanatory treatment variable is the standardized anti-tax unrest. Bond-related controls are also standardized. Standard errors are clustered at the province level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

Panel A. Patriotism measured by principal component

|                            | Collected value (in million yuan) |                    |                   |
|----------------------------|-----------------------------------|--------------------|-------------------|
|                            | All bonds                         | Voluntary bonds    | Compulsory bonds  |
|                            | 1                                 | 2                  | 3                 |
| PC patriotism 02–14 × Post | -0.006<br>(0.119)                 | 0.148**<br>(0.057) | -0.087<br>(0.064) |
| Bond controls              | Y                                 | Y                  | Y                 |
| Controls × Post            | Y                                 | Y                  | Y                 |
| Province and Year FE       | Y                                 | Y                  | Y                 |
| R-squared                  | 0.764                             | 0.729              | 0.783             |
| Observations               | 255                               | 255                | 255               |
| Mean of dependent variable | 0.544                             | 0.226              | 0.276             |

Panel B. Standardization

|                           | z collected value (in million yuan) |                   |                   |
|---------------------------|-------------------------------------|-------------------|-------------------|
|                           | All bonds                           | Voluntary bonds   | Compulsory bonds  |
|                           | 1                                   | 2                 | 3                 |
| z patriotism 02–14 × Post | -0.019<br>(0.076)                   | 0.103*<br>(0.050) | -0.090<br>(0.065) |
| Bond controls             | Y                                   | Y                 | Y                 |
| Controls × Post           | Y                                   | Y                 | Y                 |
| Province and Year FE      | Y                                   | Y                 | Y                 |
| R-squared                 | 0.764                               | 0.728             | 0.783             |
| Observations              | 255                                 | 255               | 255               |

**Table B4. Government bond subscription: Alternative investment opportunities**

This table examines the effects of anti-tax unrest on government bond issuance while controlling for the additional measure of alternative investment opportunities, measured by the local loan rate, which is the average interest rate of local loans within a province in a given year. Other model specifications and variable definitions are identical to Table 3. Panel A focuses on the full sample and Panel B excludes Zhejiang and Jiangsu provinces. Standard errors are clustered at the province level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

**Panel A. Full sample**

|                               | Collected value (in million yuan) |                    |                   |
|-------------------------------|-----------------------------------|--------------------|-------------------|
|                               | All bonds                         | Voluntary bonds    | Compulsory bonds  |
|                               | 1                                 | 2                  | 3                 |
| Anti-tax unrest $\times$ Post | 0.020<br>(0.019)                  | 0.034**<br>(0.012) | -0.007<br>(0.008) |
| Local loan rate               | -0.007<br>(0.008)                 | -0.002<br>(0.006)  | -0.003<br>(0.004) |
| Bond controls                 | Y                                 | Y                  | Y                 |
| Controls $\times$ Post        | Y                                 | Y                  | Y                 |
| Province and Year FE          | Y                                 | Y                  | Y                 |
| R-squared                     | 0.834                             | 0.728              | 0.924             |
| Observations                  | 202                               | 202                | 202               |
| Mean of dependent variable    | 0.641                             | 0.276              | 0.312             |

**Panel B. Excluding Zhejiang and Jiangsu provinces**

|                               | Collected value (in million yuan) |                    |                   |
|-------------------------------|-----------------------------------|--------------------|-------------------|
|                               | All bonds                         | Voluntary bonds    | Compulsory bonds  |
|                               | 1                                 | 2                  | 3                 |
| Anti-tax unrest $\times$ Post | 0.020<br>(0.012)                  | 0.021**<br>(0.008) | 0.001<br>(0.005)  |
| Local loan rate               | -0.007<br>(0.009)                 | -0.003<br>(0.007)  | -0.002<br>(0.004) |
| Bond controls                 | Y                                 | Y                  | Y                 |
| Controls $\times$ Post        | Y                                 | Y                  | Y                 |
| Province and Year FE          | Y                                 | Y                  | Y                 |
| R-squared                     | 0.837                             | 0.733              | 0.928             |
| Observations                  | 172                               | 172                | 172               |
| Mean of dependent variable    | 0.616                             | 0.249              | 0.343             |

Table B5. Horse race between anti-tax unrest, civic engagement, and bond subscription

This table examines the horse race between anti-tax unrest and patriotic activities in predicting banking development. Panel A examines civic engagement. The dependent variable in columns 1–4 is the number of banks in each prefecture in each year; column 5 uses its  $z$ -score. The main explanatory variable, *Anti-tax unrest*  $\times$  *Post*, is the number of anti-tax unrest events in each prefecture from 1902 through 1911 interacting with the post-1914 dummy. Civic engagements established prior to the 1914 WWI shock is measured by (i) the number of cadets at the Baoding Military Academy during 1912–1914, (ii) the number of political associations established during 1912–1914, (iii) the number of social associations established during 1912–1914, and (iv) the first principal component of the three. In column 5, all variables are standardized;  $z$  *civic engagement* is the  $z$ -score of the average  $z$ -score of the three measures. Panel B extends the horse race by additionally controlling for 1914 Patriotic bond subscriptions interacted with Post. The dependent variable is the  $z$ -score of the number of banks. Columns 1 and 2 use the standardized total collected value of the 1914 bond, while columns 3 and 4 use standardized per capita collected value; columns 2 and 4 further include  $z$  civic engagement  $\times$  Post. In both panels, standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5%, and 1%, respectively.

|                                                              | Number of banks    |                     |                     |                     | $z$ number of banks |
|--------------------------------------------------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
|                                                              | 1                  | 2                   | 3                   | 4                   | 5                   |
| Anti-tax unrest $\times$ Post                                | 0.371**<br>(0.165) | -0.012<br>(0.109)   | -0.057<br>(0.103)   | -0.193<br>(0.171)   |                     |
| $z$ anti-tax unrest $\times$ Post                            |                    |                     |                     |                     | -0.041<br>(0.063)   |
| Cadets at the Baoding Military Academy 1912–14 $\times$ Post | 0.013<br>(0.024)   |                     |                     |                     |                     |
| Political associations 1912–14 $\times$ Post                 |                    | 0.638***<br>(0.032) |                     |                     |                     |
| Social associations 1912–14 $\times$ Post                    |                    |                     | 1.718***<br>(0.245) |                     |                     |
| PC civic engagement 1912–14 $\times$ Post                    |                    |                     |                     | 2.201***<br>(0.325) |                     |
| $z$ civic engagement 1912–14 $\times$ Post                   |                    |                     |                     |                     | 0.535***<br>(0.132) |
| Controls $\times$ Post                                       | Y                  | Y                   | Y                   | Y                   | Y                   |
| Prefecture and Year FE                                       | Y                  | Y                   | Y                   | Y                   | Y                   |
| R-squared                                                    | 0.755              | 0.789               | 0.791               | 0.794               | 0.786               |
| Observations                                                 | 4290               | 4290                | 3930                | 4290                | 4290                |

Table B5. **Horse race between anti-tax unrest, civic engagement, and bond subscription (cont')**

Panel B. Civic engagement and bond subscription

|                                                             | <i>z</i> number of banks |                    |                     |                     |
|-------------------------------------------------------------|--------------------------|--------------------|---------------------|---------------------|
|                                                             | 1                        | 2                  | 3                   | 4                   |
| <i>z</i> collected value 1914 bond $\times$ Post            | 0.583***<br>(0.185)      | 0.319<br>(0.209)   |                     |                     |
| <i>z</i> collected value per capita 1914 bond $\times$ Post |                          |                    | 0.607***<br>(0.079) | 0.469***<br>(0.140) |
| <i>z</i> civic engagement 1912–14 $\times$ Post             |                          | 0.314*<br>(0.188)  |                     | 0.178<br>(0.150)    |
| <i>z</i> anti-tax unrest $\times$ Post                      | -0.177*<br>(0.107)       | -0.144*<br>(0.079) | -0.102**<br>(0.049) | -0.110**<br>(0.047) |
| Controls $\times$ Post                                      | Y                        | Y                  | Y                   | Y                   |
| Prefecture and Year FE                                      | Y                        | Y                  | Y                   | Y                   |
| R-squared                                                   | 0.785                    | 0.790              | 0.794               | 0.795               |
| Observations                                                | 4290                     | 4290               | 4290                | 4290                |

**Table B6. Reverse causality: Anti-tax unrest and modern banks, 1902–1911**

This table addresses the issue of reverse causality by analyzing panel data on banks and anti-tax unrest events from 1902 through 1911. Columns 1 and 2 examine whether modern banks promote anti-tax unrest. The dependent variable is the number of anti-tax unrest events, while the explanatory variable is the number of banks in each prefecture in each year, examined both in the current year (column 1) and with a one-year lag (column 2). Columns 3 and 4 examine whether anti-tax unrest instantly influenced the number of banks. The dependent variable is the number of banks, and the explanatory variable is the number of anti-tax unrest events in each prefecture in each year, examined both in the current year (column 3) and with a one-year lag (column 4). The number of observations in column 4 shrank because the number of anti-tax unrest events is not available for the year 1901. We control for prefecture and year fixed effects. The analysis encompasses 286 prefectures from 18 provinces within China proper. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                          | Anti-tax unrest  |                   | Number of banks  |                   |
|--------------------------|------------------|-------------------|------------------|-------------------|
|                          | 1                | 2                 | 3                | 4                 |
| Number of banks          | 0.023<br>(0.020) |                   |                  |                   |
| Number of banks (lagged) |                  | -0.002<br>(0.022) |                  |                   |
| Anti-tax unrest          |                  |                   | 0.042<br>(0.041) |                   |
| Anti-tax unrest (lagged) |                  |                   |                  | -0.031<br>(0.051) |
| Prefecture and Year FE   | Y                | Y                 | Y                | Y                 |
| R-squared                | 0.174            | 0.173             | 0.618            | 0.653             |
| Observations             | 2,860            | 2,860             | 2,860            | 2,574             |

Table B7. **Exclusion restriction: Reduced-form regression**

This table provides the reduced-form regression. The dependent variable is the number of modern Chinese banks in each prefecture for each year. In column 1, we regress the number of banks directly on the instrumental variable  $Massacres \times Post$ , where *Massacres* refer to the number of massacres occurring in each prefecture from 1644 through 1661. In column 2, we further include the explanatory variable of interest,  $Anti-tax\ unrest \times Post$ , to provide a horse-race estimation. The controls are identical to those in column 2 of Table 4. Columns 3-4 use standardized variables. Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures, covering the period from 1912 through 1926. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                                           | Number of banks    |                   | <i>z</i> number of banks |                     |
|-------------------------------------------|--------------------|-------------------|--------------------------|---------------------|
|                                           | 1                  | 2                 | 3                        | 4                   |
| Massacres $\times$ Post                   | 1.151**<br>(0.444) | 0.576<br>(0.456)  |                          |                     |
| Anti-tax unrest $\times$ Post             |                    | 0.315*<br>(0.160) |                          |                     |
| <i>z</i> patriotism 1902–14 $\times$ Post |                    |                   | 0.132**<br>(0.051)       | -0.090<br>(0.067)   |
| <i>z</i> massacres $\times$ Post          |                    |                   |                          | 0.542***<br>(0.155) |
| Controls $\times$ Post                    | Y                  | Y                 | Y                        | Y                   |
| Prefecture and Year FE                    | Y                  | Y                 | Y                        | Y                   |
| R-squared                                 | 0.754              | 0.755             | 0.754                    | 0.782               |
| Observations                              | 4,290              | 4,290             | 4,290                    | 4,290               |

Table B8. **Placebo IV: Massacres by late-Ming peasant revolt leaders**

This table reports the first stage of placebo IV. The dependent variable, *Anti-tax unrest*  $\times$  *Post*, is instrumented by *Non-Qing massacres*  $\times$  *Post*, or massacres conducted by non-Qing forces, primarily those led by historical peasant revolt leaders Li Zicheng (1606–1645) and Zhang Xianzhong (1606–1647) and their followers between 1644 and 1661. Model specifications are identical to Panel B of Table 5. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                                     | Anti-tax unrest $\times$ Post |                   |                       | $z$ patriotism 1902–14 $\times$ Post |
|-------------------------------------|-------------------------------|-------------------|-----------------------|--------------------------------------|
|                                     | (1)                           | (2)               | (3)                   | (4)                                  |
| Non-Qing massacres $\times$ Post    | 0.391*<br>(0.236)             | -0.009<br>(0.252) | 0.029<br>(0.260)      |                                      |
| $z$ non-Qing massacre $\times$ Post |                               |                   |                       | -0.017<br>(0.061)                    |
| Controls $\times$ Post              |                               | Y                 | Y                     | Y                                    |
| Prefecture and Year FE              | Y                             | Y                 | Y                     | Y                                    |
| R-squared                           | 0.798                         | 0.840             | 0.829                 | 0.825                                |
| Observations                        | 4,290                         | 4,290             | 3,930                 | 4,290                                |
| Sample                              | Full                          | Full              | Excl. JS &<br>ZJ pro. | Full                                 |

Table B9. **Land tax, commercial tax, and anti-tax unrest**

This table examines whether the level of tax could predict anti-tax unrest. In Panel A, the dependent variable is the number of anti-tax unrest events from 1902 through 1911 in each prefecture. The explanatory variables are the logarithm of per capita land tax in 1820 in each prefecture (column 1) and the number of commercial tax bureaus in each prefecture (column 2). In Panel B, the dependent variable is the number of modern banks in each prefecture for each year. The explanatory variables include both Anti-tax unrest and the measures of tax burden interacting with the post-1914 dummy. In both panels, control variables include the logarithm of prefectural population in 1910, the logarithm of prefectural land area, the logarithm of the prefecture's distance to the coast, the logarithm of the river density within each prefecture, the logarithm of the prefectural altitude, the ruggedness of the terrain, and the logarithm of the prefecture's distance to capital (Beijing). The analysis encompasses 286 prefectures from 18 provinces within China proper. Standard errors are robust and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

Panel A. Anti-tax unrest explained by tax burden

|                                   | Anti-tax unrest    |                     |
|-----------------------------------|--------------------|---------------------|
|                                   | 1                  | 2                   |
| Land tax per capita in 1820 (log) | 2.893**<br>(1.344) |                     |
| Commercial tax bureaus            |                    | 0.258***<br>(0.052) |
| Controls                          | Y                  | Y                   |
| R-squared                         | 0.204              | 0.334               |
| Observations                      | 286                | 286                 |

Panel B. Banks explained by anti-tax unrest and tax burden

|                                                 | Number of banks     |                    |
|-------------------------------------------------|---------------------|--------------------|
|                                                 | 1                   | 2                  |
| Anti-tax unrest $\times$ Post                   | 0.395***<br>(0.141) | 0.300**<br>(0.121) |
| Land tax per capita in 1820 (log) $\times$ Post | 0.781<br>(2.738)    |                    |
| Commercial tax bureaus $\times$ Post            |                     | 0.145<br>(0.106)   |
| Controls $\times$ Post                          | Y                   | Y                  |
| Prefecture and Year FE                          | Y                   | Y                  |
| R-squared                                       | 0.755               | 0.756              |
| Observations                                    | 4,290               | 4,290              |

Table B10. **The effects of anti-tax unrest on official and private banks**

This table examines the effects of anti-tax unrest on official banks (Panel A) and private banks (Panel B). The same regression models are used in both panels, differing only in their dependent variables. In Panel A, the dependent variable is the number of official banks, which were initiated by governmental bodies. In Panel B, the dependent variable is the number of private banks, which were fully owned by private businessmen. The explanatory variable, Anti-tax unrest  $\times$  Post, represents the number of anti-tax unrest events in each county from 1902 through 1911, interacted with a post-1914 dummy. Columns 1–3 report the regression results using OLS, while columns 4–6 report results using the IV-2SLS approach, where *Anti-tax unrest  $\times$  Post* is instrumented by *Massacres  $\times$  Post*. Here, *Massacres* refer to the number occurring in each county from 1644 through 1661. Control variables are added progressively across columns: baseline controls match those in Table 4, column 2 with additional controls as specified in Figure 5. Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures, covering the period from 1912 through 1926. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

Panel A. Official banks

|                               | Number of official banks |                     |                     |                     |                  |                   |
|-------------------------------|--------------------------|---------------------|---------------------|---------------------|------------------|-------------------|
|                               | OLS                      |                     |                     | IV-2SLS             |                  |                   |
|                               | 1                        | 2                   | 3                   | 4                   | 5                | 6                 |
| Anti-tax unrest $\times$ Post | 0.194***<br>(0.038)      | 0.135***<br>(0.047) | 0.160***<br>(0.049) | 0.195***<br>(0.073) | 0.145<br>(0.092) | 0.167*<br>(0.092) |
| Controls $\times$ Post        |                          | Y                   | Y                   |                     | Y                | Y                 |
| Additional controls           |                          |                     | Y                   |                     |                  | Y                 |
| Year and Prefecture FE        | Y                        | Y                   | Y                   | Y                   | Y                | Y                 |
| R-squared                     | 0.821                    | 0.836               | 0.843               |                     |                  |                   |
| Kleibergen–Paap F-stat.       |                          |                     |                     | 16.182              | 10.194           | 8.905             |
| Observations                  | 4,290                    | 4,290               | 4,290               | 4,290               | 4,290            | 4,290             |
| Mean of dep. var.             | 1.171                    | 1.171               | 1.171               | 1.171               | 1.171            | 1.171             |

Panel B. Private banks

|                               | Number of private banks |                   |                   |                    |                   |                   |
|-------------------------------|-------------------------|-------------------|-------------------|--------------------|-------------------|-------------------|
|                               | OLS                     |                   |                   | IV-2SLS            |                   |                   |
|                               | 1                       | 2                 | 3                 | 4                  | 5                 | 6                 |
| Anti-tax unrest $\times$ Post | 0.372**<br>(0.170)      | 0.243*<br>(0.128) | 0.163*<br>(0.089) | 0.510**<br>(0.248) | 0.444*<br>(0.245) | 0.369*<br>(0.214) |
| Controls $\times$ Post        |                         | Y                 | Y                 |                    | Y                 | Y                 |
| Additional controls           |                         |                   | Y                 |                    |                   | Y                 |
| Year and Prefecture FE        | Y                       | Y                 | Y                 | Y                  | Y                 | Y                 |
| R-squared                     | 0.622                   | 0.631             | 0.640             |                    |                   |                   |
| Kleibergen–Paap F-stat.       |                         |                   |                   | 16.182             | 10.194            | 8.905             |
| Observations                  | 4,290                   | 4,290             | 4,290             | 4,290              | 4,290             | 4,290             |
| Mean of dep. var.             | 0.692                   | 0.692             | 0.692             | 0.692              | 0.692             | 0.692             |

Table B11. **Official and private banks**

This table aims to provide suggestive evidence of the penetration of official banks compared to that of private banks. In column 1, the dependent variable is the number of official banks, and the explanatory variables are *Anti-tax unrest*  $\times$  *Post* and the number of private banks (lagged one year). In column 2, the dependent variable is the number of private banks, and the explanatory variables are *Anti-tax unrest*  $\times$  *Post* and the number of official banks (lagged one year). The regressions include baseline controls, as well as the additional controls identical to those in Table 4, column 2 and in Figure 5. Prefecture and year fixed effects are included. The analysis encompasses 286 prefectures, covering the period from 1912 through 1926. Standard errors are clustered at the prefecture level and are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at 10%, 5% and 1%, respectively.

|                               | Official banks     | Private banks       |
|-------------------------------|--------------------|---------------------|
|                               | 1                  | 2                   |
| Anti-tax unrest $\times$ Post | 0.094*<br>(0.052)  |                     |
| Private banks (lagged)        | 0.115**<br>(0.047) | 0.182<br>(0.123)    |
| Official banks (lagged)       |                    | 0.540***<br>(0.187) |
| Controls $\times$ Post        | Yes                | Yes                 |
| Additional controls           | Yes                | Yes                 |
| Prefecture and Year FE        | Yes                | Yes                 |
| R-squared                     | 0.845              | 0.648               |
| Observations                  | 4,290              | 4,290               |

Table B12. Summary statistics for bank level variables

|                             | Obs. | Mean   | S.D.   | Min    | Max     |
|-----------------------------|------|--------|--------|--------|---------|
| Exposure to anti-tax unrest | 114  | 0.223  | 0.685  | 0      | 6       |
| ROA (%)                     | 114  | 3.003  | 1.903  | -0.592 | 9.861   |
| Bond-to-asset ratio (%)     | 114  | 8.446  | 4.908  | 0.272  | 27.594  |
| Total assets (million yuan) | 114  | 43.771 | 85.101 | 1.844  | 505.815 |
| Leverage ratio (%)          | 114  | 65.893 | 12.295 | 31.171 | 92.914  |
| Cash ratio (%)              | 114  | 7.293  | 4.611  | 1.128  | 23.565  |

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